Metabolite Reorganization Results

Christian Anderson

26 February 2026

Contents

Node/Edge Overlaps

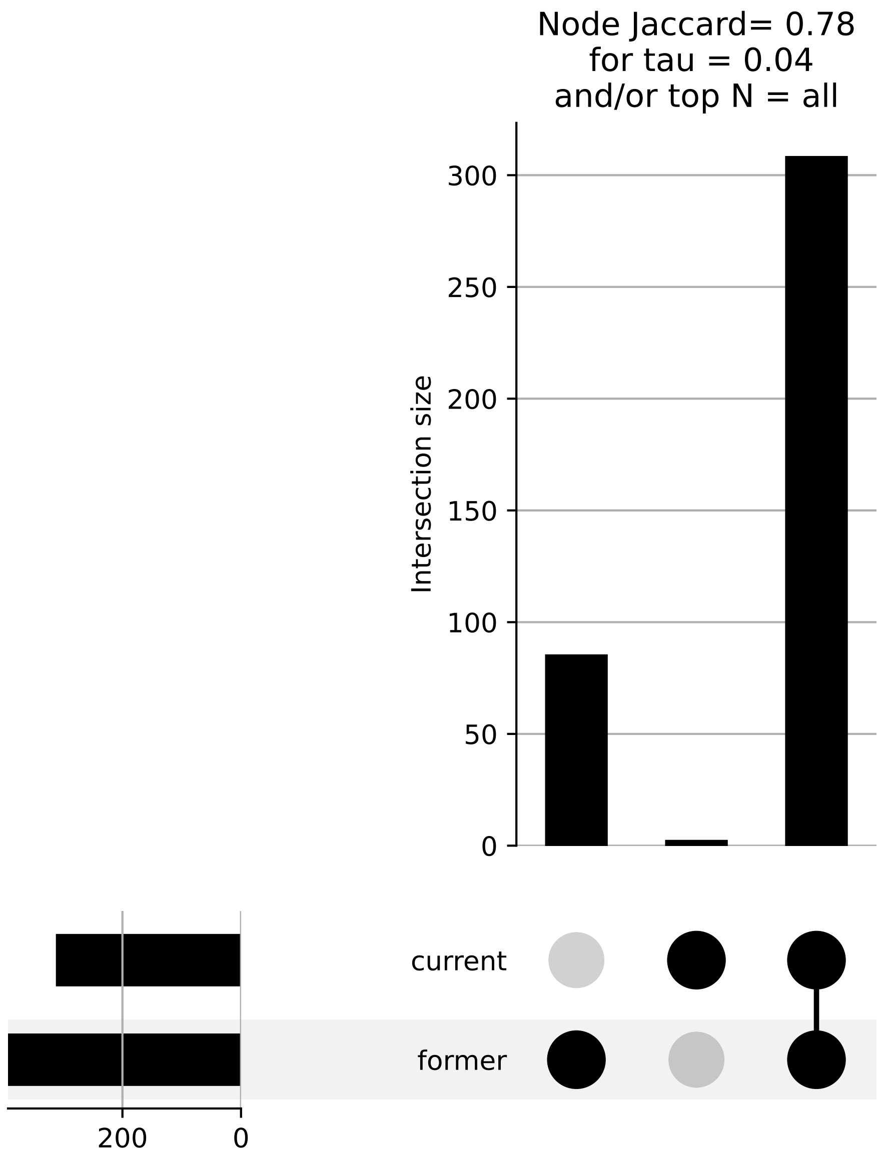


Figure 1: Node intersection and difference

results/005/node-upset.pdf

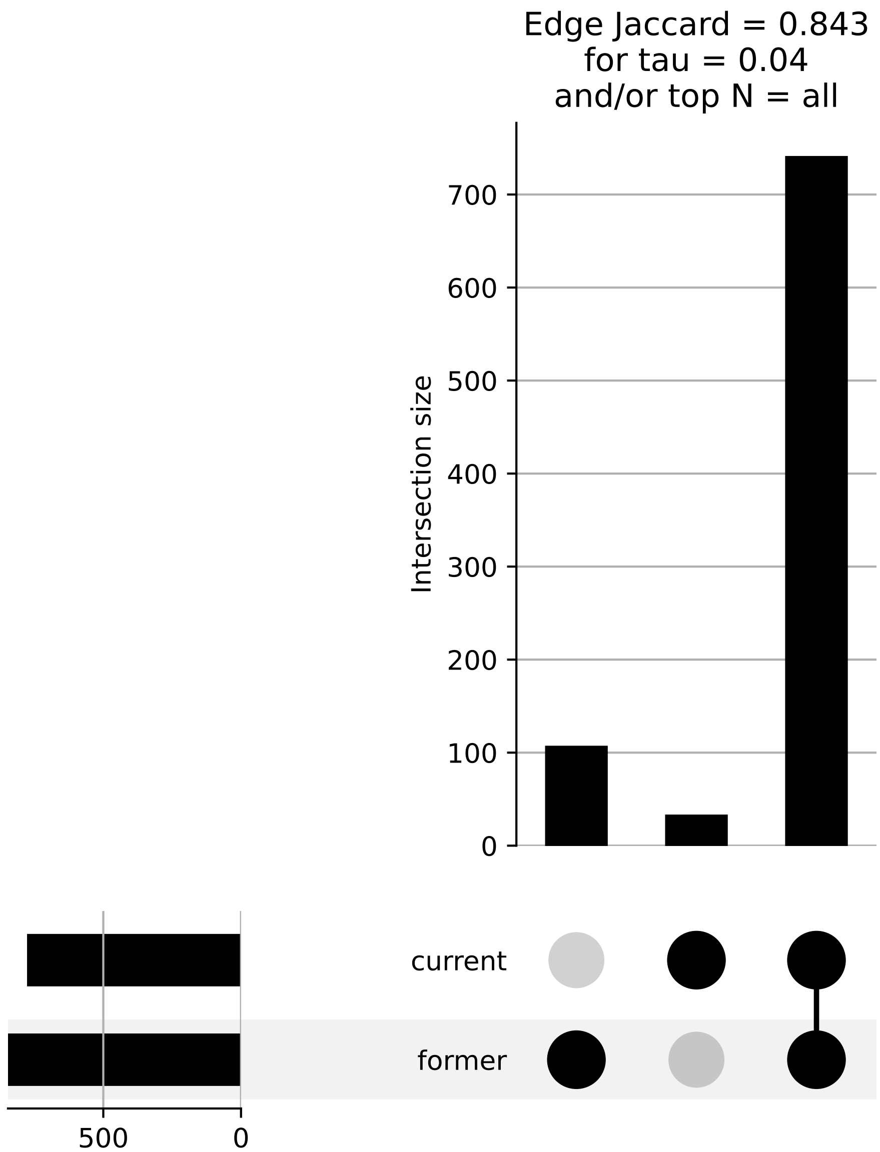


Figure 2: Edge intersection and difference.

results/005/edge-upset.pdf

Current Smokers

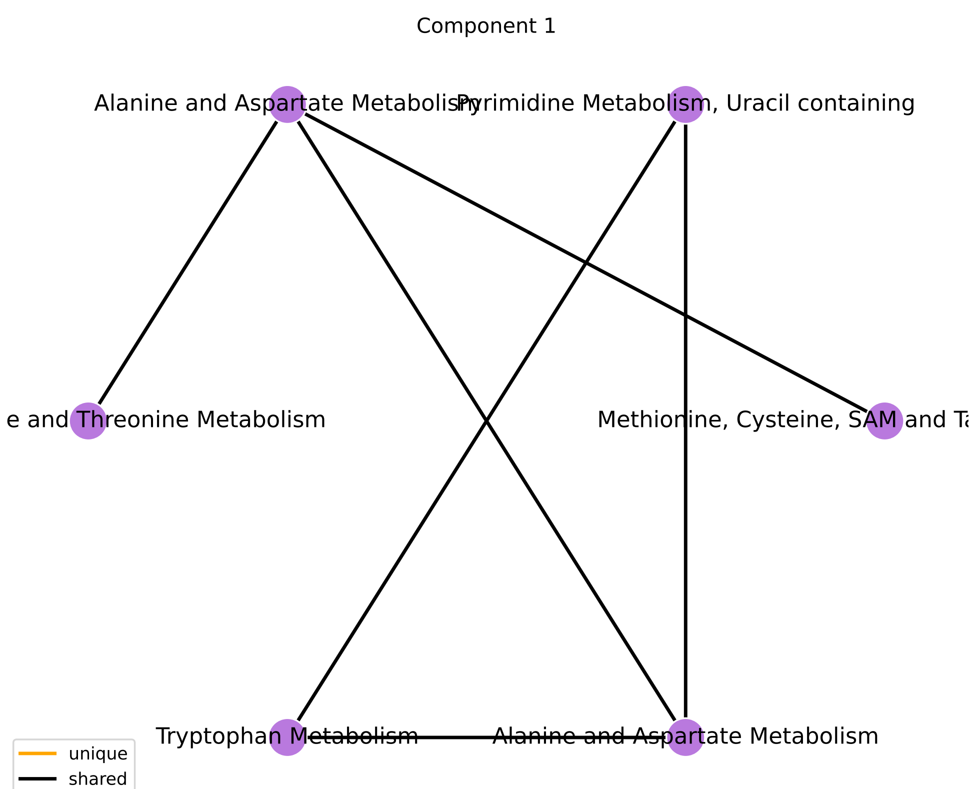
Component 1

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Amino Acid	180	5	8.199	2.104	0.00989	0.089
Nucleotide	32	2	9.536	2.255	0.0318	0.143

results/006/enrichment/curr-components/1-superpathway.csv

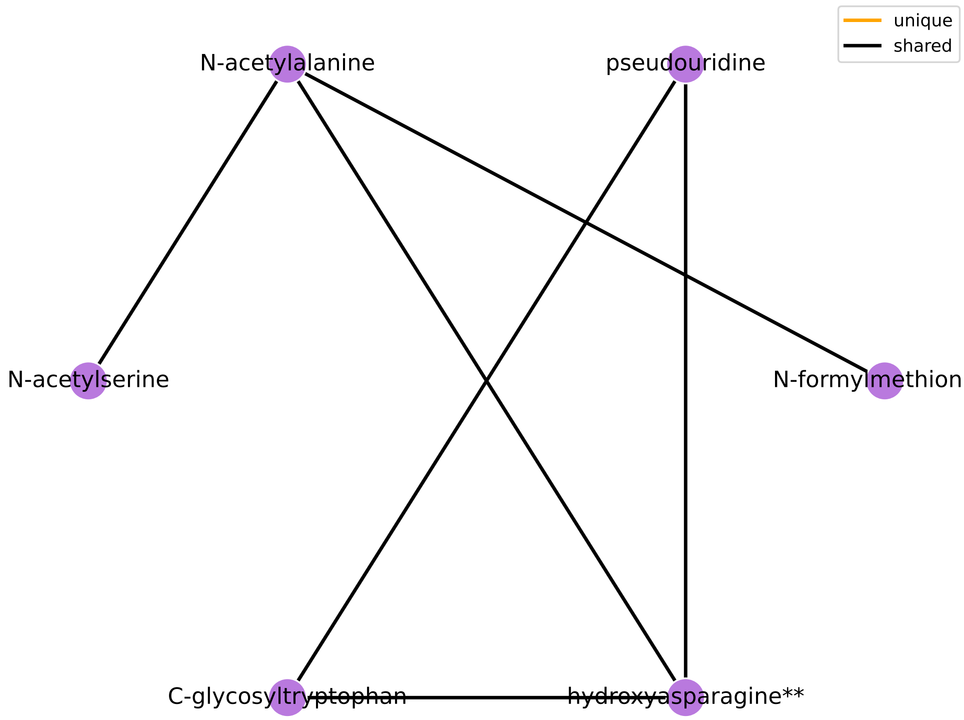
Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Alanine and Aspartate Metabolism	8	2	47.866	3.868	0.002	0.165
Pyrimidine Metabolism, Uracil containing	10	2	36.068	3.585	0.00318	0.165
Glycine, Serine and Threonine Metabolism	10	1	13.562	2.607	0.0891	1
Tryptophan Metabolism	16	1	8.114	2.094	0.139	1
Methionine, Cysteine, SAM and Taurine Metabolism	17	1	7.601	2.028	0.147	1

results/006/enrichment/curr-components/1-subpathway.csv



results/005_1/component-graphs/current/sub_pathway-0.pdf

Component 1



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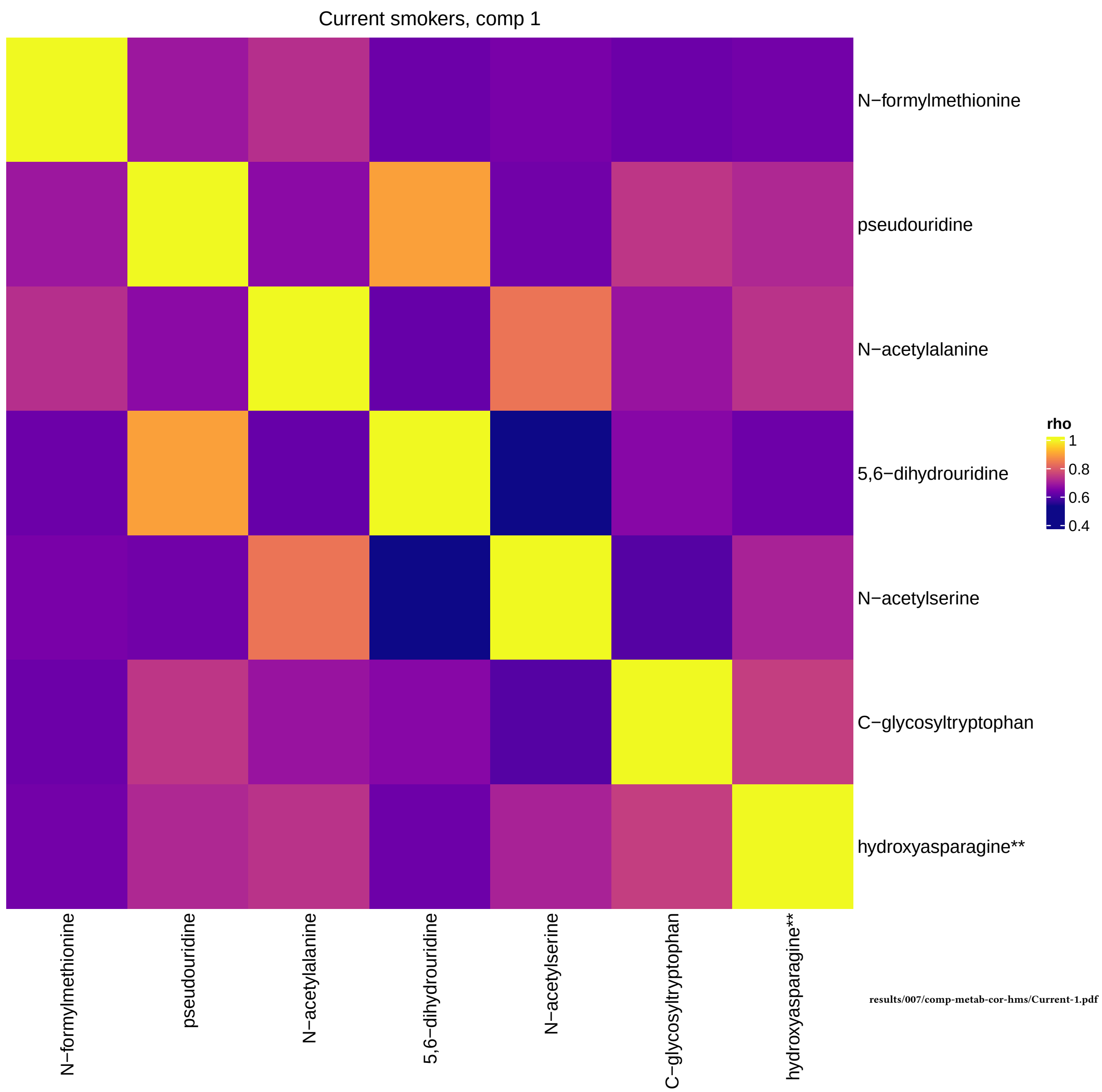


results/007/comp-metab-cor-hms/Current-1.pdf

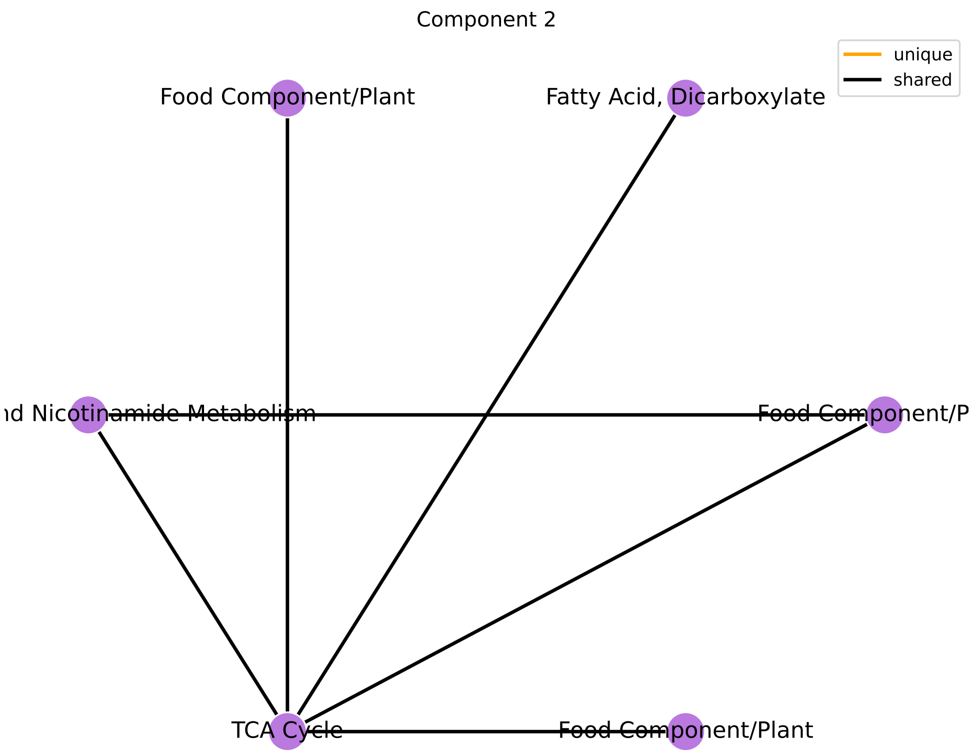
Component 2

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Xenobiotics	96	4	9.501	2.251	0.00627	0.0564
Energy	10	1	13.562	2.607	0.0891	0.401
Cofactors and Vitamins	25	1	5.026	1.615	0.21	0.63
Lipid	363	1	0.179	-1.72	0.99	1

results/006/enrichment/curr-components/2-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Food Component/Plant	38	3	15.174	2.72	0.00355	0.369
Nicotinate and Nicotinamide Metabolism	5	1	30.252	3.41	0.0454	1
TCA Cycle	9	1	15.248	2.724	0.0805	1
Chemical	18	1	7.145	1.966	0.155	1
Fatty Acid, Dicarboxylate	23	1	5.494	1.704	0.195	1

results/006/enrichment/curr-components/2-subpathway.csv



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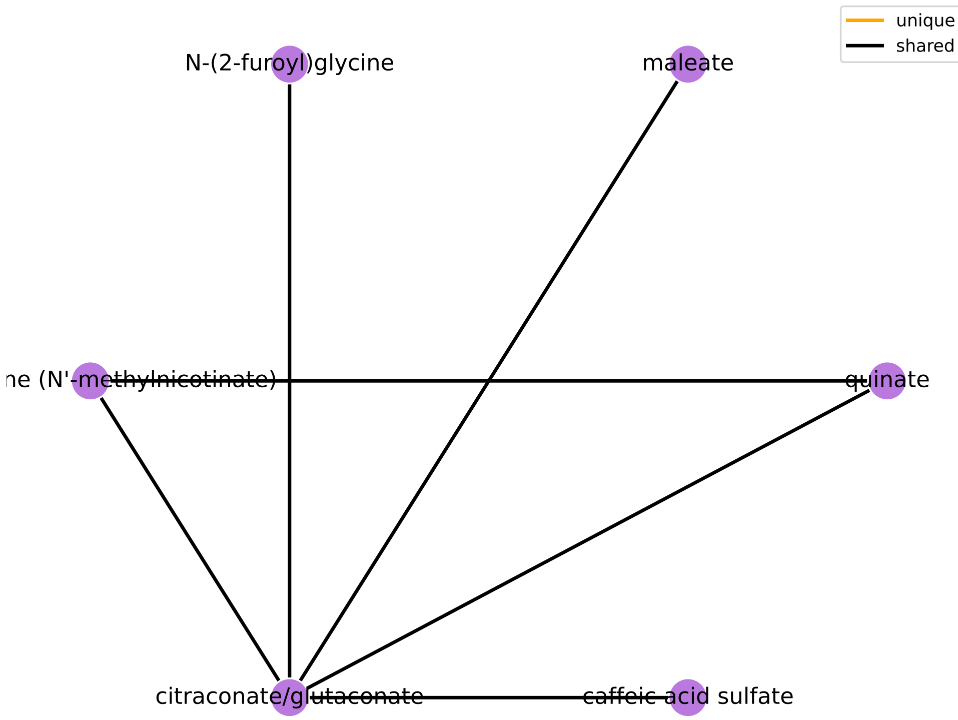
Component 2



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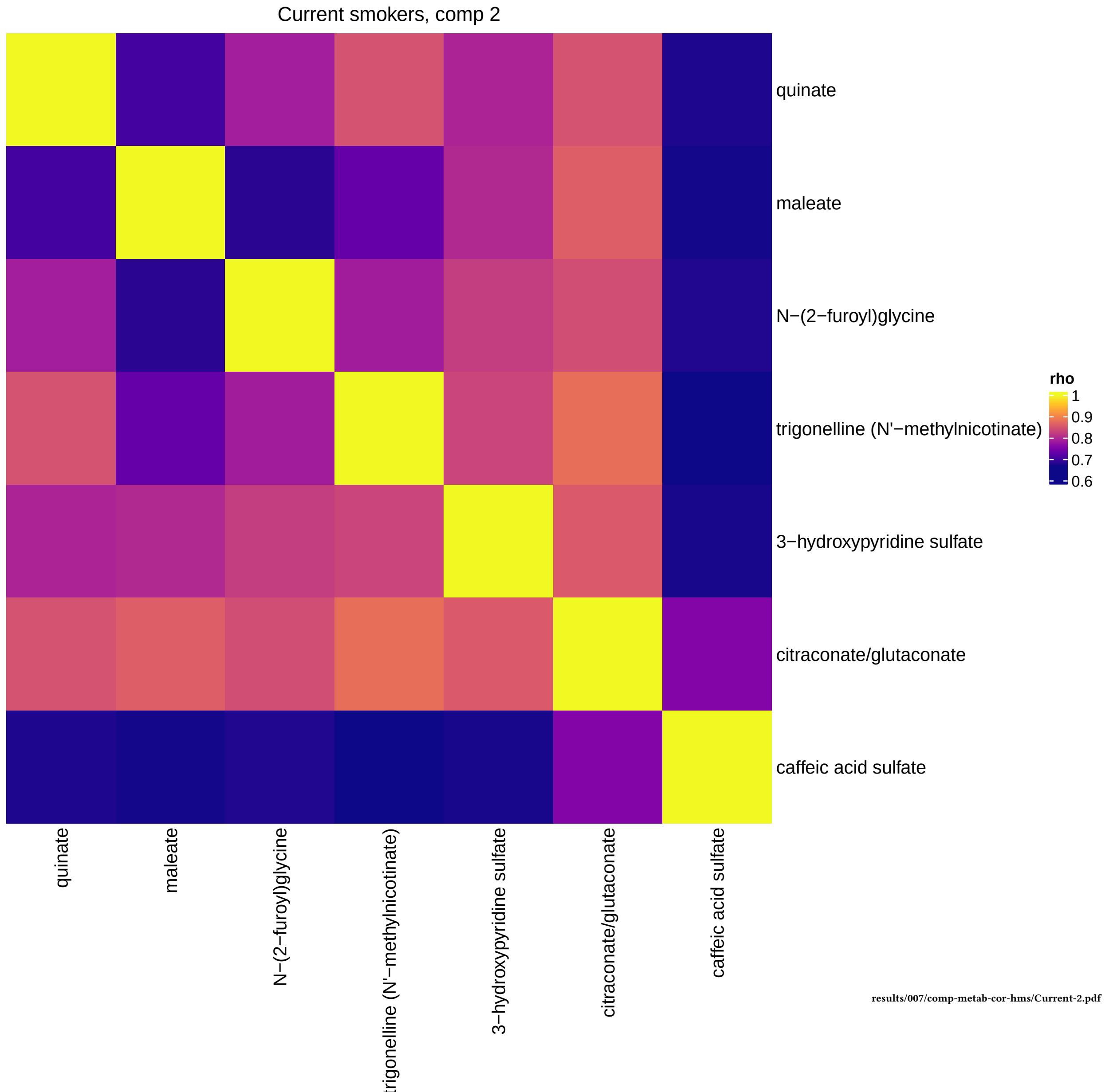

Current smokers, comp 2
quinate
maleate
N−(2−furoyl)glycine
trigonelline (N'−methylnicotinate)
3−hydroxypyridine sulfate
citraconate/glutaconate
caffeic acid sulfate
rho
1
0.9
0.8
0.7
0.6
results/007/comp-metab-cor-hms/Current-2.pdf

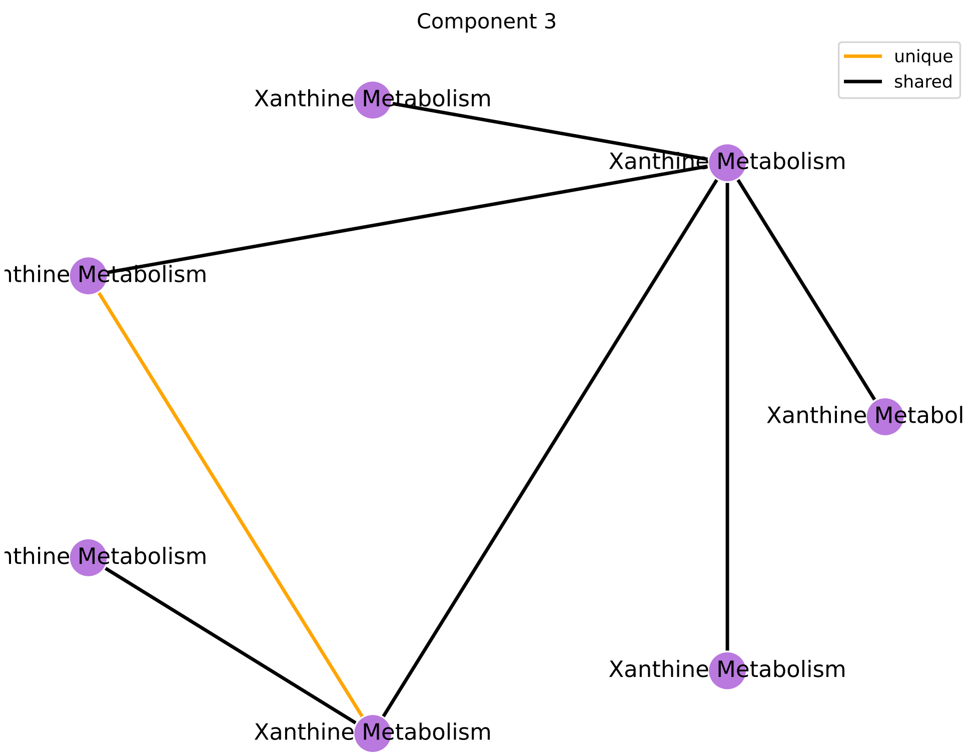
Component 3

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Xenobiotics	96	8	Inf	Inf	5.09e-08	4.58e-07

results/006/enrichment/curr-components/3-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Xanthine Metabolism	14	8	Inf	Inf	1.15e-15	1.2e-13

results/006/enrichment/curr-components/3-subpathway.csv



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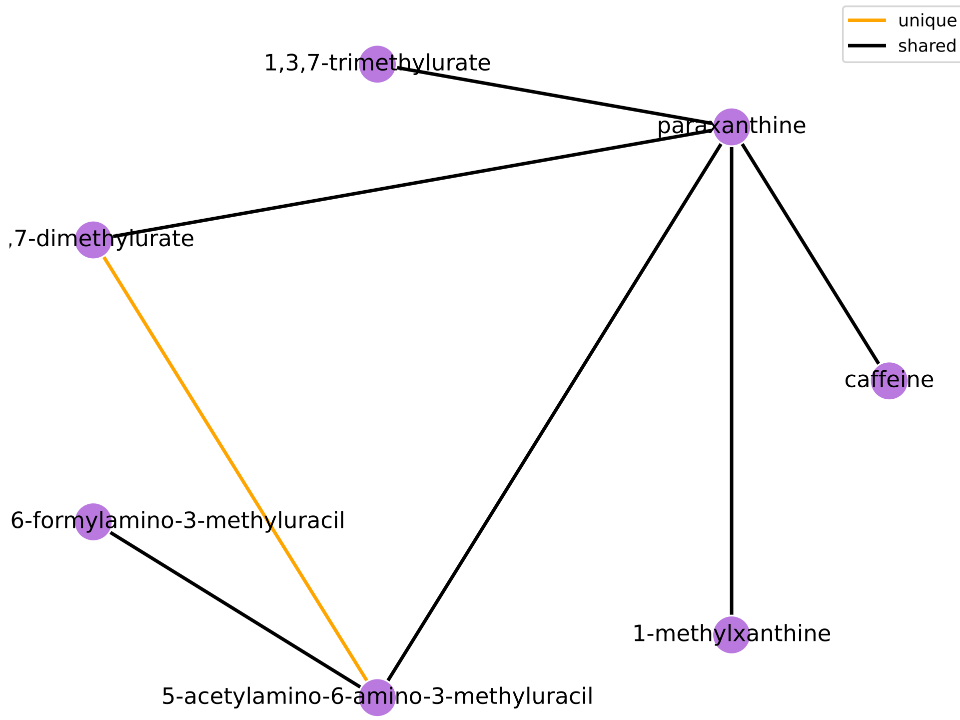
Component 3



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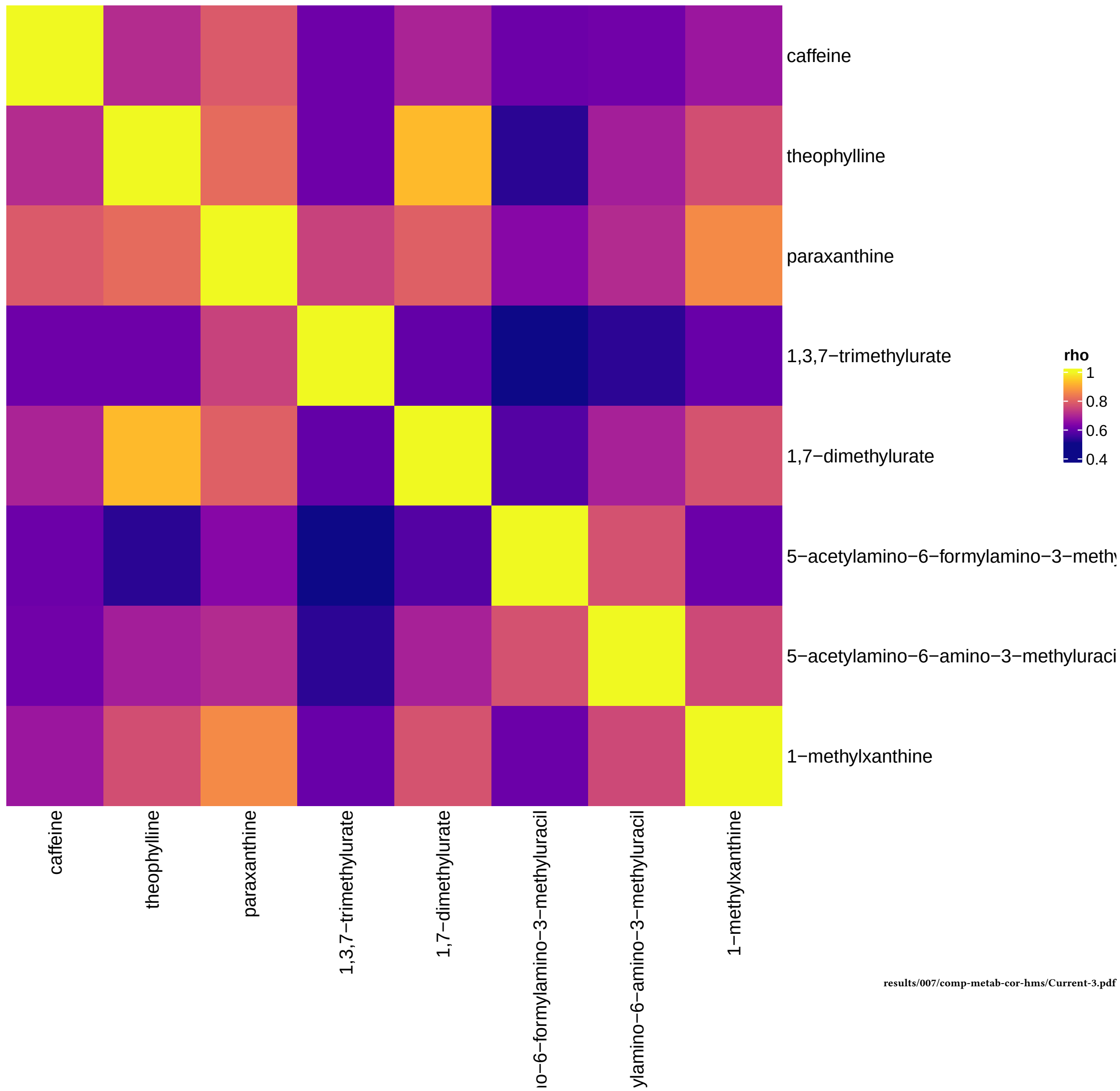
Current smokers, comp 3



results/007/comp-metab-cor-hms/Current-3.pdf

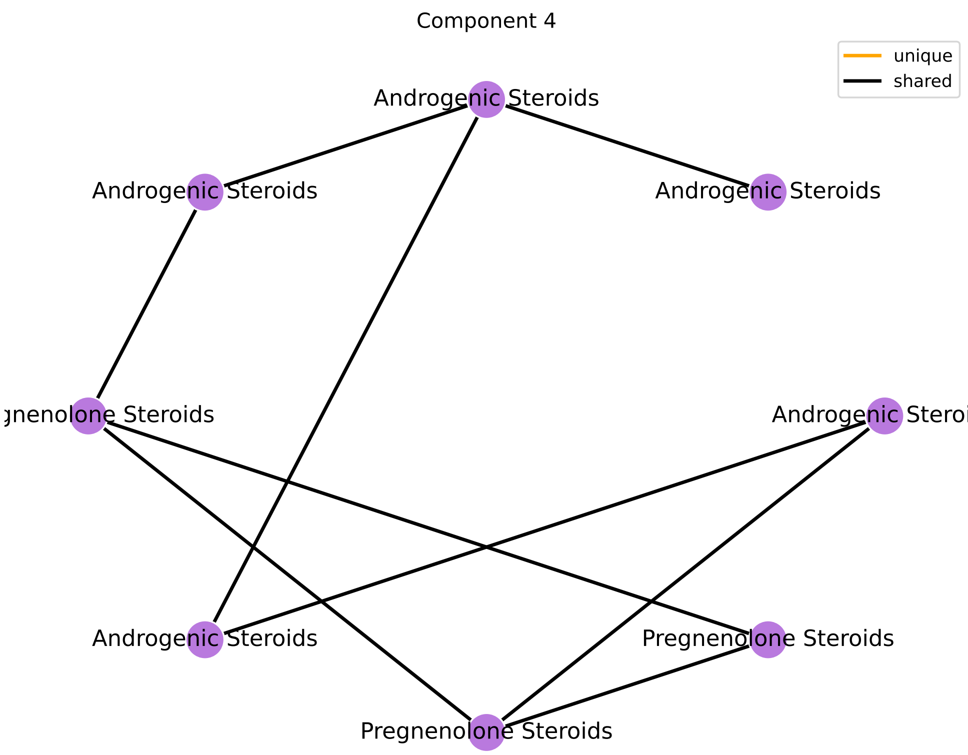
Component 4

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	9	Inf	Inf	0.00126	0.0113

results/006/enrichment/curr-components/4-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Pregnenolone Steroids	4	4	Inf	Inf	9.23e-09	9.6e-07
Androgenic Steroids	14	5	97.826	4.583	1.18e-07	6.14e-06

results/006/enrichment/curr-components/4-subpathway.csv



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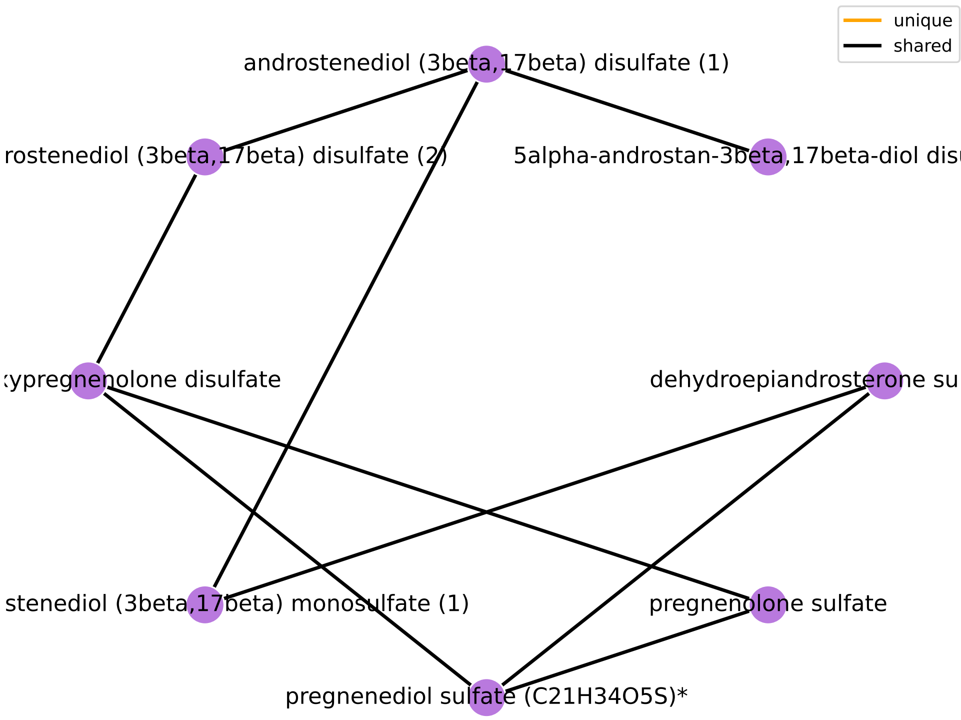
Component 4



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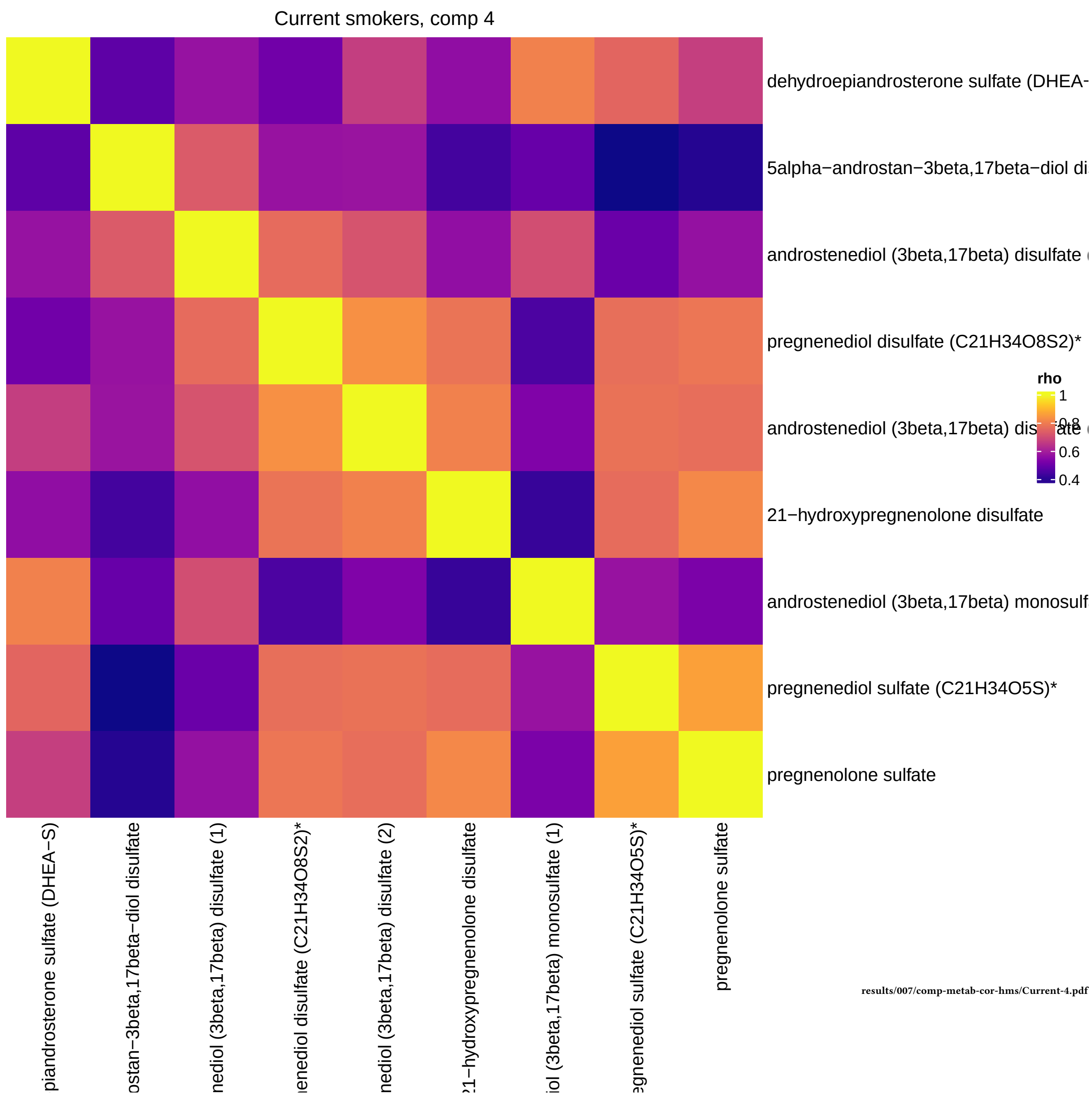

Current smokers, comp 4
dehydroepiandrosterone sulfate (DHEA-
5alpha−androstan−3beta,17beta−diol di
androstenediol (3beta,17beta) disulfate
pregnenediol disulfate (C21H34O8S2)*
androstenediol (3beta,17beta) disulfate
21−hydroxypregnenolone disulfate
androstenediol (3beta,17beta) monosulf
pregnenediol sulfate (C21H34O5S)*
pregnenolone sulfate
rho
1
0.8
0.6
0.4
piandrosterone sulfate (DHEA−S)
ostan−3beta,17beta−diol disulfate
nediol (3beta,17beta) disulfate (1)
enediol disulfate (C21H34O8S2)*
nediol (3beta,17beta) disulfate (2)
21−hydroxypregnenolone disulfate
iol (3beta,17beta) monosulfate (1)
egnenediol sulfate (C21H34O5S)*
pregnenolone sulfate
results/007/comp-metab-cor-hms/Current-4.pdf

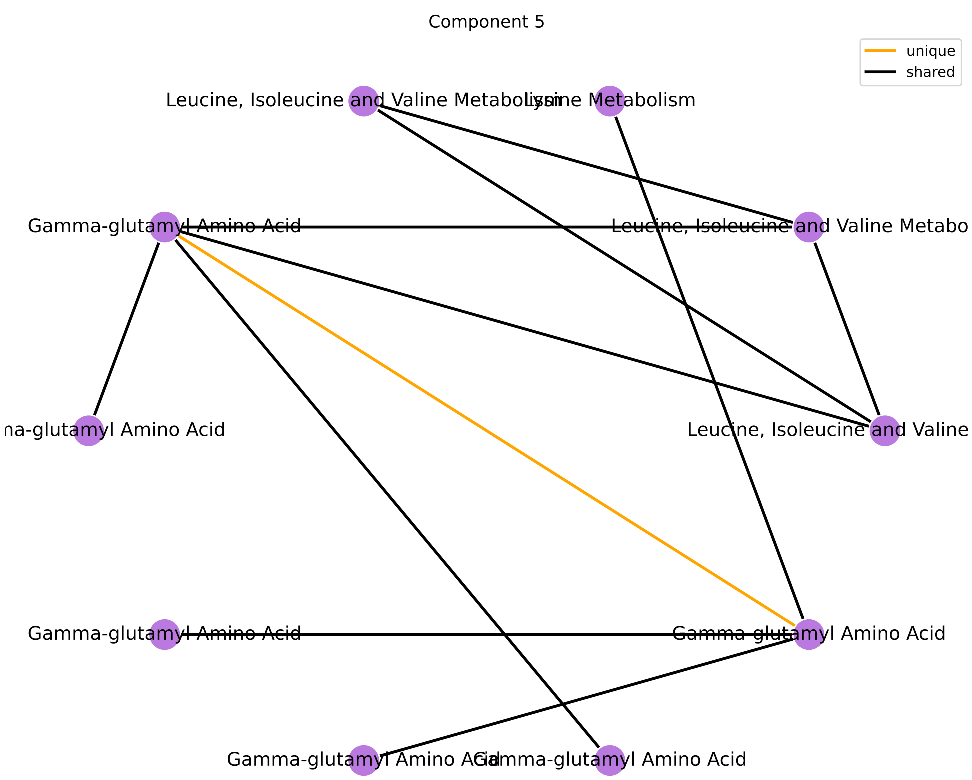
Component 5

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Peptide	25	6	44.83	3.803	2.84e-07	2.56e-06
Amino Acid	180	5	2.719	1	0.094	0.423

results/006/enrichment/curr-components/5-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Gamma-glutamyl Amino Acid	17	6	77.358	4.348	2.08e-08	2.16e-06
Leucine, Isoleucine and Valine Metabolism	31	3	9.552	2.257	0.00819	0.426
Lysine Metabolism	12	1	6.647	1.894	0.162	1
Methionine, Cysteine, SAM and Taurine Metabolism	17	1	4.549	1.515	0.222	1

results/006/enrichment/curr-components/5-subpathway.csv



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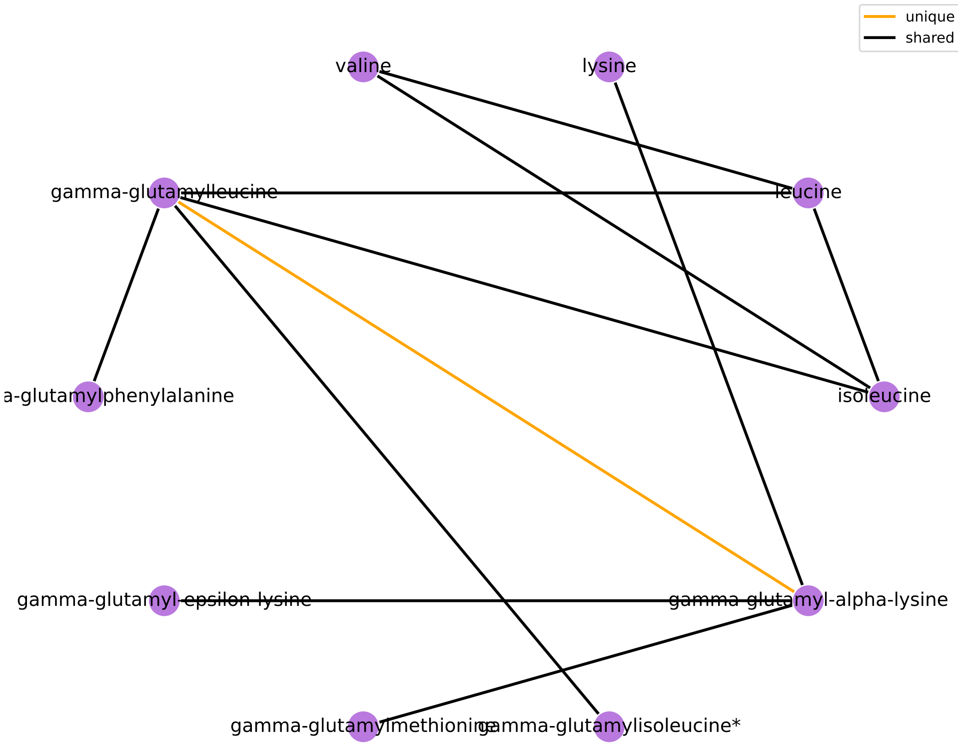
Component 5



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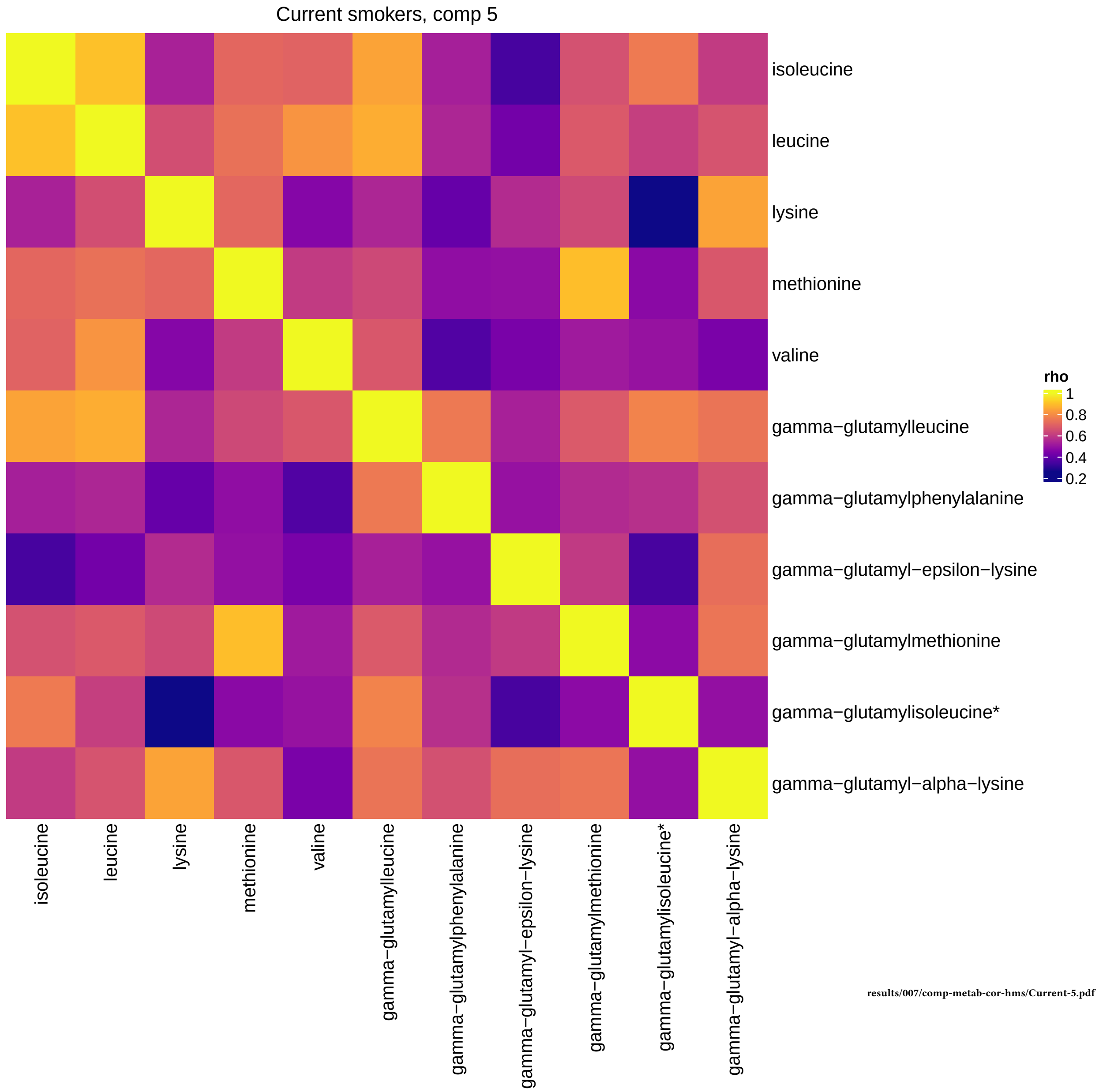


results/007/comp-metab-cor-hms/Current-5.pdf

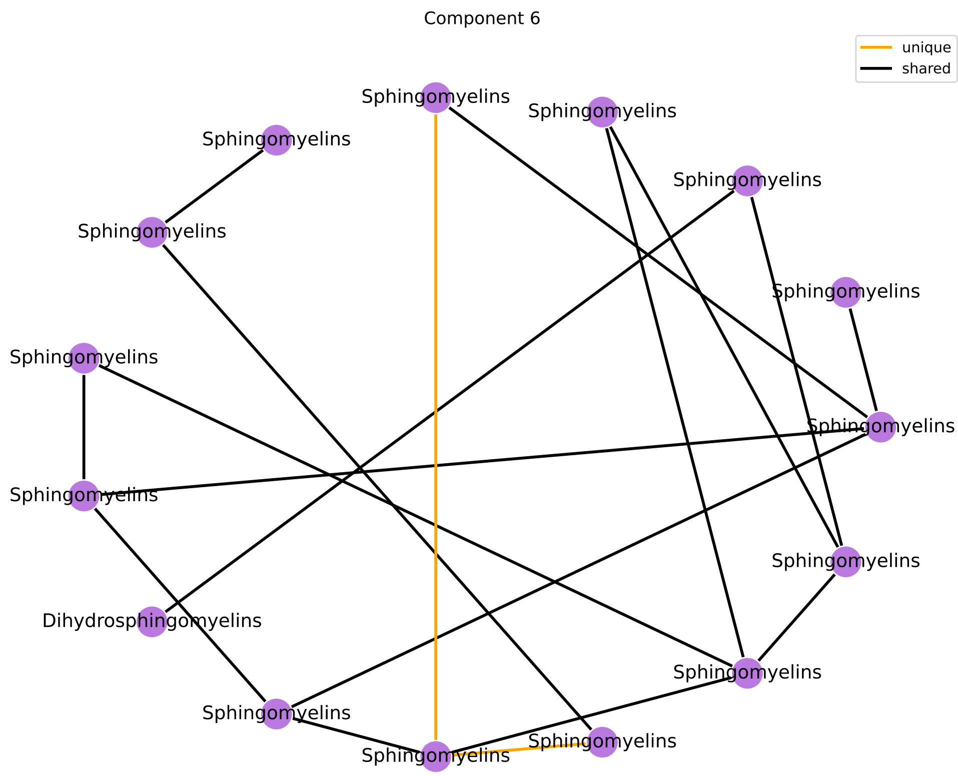
Component 6

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	17	Inf	Inf	3e-06	2.7e-05

results/006/enrichment/curr-components/6-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Sphingomyelins	28	16	913.538	6.817	1.05e-24	1.09e-22
Dihydrosphingomyelins	5	1	11.394	2.433	0.107	1

results/006/enrichment/curr-components/6-subpathway.csv



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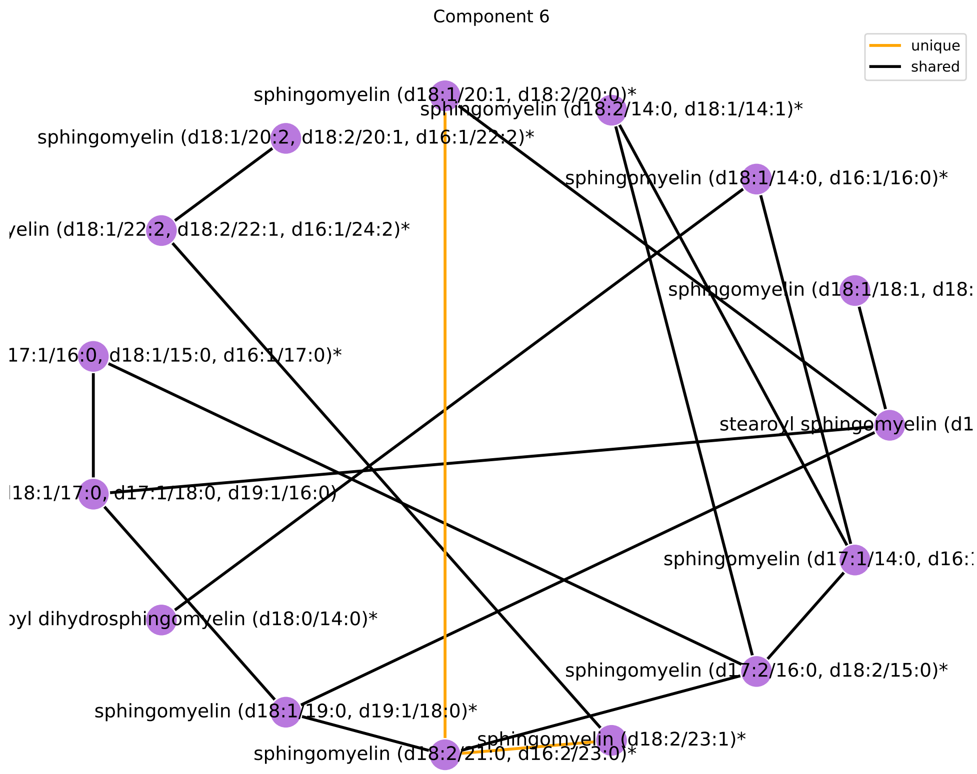

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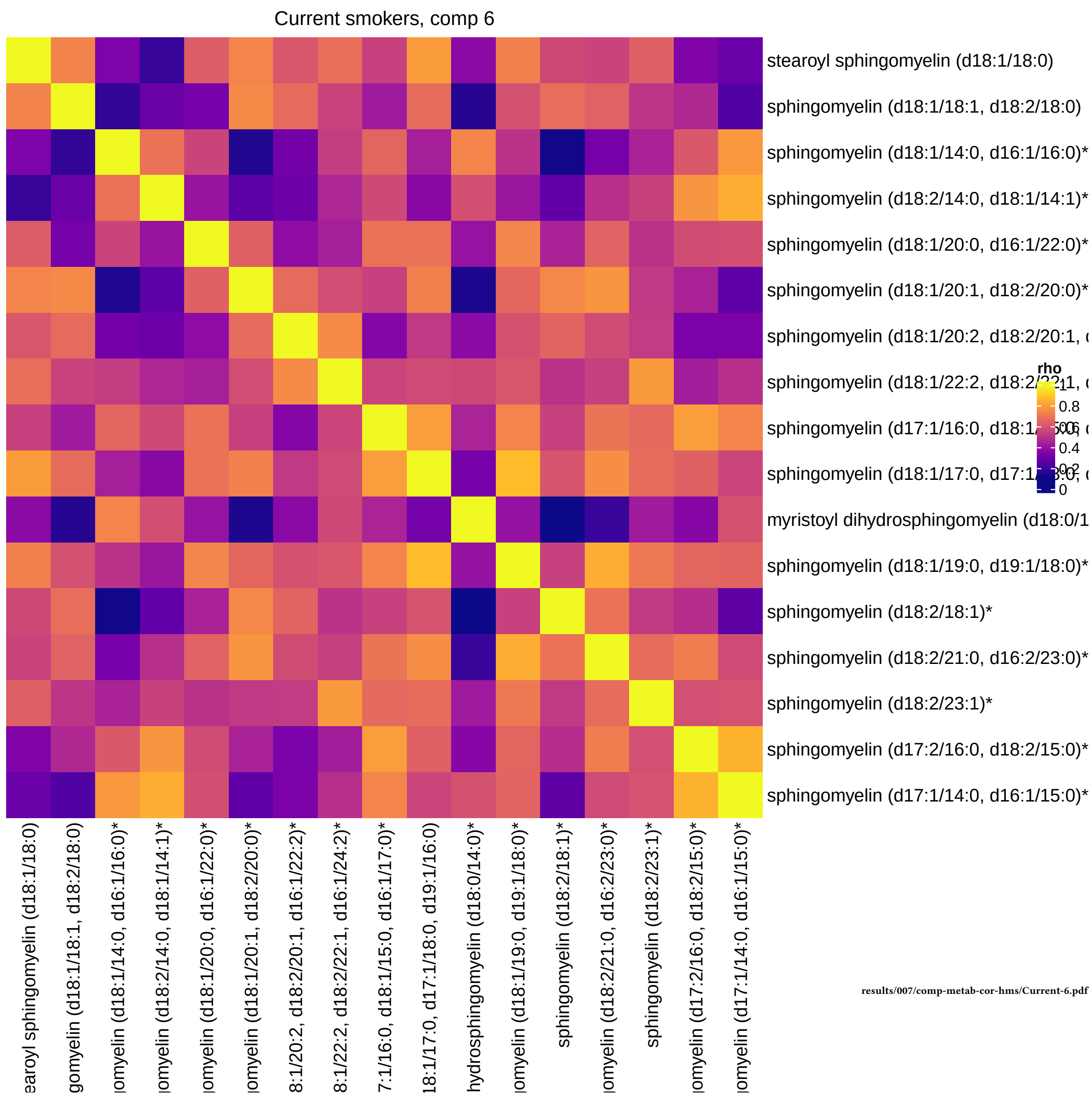


results/007/comp-metab-cor-hms/Current-6.pdf

Component 7

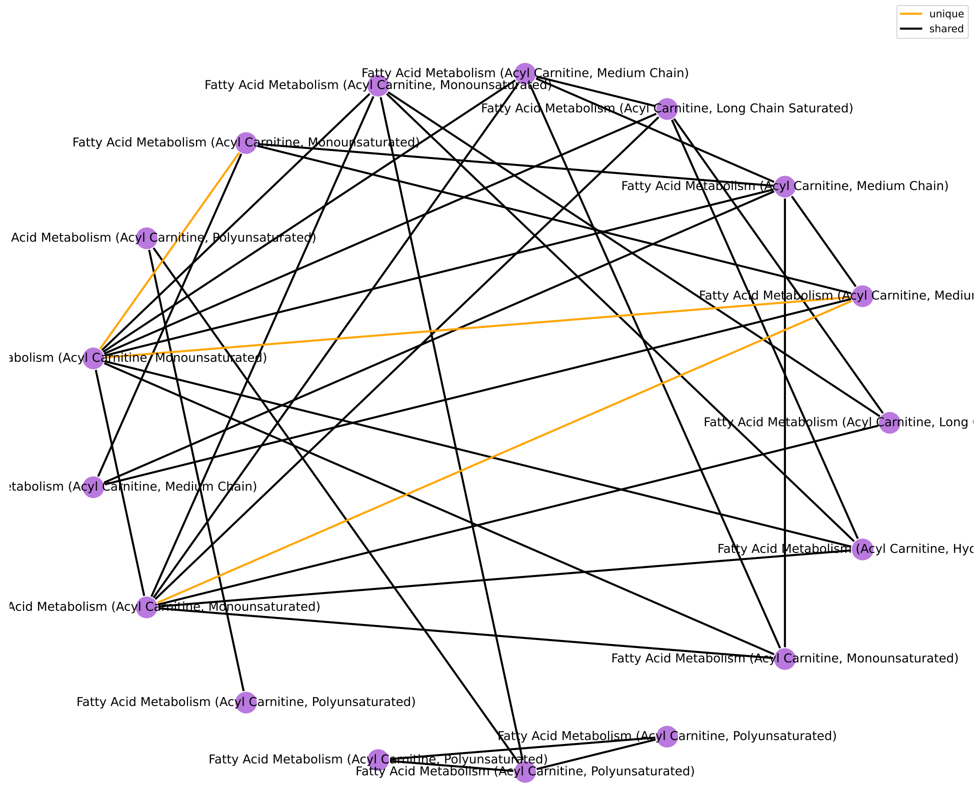
Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	18	Inf	Inf	1.4e-06	1.26e-05

results/006/enrichment/curr-components/7-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Fatty Acid Metabolism (Acyl Carnitine, Medium Chain)	5	5	Inf	Inf	4.16e-09	2.16e-07
Fatty Acid Metabolism (Acyl Carnitine, Polyunsaturated)	5	5	Inf	Inf	4.16e-09	2.16e-07
Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated)	8	5	90.526	4.506	2.23e-07	7.73e-06
Fatty Acid Metabolism (Acyl Carnitine, Long Chain Saturated)	8	2	15.084	2.714	0.0137	0.356
Fatty Acid Metabolism (Acyl Carnitine, Hydroxy)	3	1	21.318	3.06	0.0697	1

results/006/enrichment/curr-components/7-subpathway.csv

Component 7



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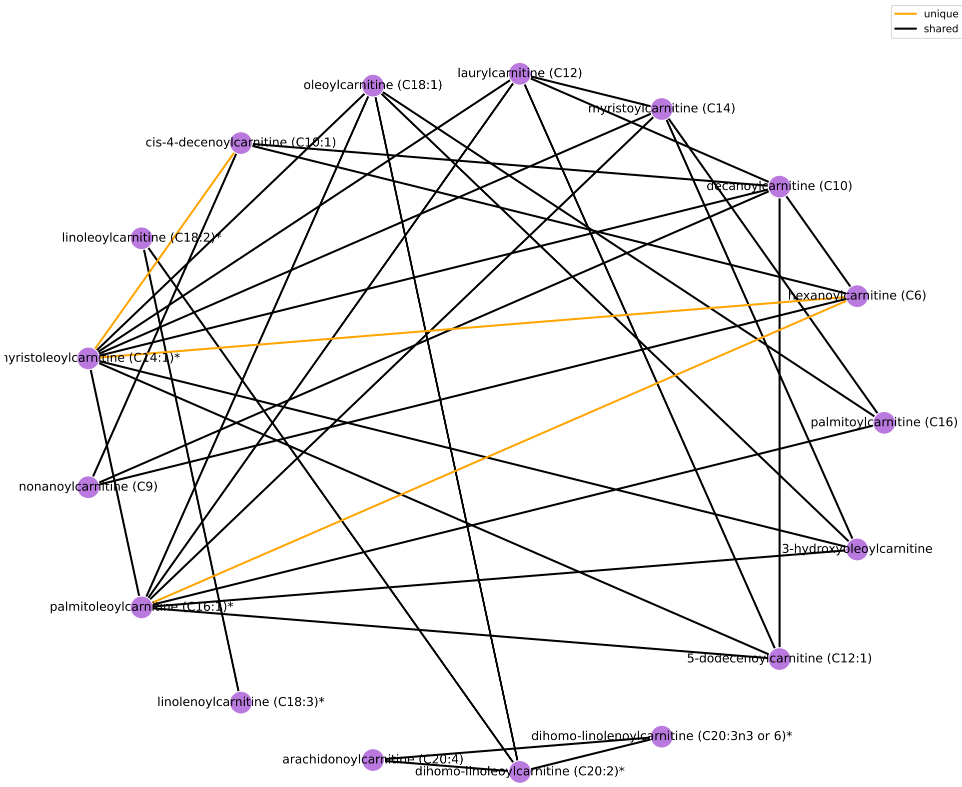
Component 7



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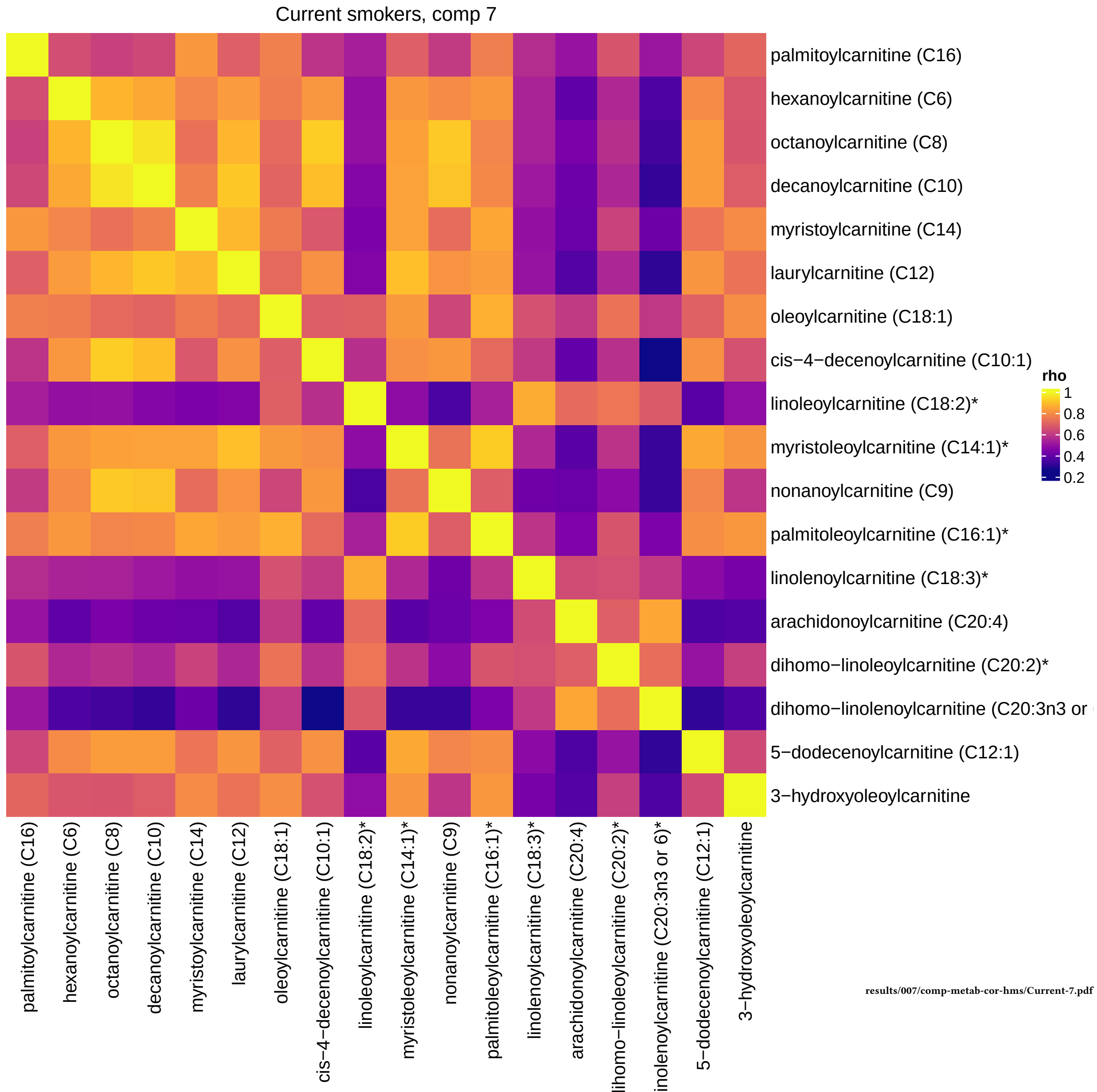


results/007/comp-metab-cor-hms/Current-7.pdf

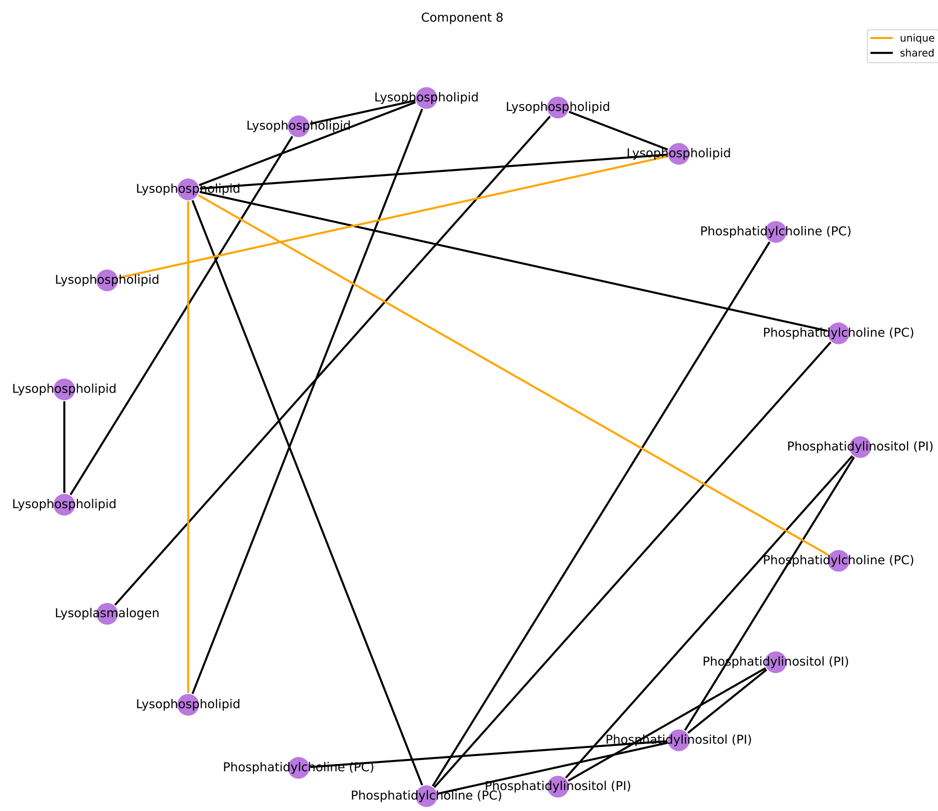
Component 8

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	20	Inf	Inf	3.04e-07	2.74e-06

results/006/enrichment/curr-components/8-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lysophospholipid	24	10	50.402	3.92	1.88e-11	1.96e-09
Phosphatidylinositol (PI)	6	4	88.541	4.483	5.15e-06	0.000268
Phosphatidylcholine (PC)	19	5	17.012	2.834	6.92e-05	0.0024
Lysoplasmalogen	3	1	19.06	2.948	0.0772	1

results/006/enrichment/curr-components/8-subpathway.csv



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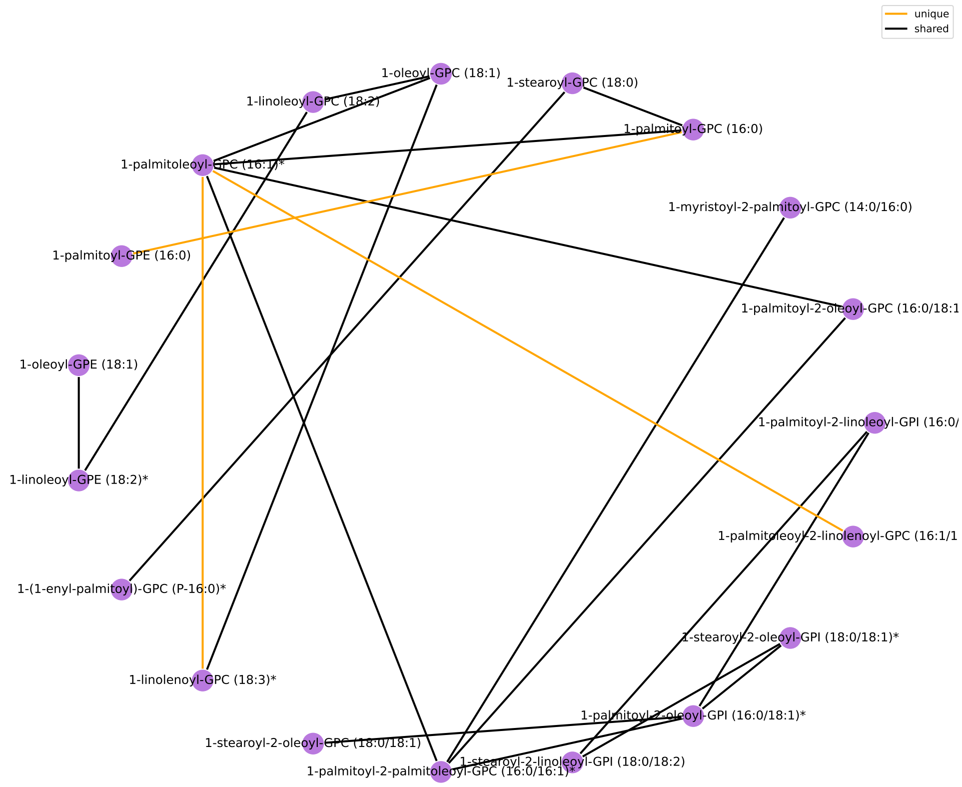
Component 8



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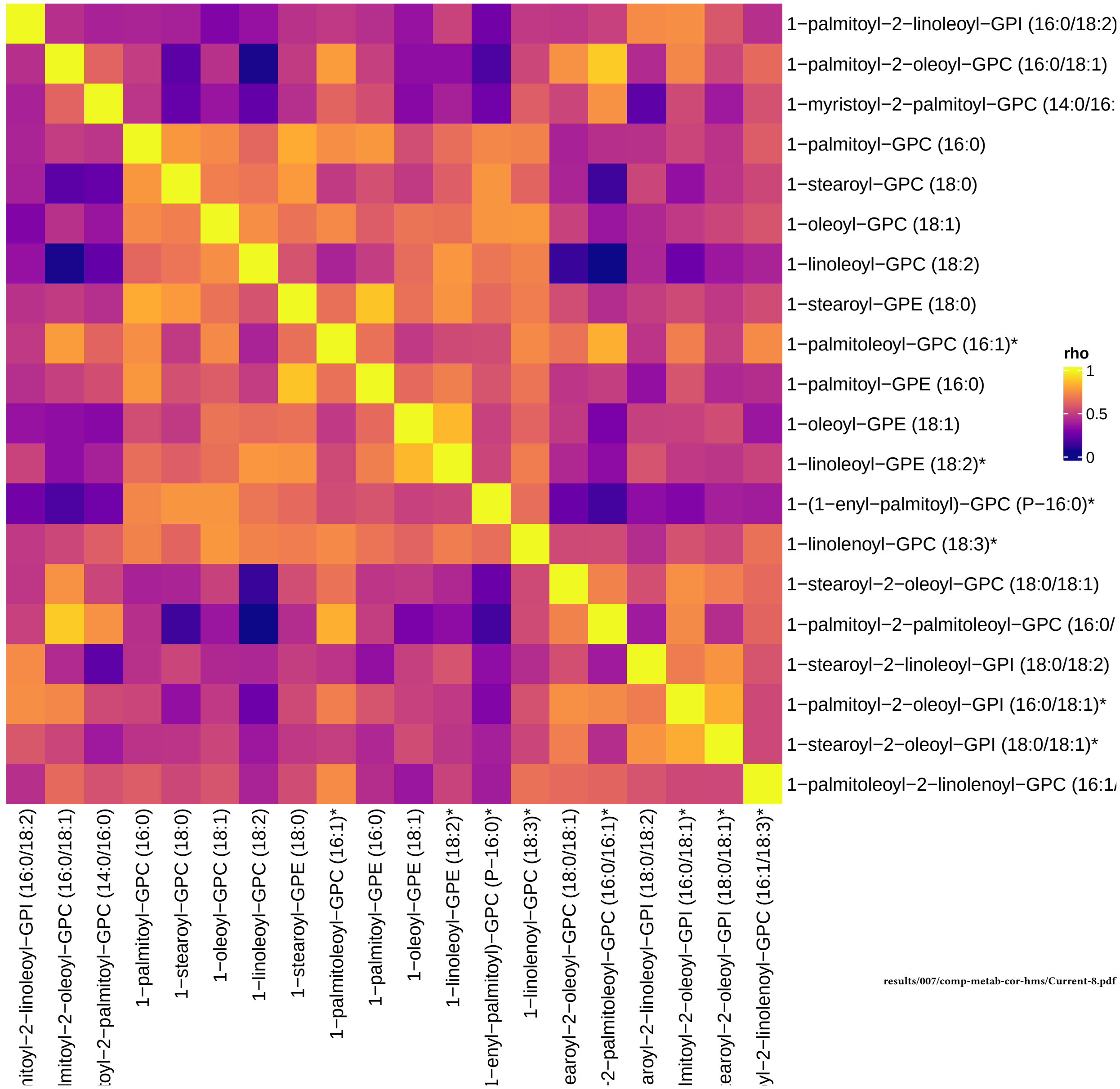
Current smokers, comp 8



results/007/comp-metab-cor-hms/Current-8.pdf

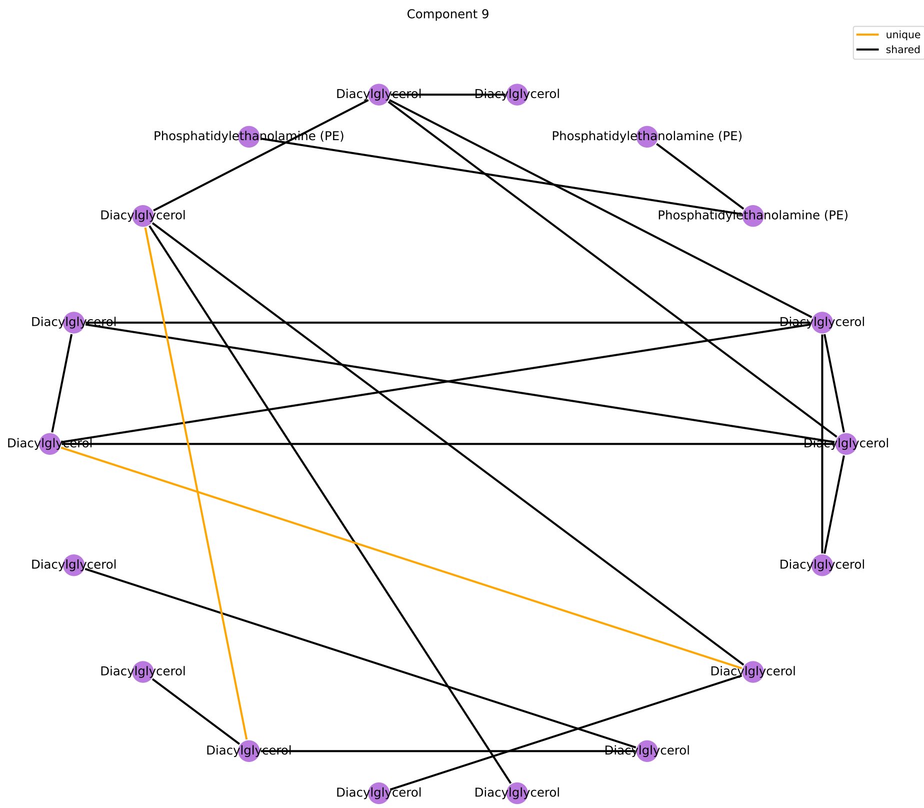
Component 9

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	21	Inf	Inf	1.41e-07	1.27e-06

results/006/enrichment/curr-components/9-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Diacylglycerol	19	15	419.751	6.04	1.96e-23	2.04e-21
Phosphatidylethanolamine (PE)	11	6	56.82	4.04	8.9e-08	4.63e-06

results/006/enrichment/curr-components/9-subpathway.csv



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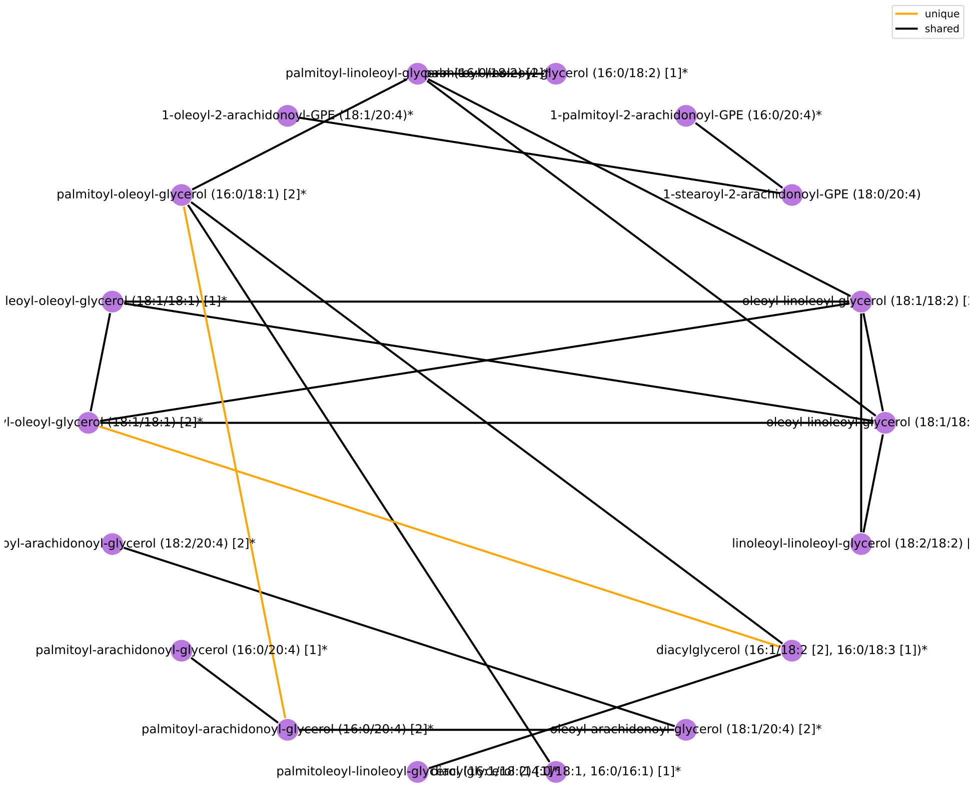
Component 9



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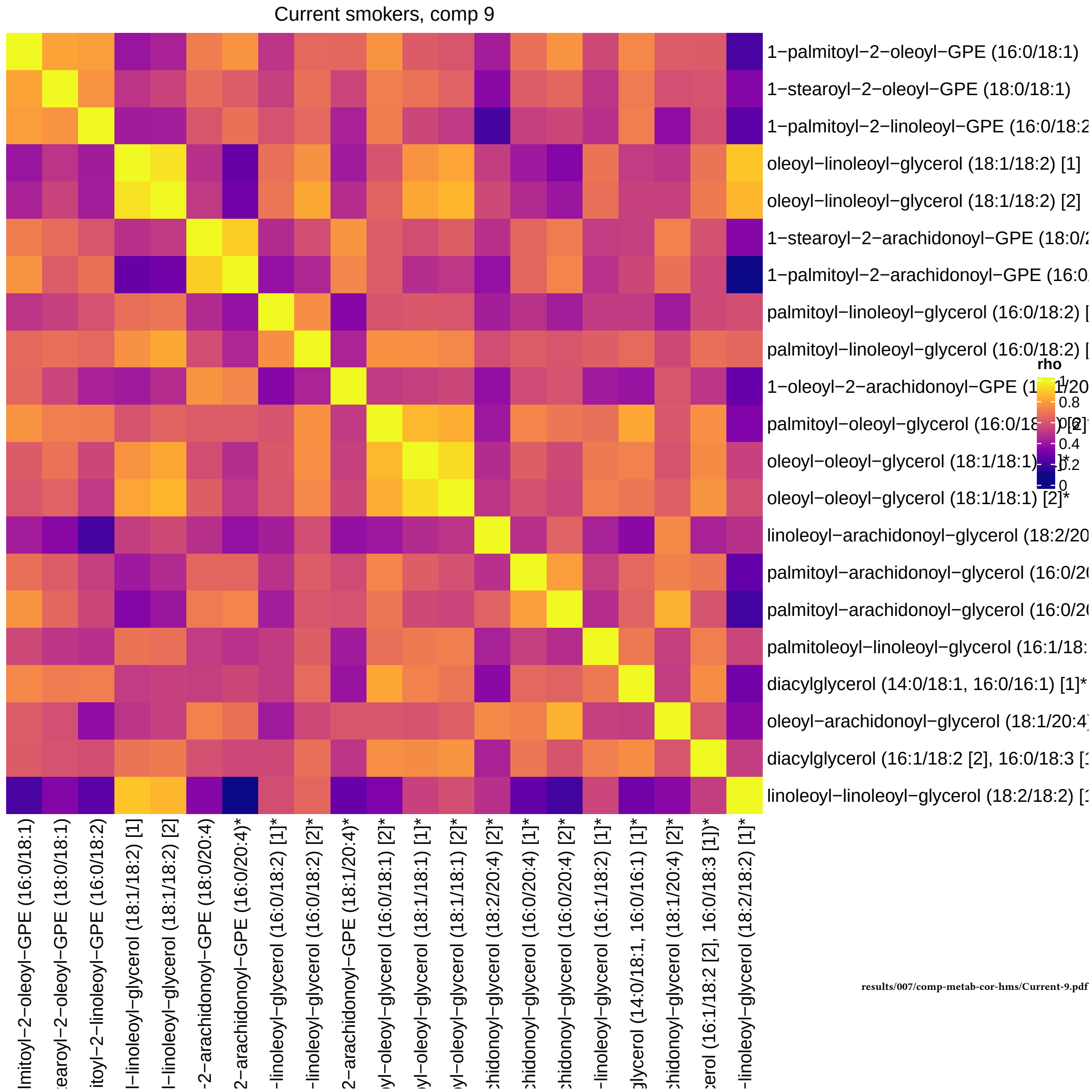


results/007/comp-metab-cor-hms/Current-9.pdf

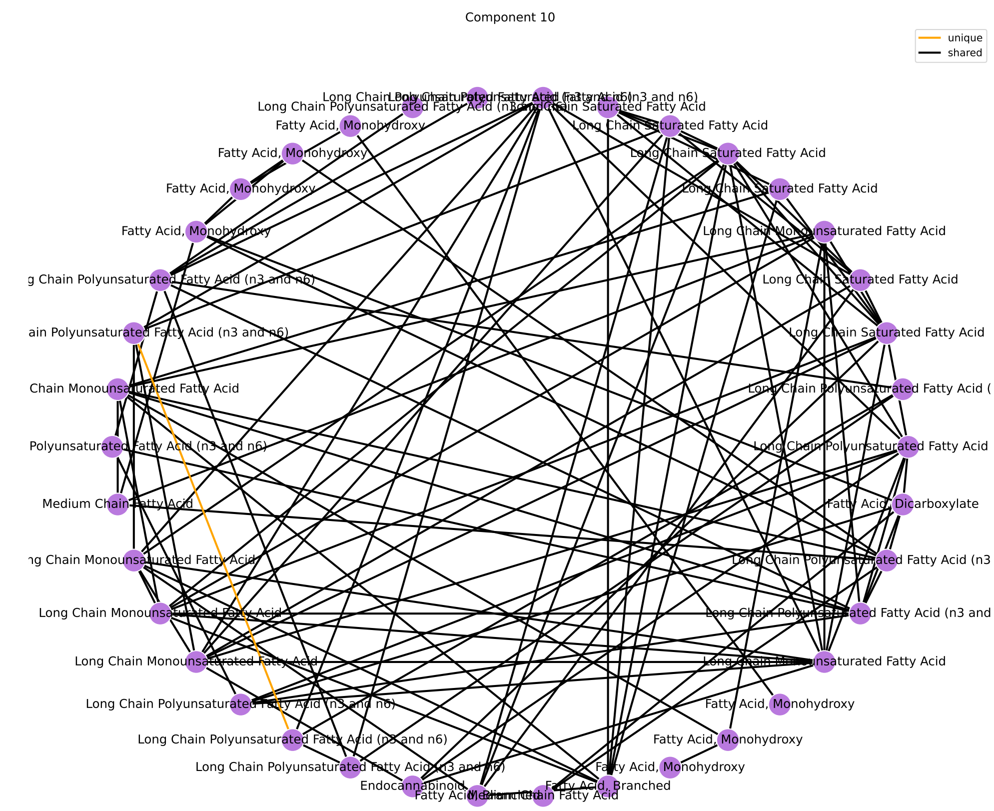
Component 10

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	40	Inf	Inf	4.96e-14	4.46e-13

results/006/enrichment/curr-components/10-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Long Chain Polyunsaturated Fatty Acid (n3 and n6)	14	14	Inf	Inf	1.1e-19	1.14e-17
Long Chain Monounsaturated Fatty Acid	7	6	123.581	4.817	1e-07	3.47e-06
Long Chain Saturated Fatty Acid	7	6	123.581	4.817	1e-07	3.47e-06
Fatty Acid, Monohydroxy	20	7	11.405	2.434	3.12e-05	0.000811
Fatty Acid, Branched	2	2	Inf	Inf	0.00272	0.0566
Ketone Bodies	1	1	Inf	Inf	0.0528	0.915
Medium Chain Fatty Acid	8	2	6.212	1.826	0.0622	0.924
Endocannabinoid	5	1	4.559	1.517	0.238	1
Fatty Acid, Dicarboxylate	23	1	0.811	-0.209	0.718	1

results/006/enrichment/curr-components/10-subpathway.csv



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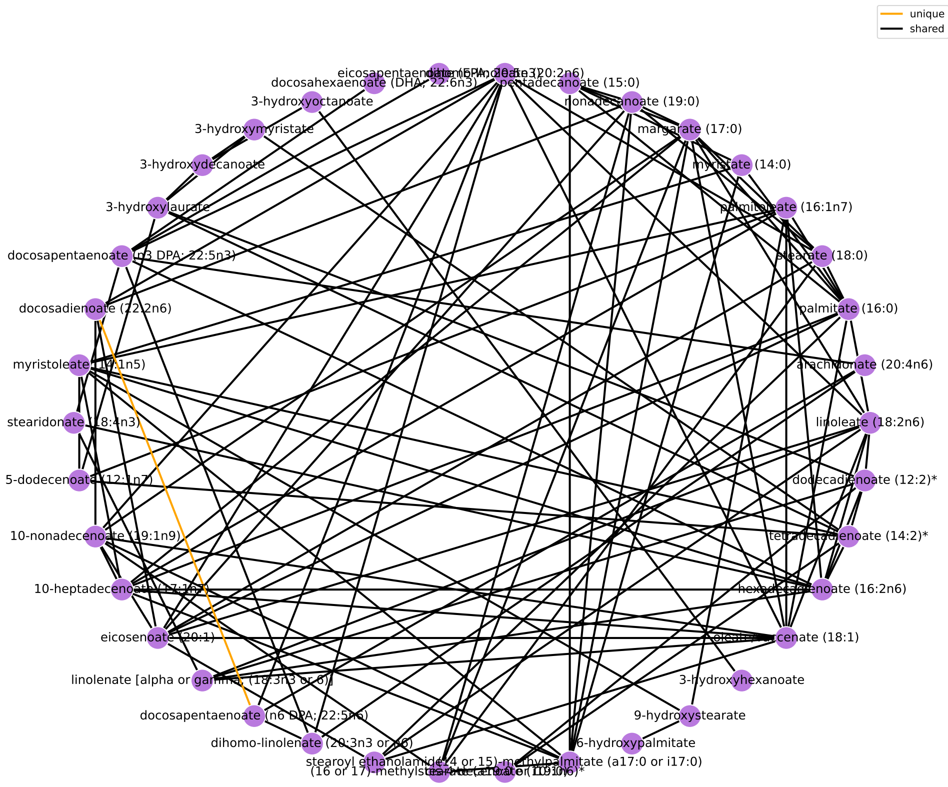
Component 10



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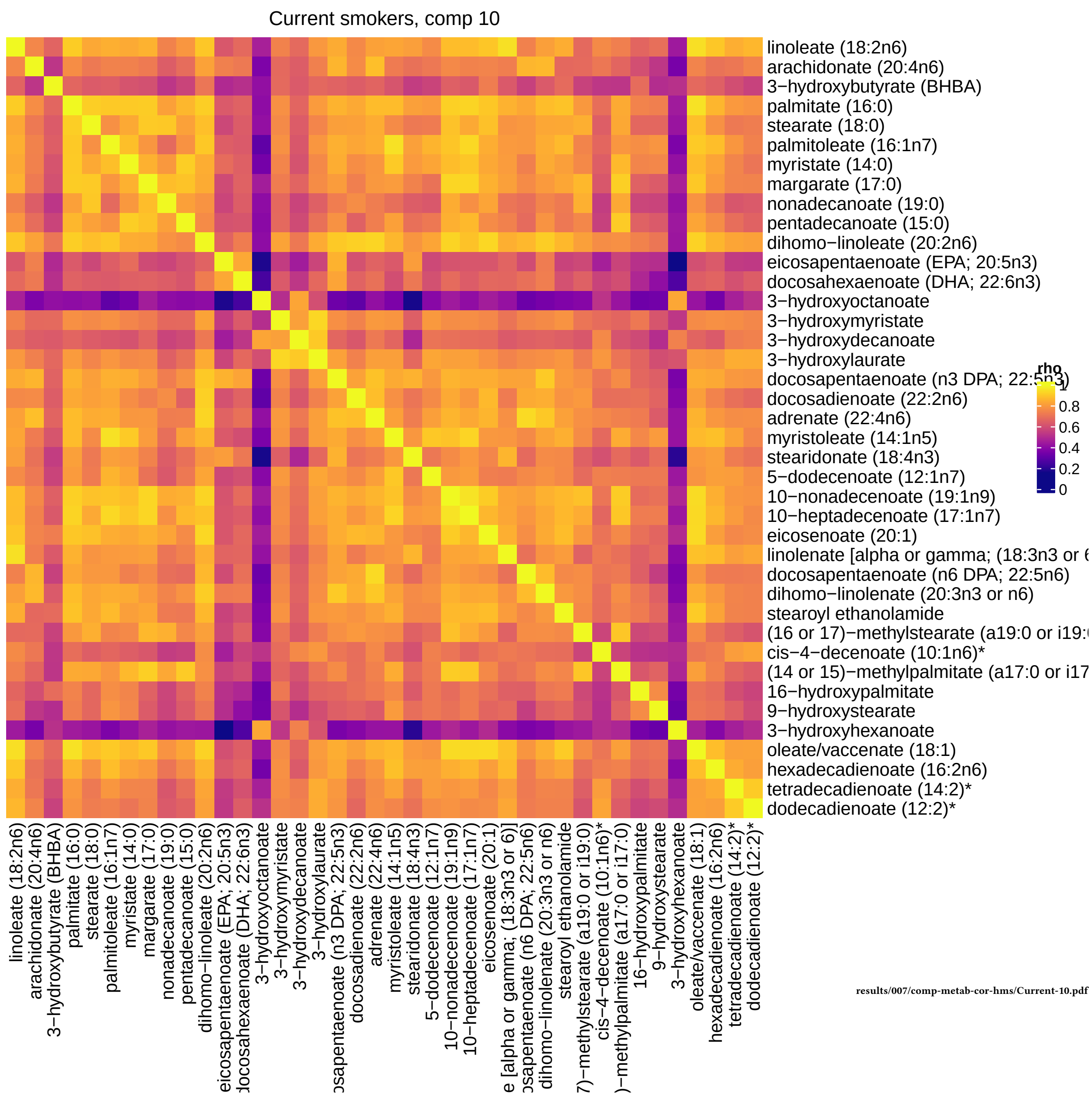


results/007/comp-metab-cor-hms/Current-10.pdf

Unique To Current Smokers

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	2	Inf	Inf	0.229	1

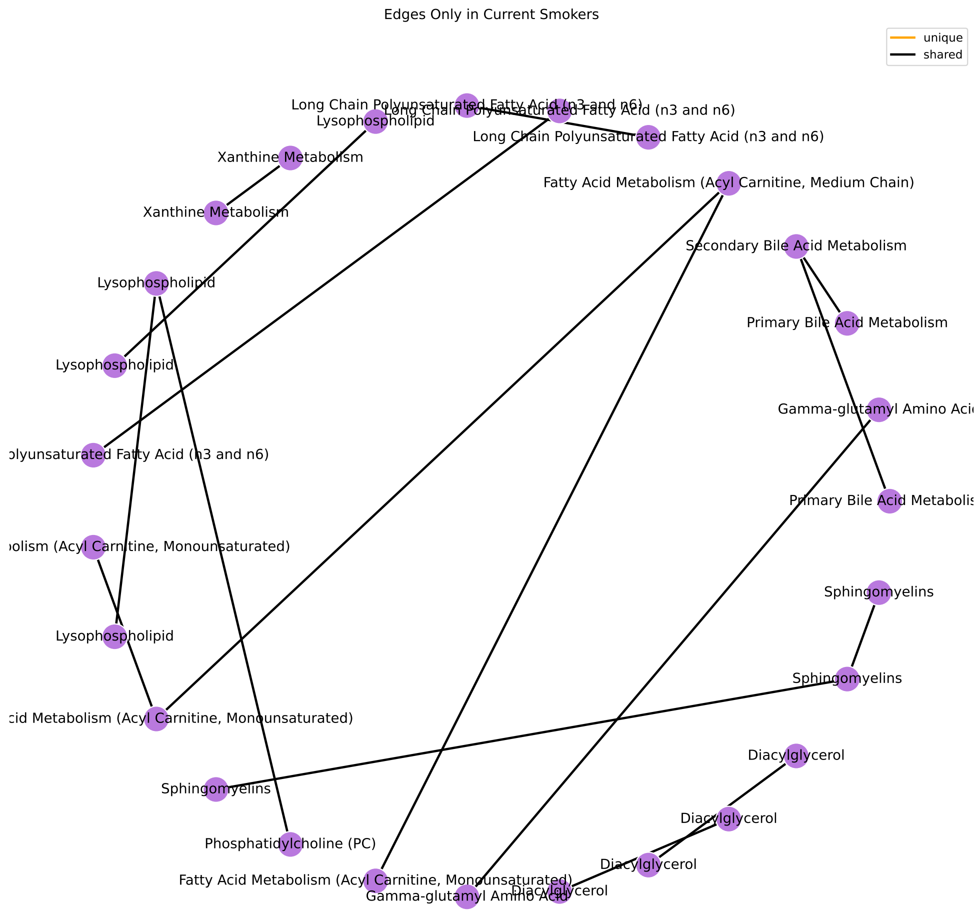
results/006/enrichment/curr-unique-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Ketone Bodies	1	1	Inf	Inf	0.00264	0.275
Sphingomyelins	28	1	26.532	3.278	0.0726	1

results/006/enrichment/curr-unique-subpathway.csv

results/005_1/sub_pathway-node-only-curr-graph.pdf

results/005_1/chemical_name-node-only-curr-graph.pdf



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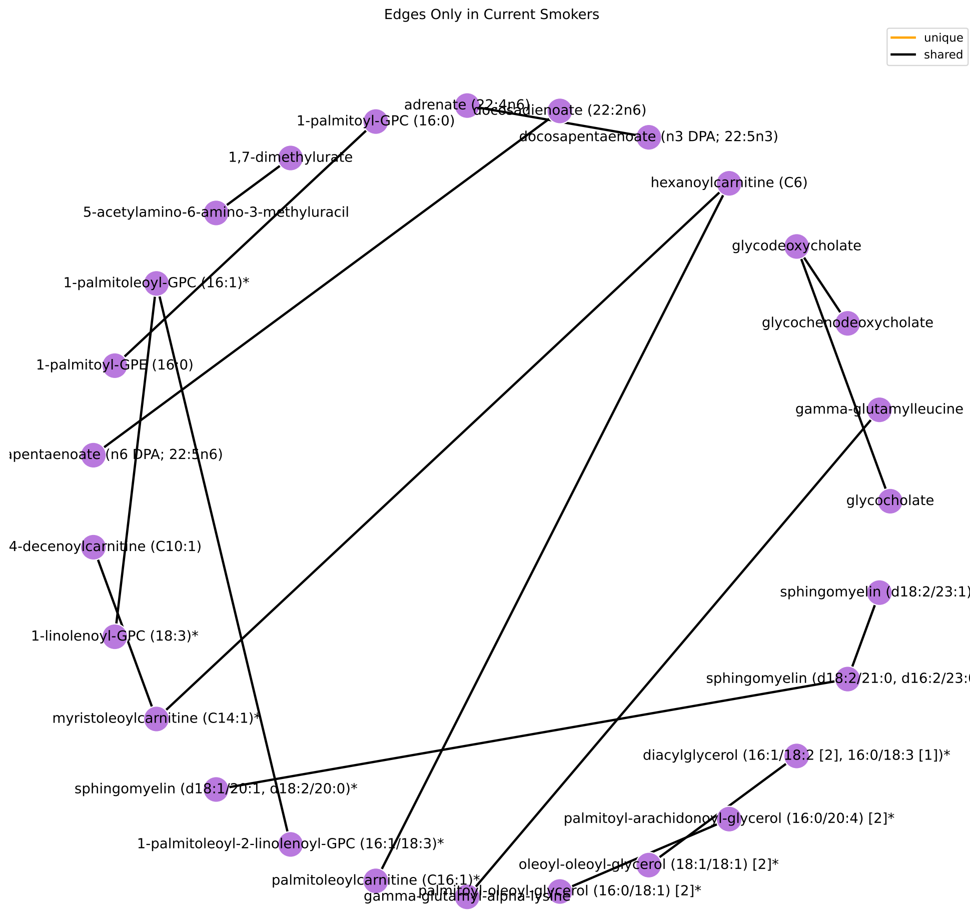


results/005_1/chemical_name-edge-only-curr-graph.pdf

Former Smokers

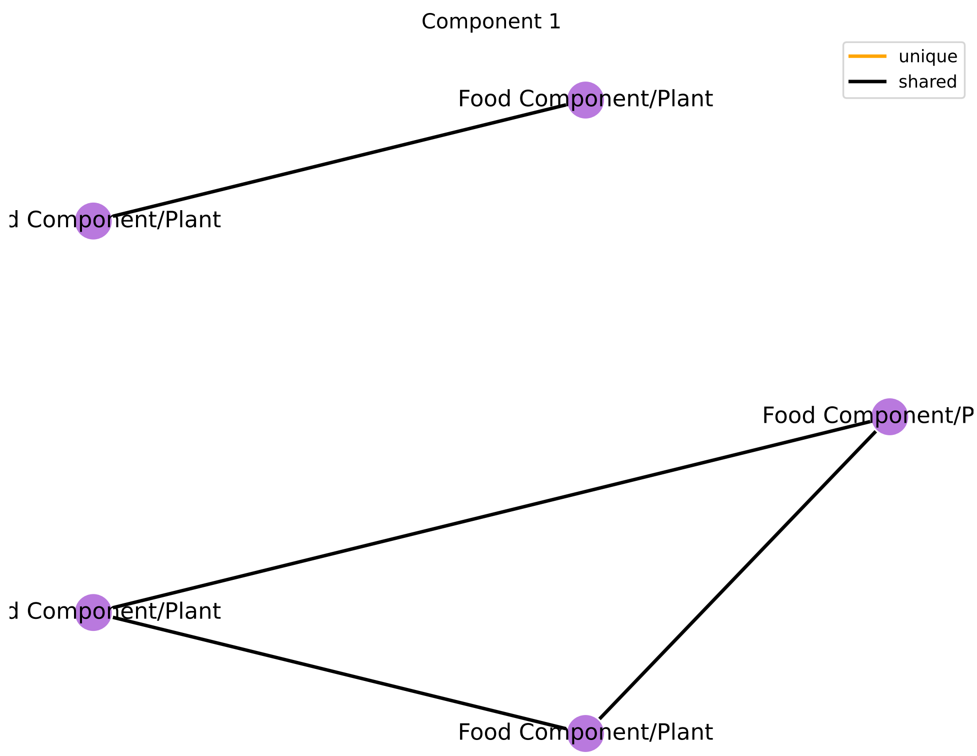
Component 1

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Xenobiotics	96	6	Inf	Inf	3.59e-06	3.23e-05

results/006/enrichment/form-components/1-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Food Component/Plant	38	6	Inf	Inf	1.07e-08	1.11e-06

results/006/enrichment/form-components/1-subpathway.csv



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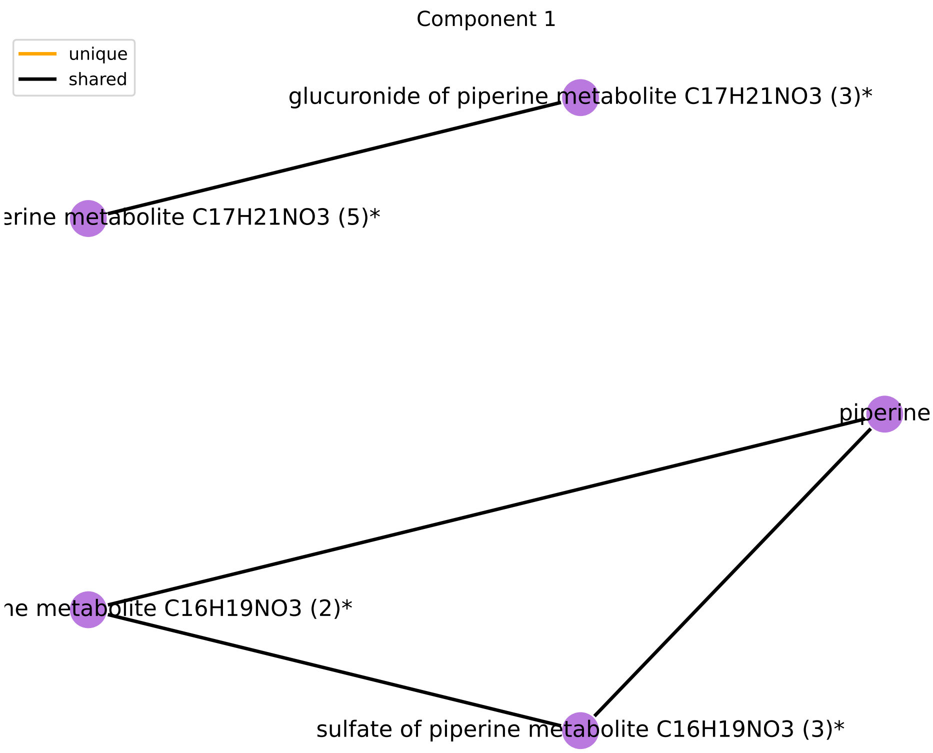

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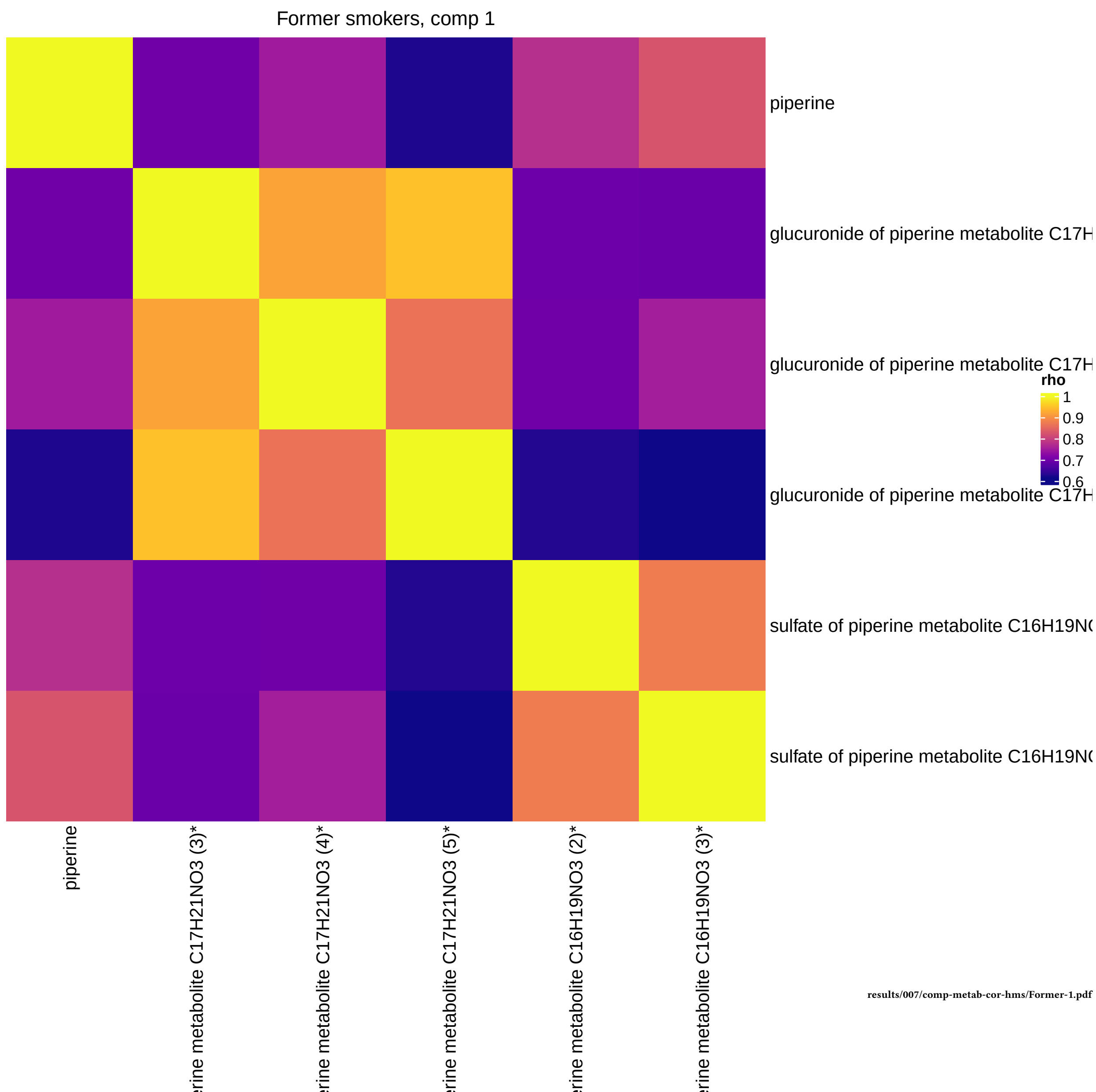


results/007/comp-metab-cor-hms/Former-1.pdf

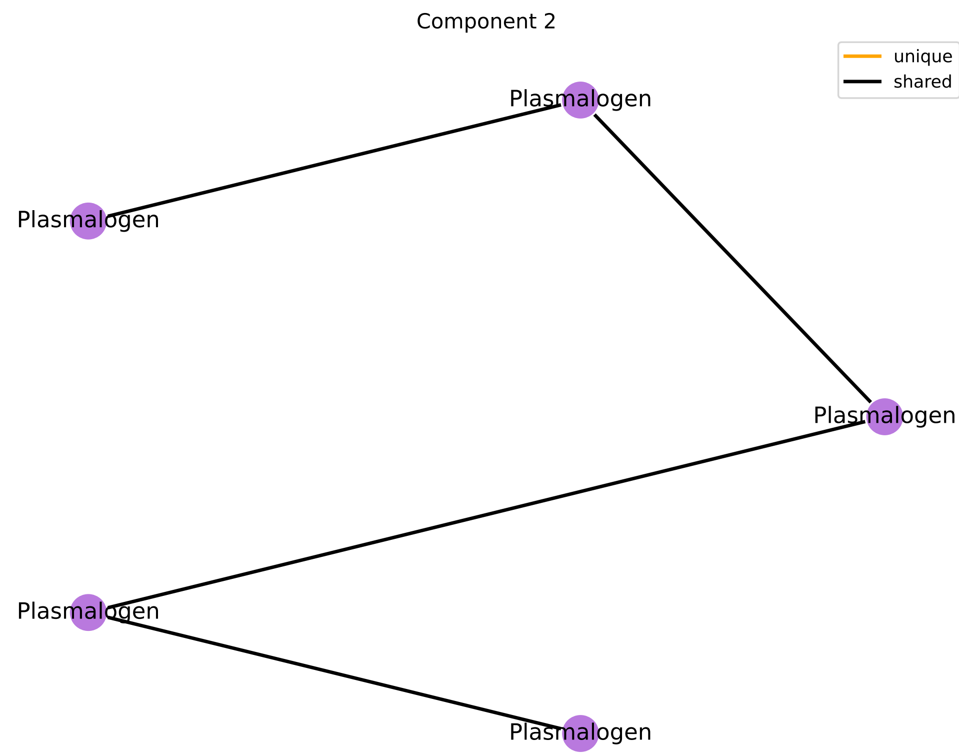
Component 2

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	6	Inf	Inf	0.0118	0.106

results/006/enrichment/form-components/2-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Plasmalogen	11	6	Inf	Inf	1.79e-12	1.86e-10

results/006/enrichment/form-components/2-subpathway.csv



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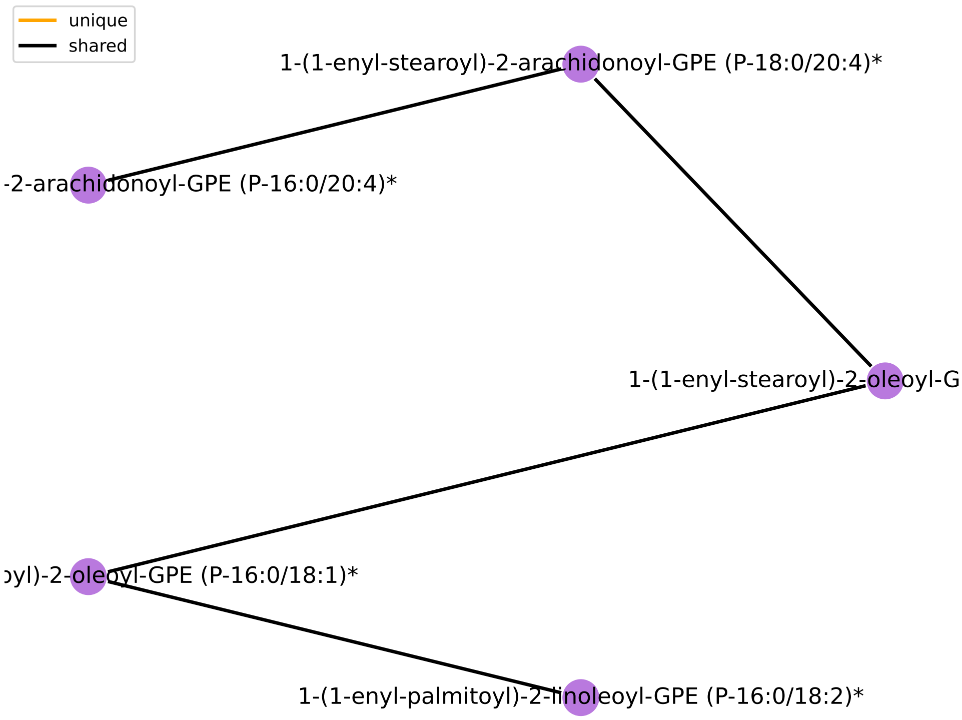
Component 2



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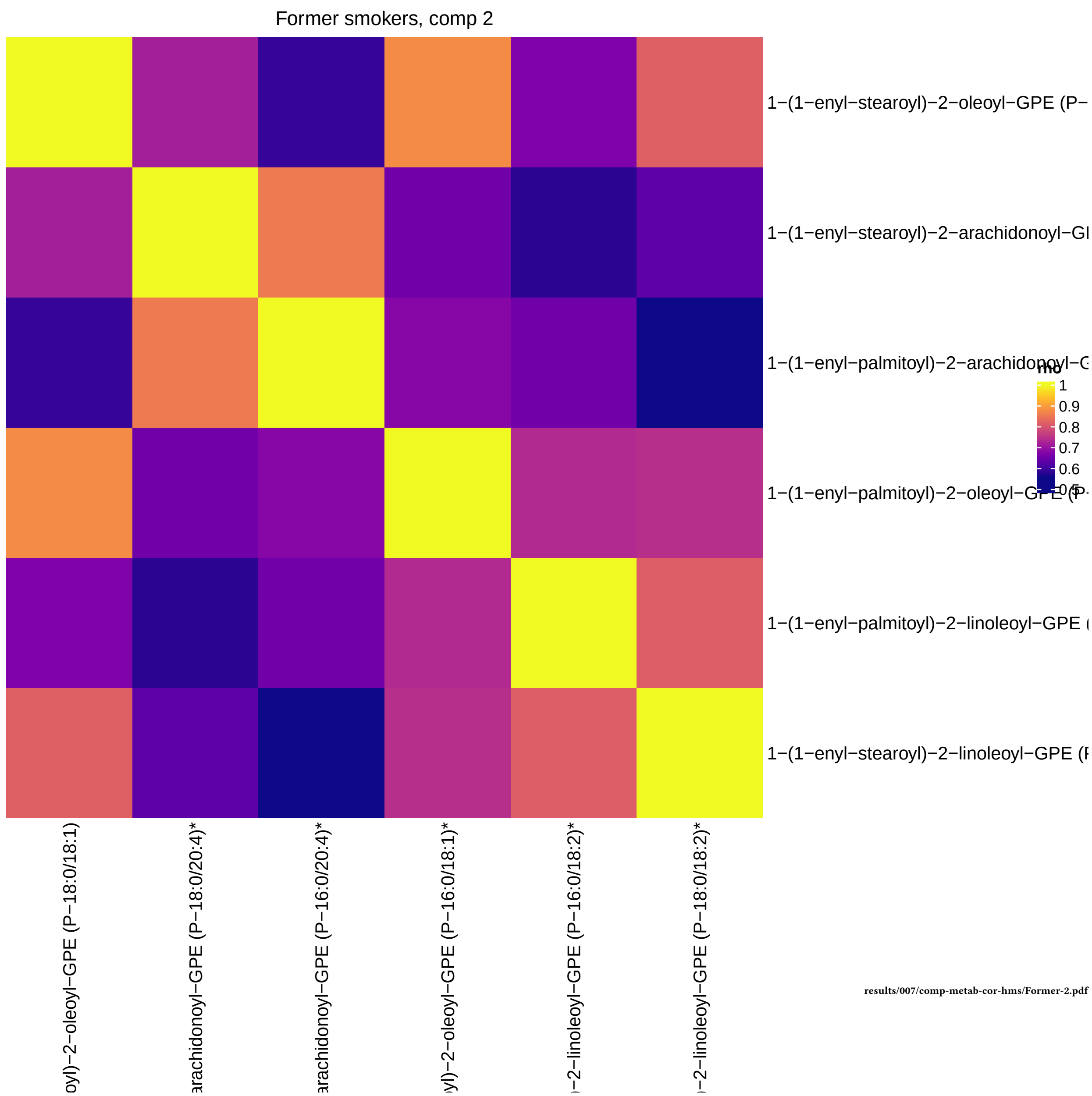


results/007/comp-metab-cor-hms/Former-2.pdf

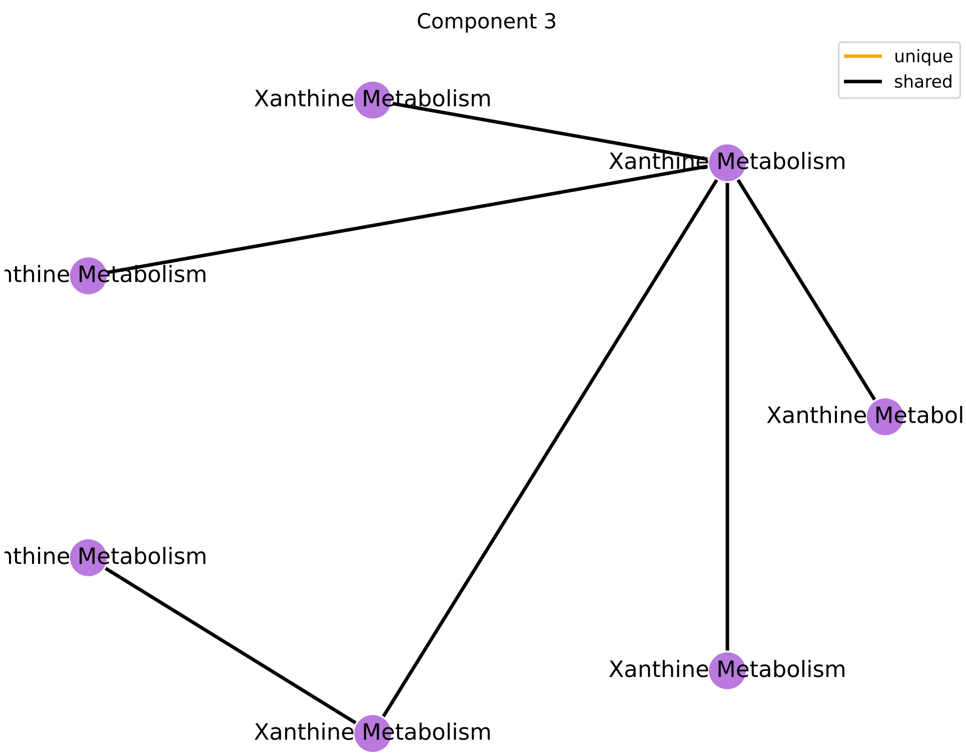
Component 3

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Xenobiotics	96	8	Inf	Inf	5.09e-08	4.58e-07

results/006/enrichment/form-components/3-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Xanthine Metabolism	14	8	Inf	Inf	1.15e-15	1.2e-13

results/006/enrichment/form-components/3-subpathway.csv



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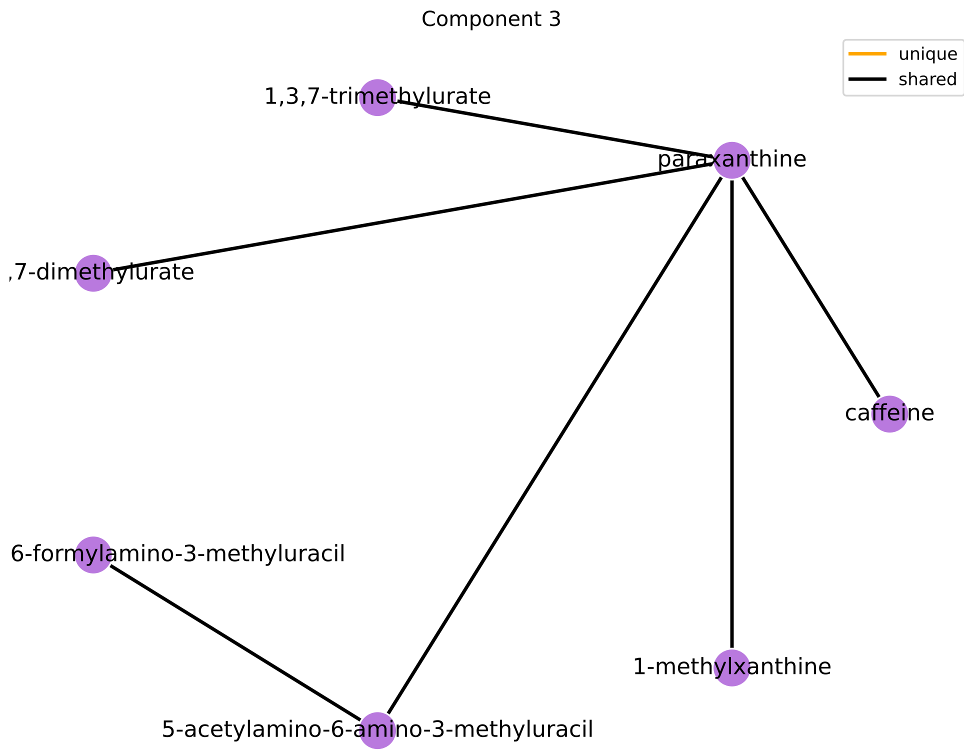

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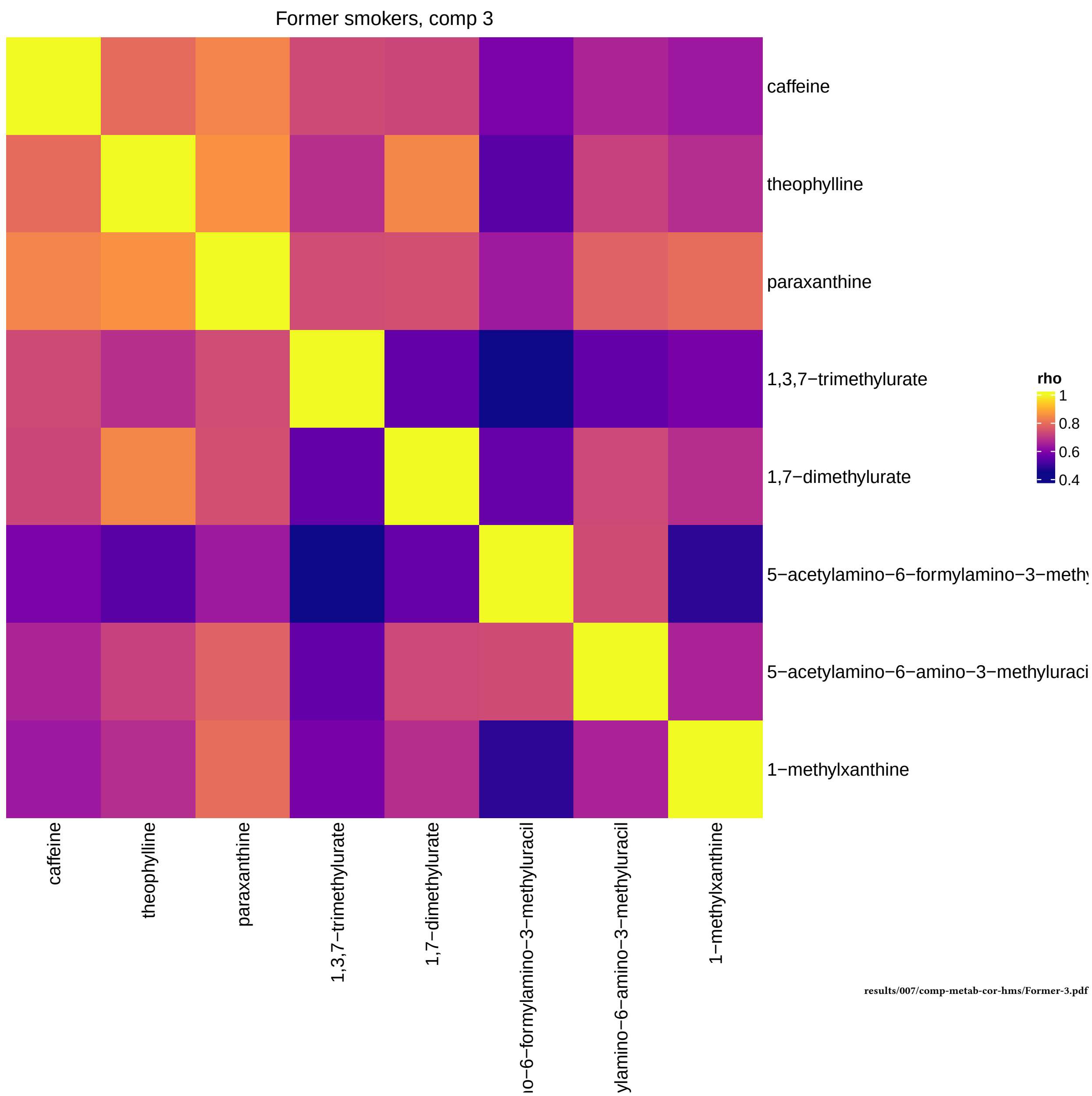


results/007/comp-metab-cor-hms/Former-3.pdf

Component 4

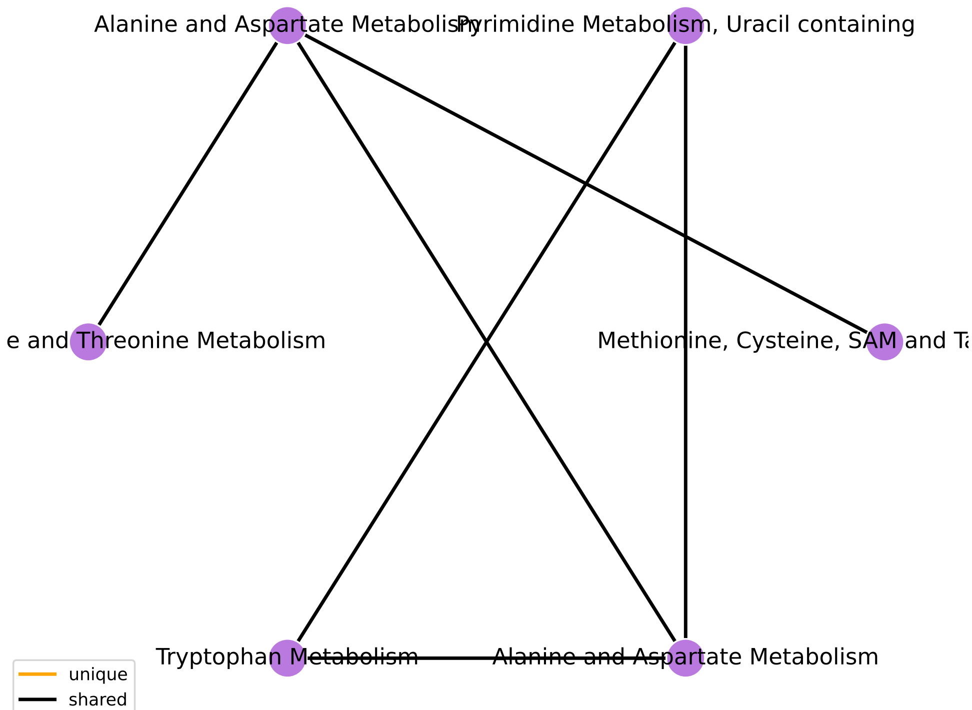
Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Nucleotide	32	4	14.512	2.675	0.000704	0.00634
Amino Acid	180	7	5.789	1.756	0.00524	0.0236

results/006/enrichment/form-components/4-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Alanine and Aspartate Metabolism	8	2	26.804	3.289	0.00512	0.421
Pyrimidine Metabolism, Uracil containing	10	2	20.157	3.004	0.00809	0.421
Methionine, Cysteine, SAM and Taurine Metabolism	17	2	10.738	2.374	0.0231	0.749
Purine Metabolism, Guanine containing	2	1	71.303	4.267	0.0288	0.749
Purine Metabolism, Adenine containing	4	1	24.264	3.189	0.0569	1
Glycine, Serine and Threonine Metabolism	10	1	8.134	2.096	0.137	1
Glutamate Metabolism	11	1	7.318	1.99	0.149	1
Tryptophan Metabolism	16	1	4.857	1.58	0.21	1

results/006/enrichment/form-components/4-subpathway.csv

Component 4



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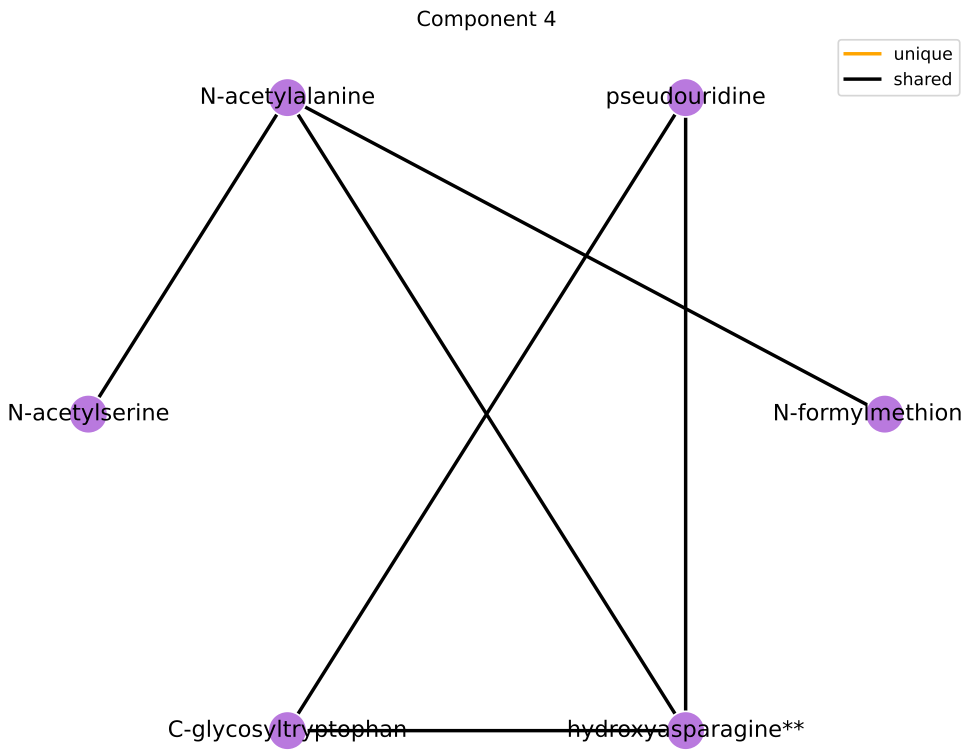

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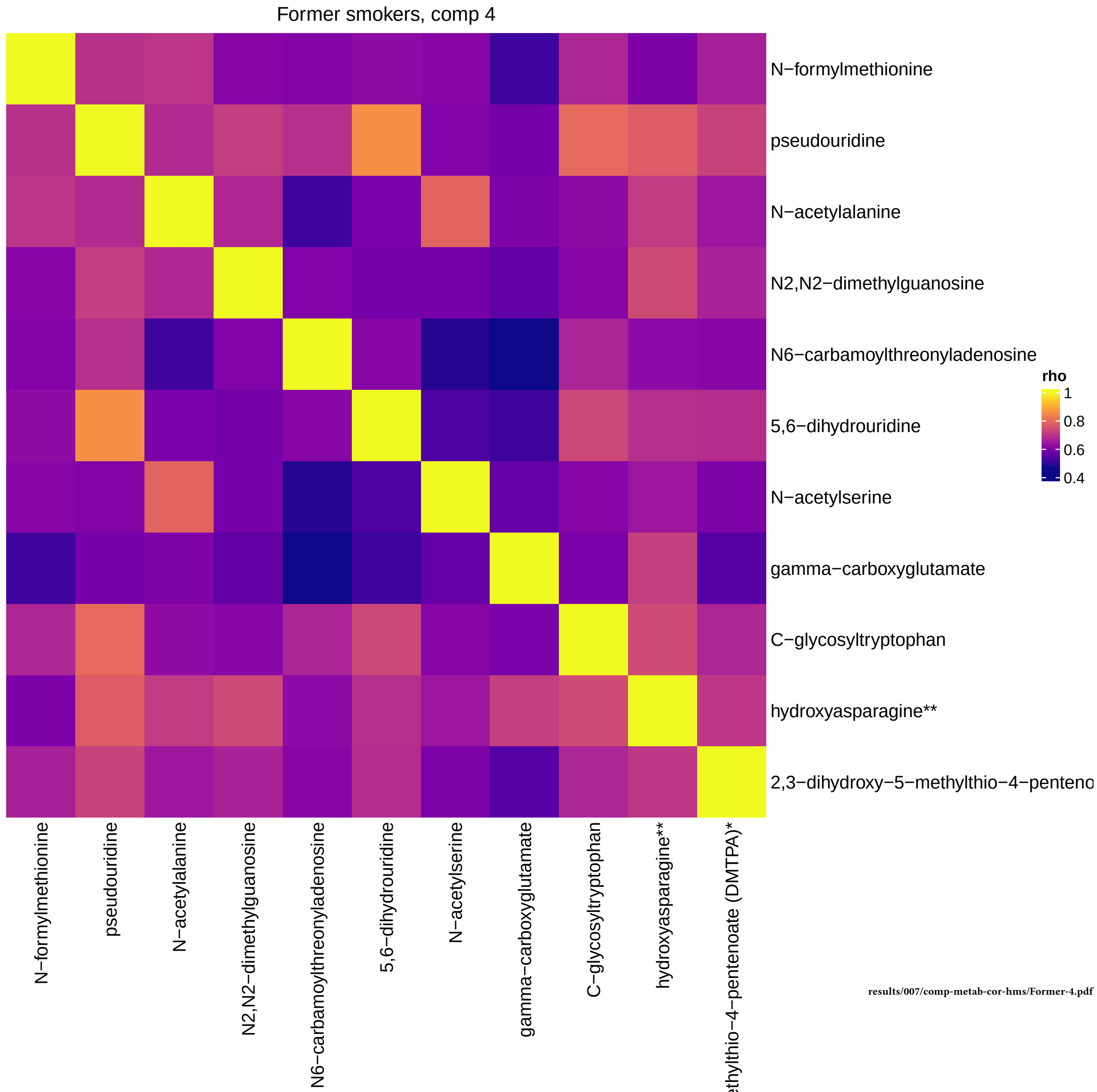

Former smokers, comp 4
N−formylmethionine
pseudouridine
N−acetylalanine
N2,N2−dimethylguanosine
N6−carbamoylthreonyladenosine
5,6−dihydrouridine
N−acetylserine
gamma−carboxyglutamate
C−glycosyltryptophan
hydroxyasparagine**
2,3−dihydroxy−5−methylthio−4−penteno
rho
1
0.8
0.6
0.4
ethylthio−4−pentenoate (DMTPA)*
results/007/comp-metab-cor-hms/Former-4.pdf

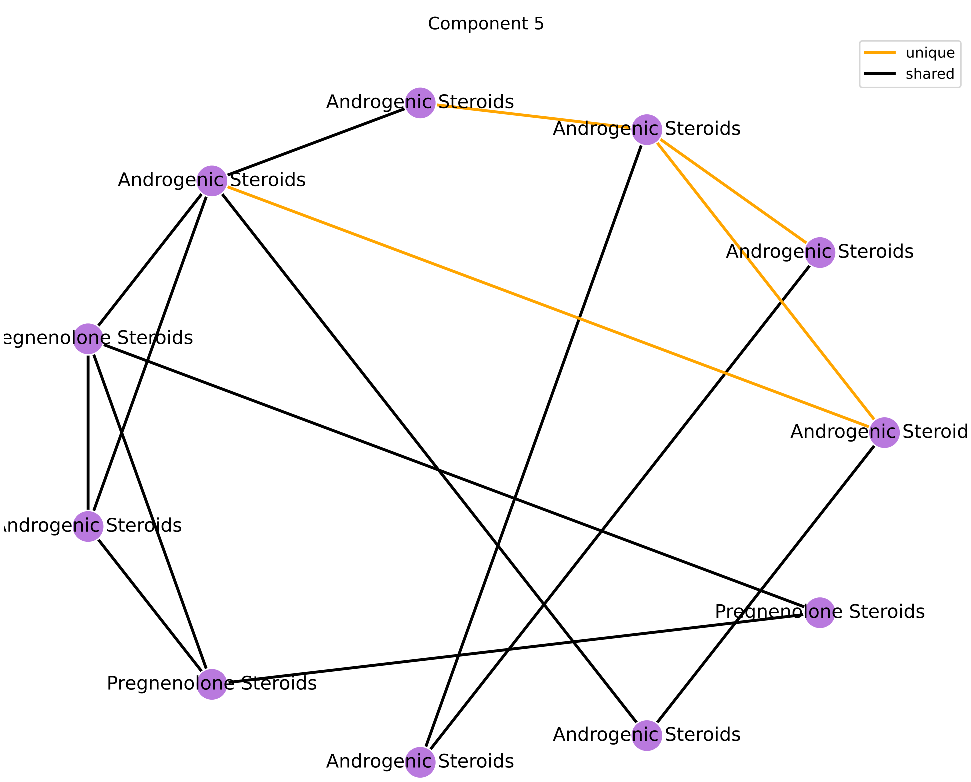
Component 5

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	13	Inf	Inf	6.22e-05	0.00056

results/006/enrichment/form-components/5-superpathway.csv

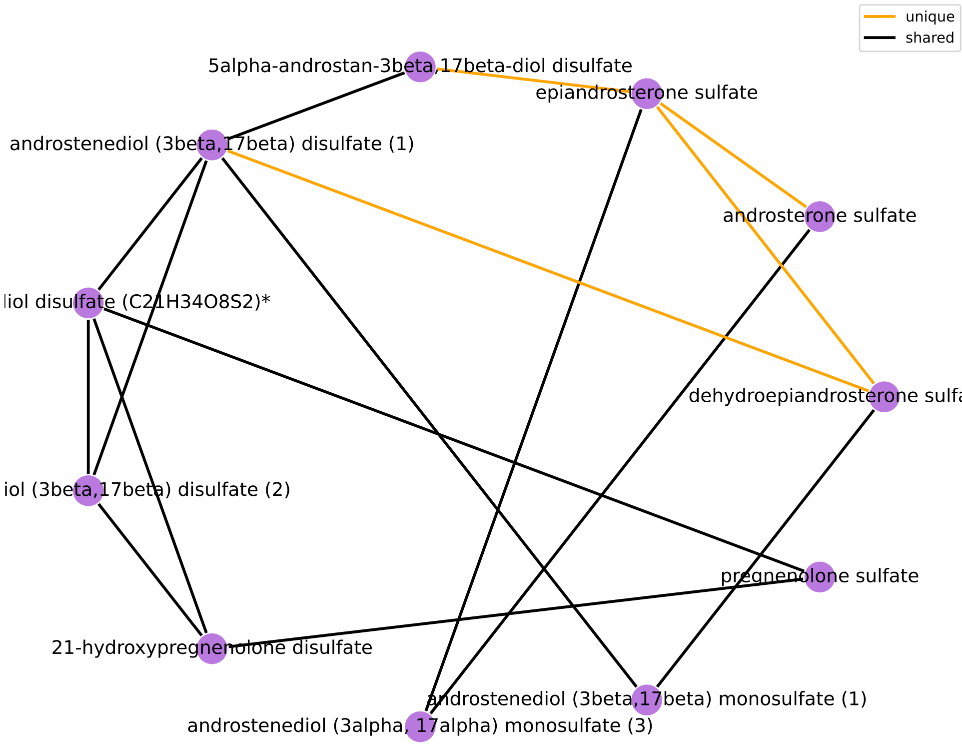
Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Androgenic Steroids	14	9	303.398	5.715	6.44e-15	6.7e-13
Pregnenolone Steroids	4	4	Inf	Inf	5.24e-08	2.72e-06

results/006/enrichment/form-components/5-subpathway.csv



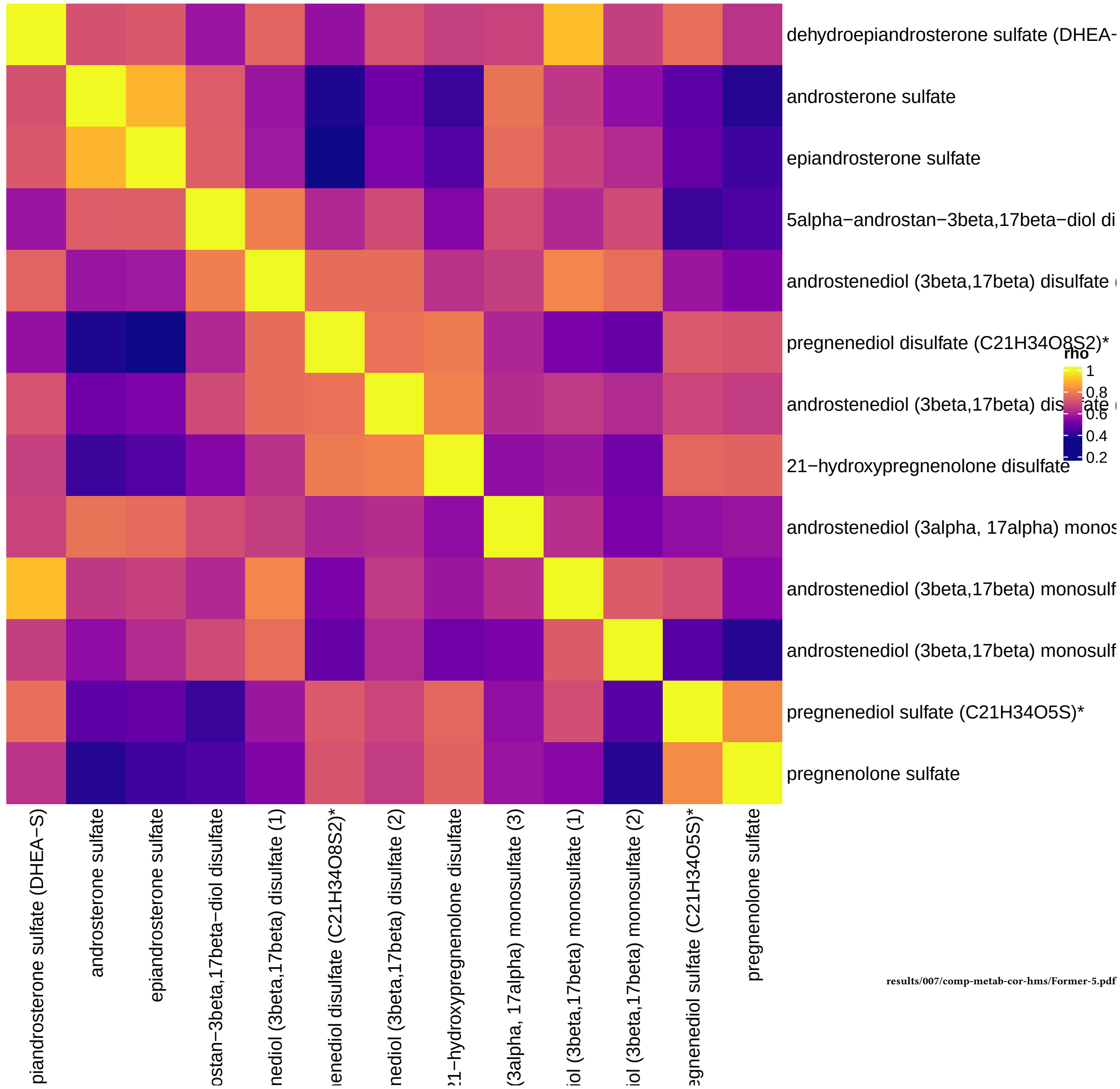
results/005_1/component-graphs/former/sub_pathway-4.pdf

Component 5



results/005_1/component-graphs/former/chemical_name-4.pdf

Former smokers, comp 5



results/007/comp-metab-cor-hms/Former-5.pdf

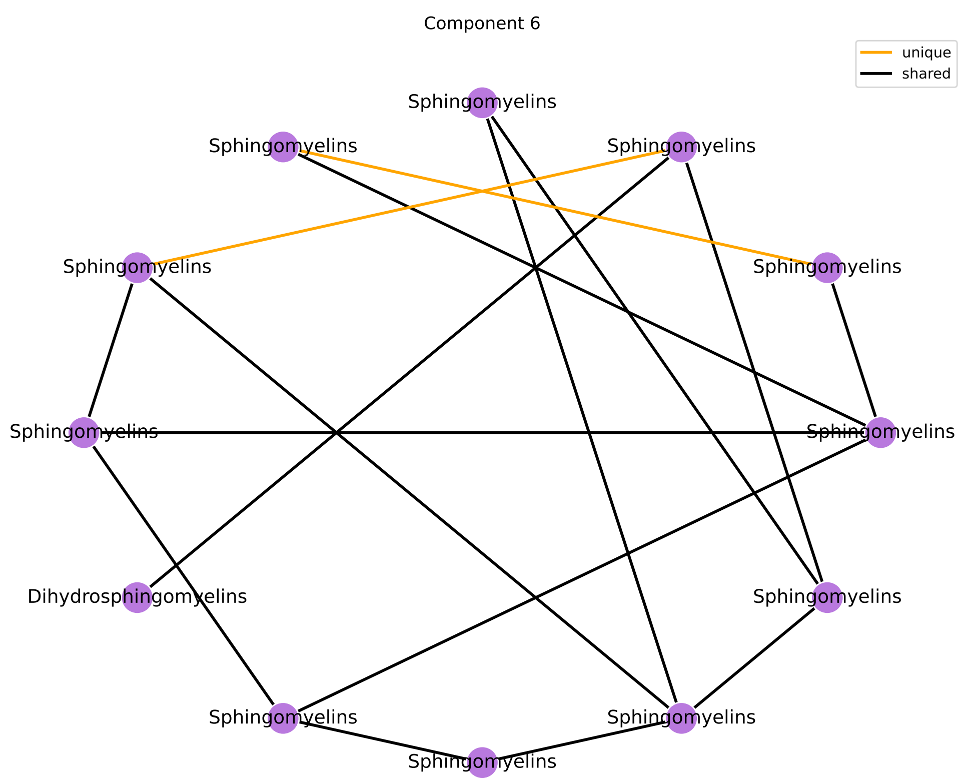
Component 6

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	13	Inf	Inf	6.22e-05	0.00056

results/006/enrichment/form-components/6-superpathway.csv

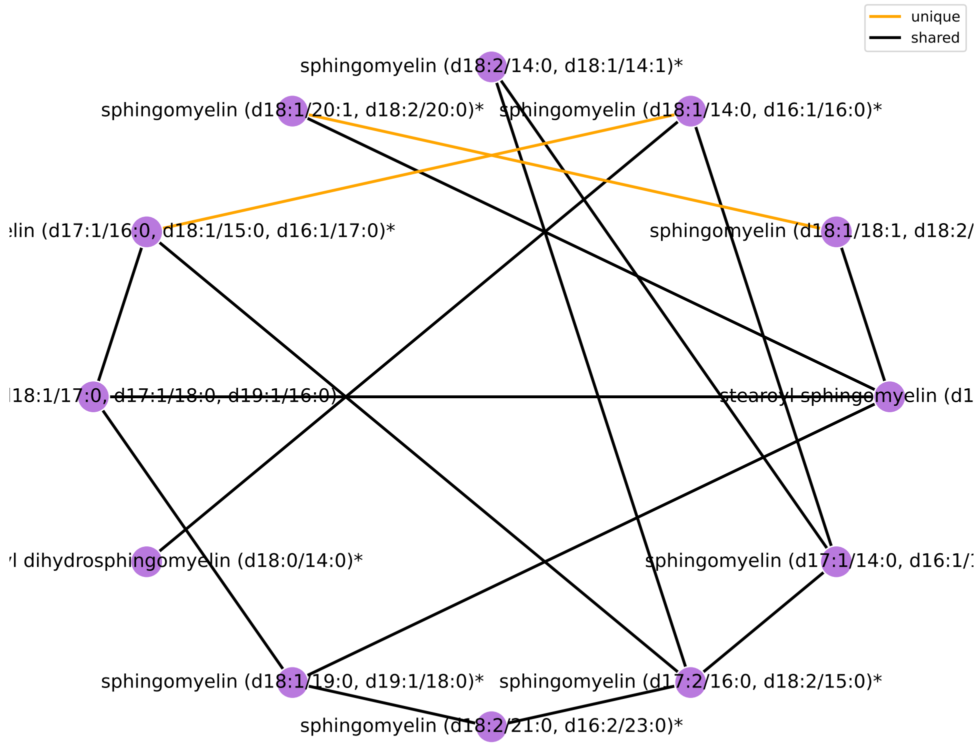
Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Sphingomyelins	28	12	518.406	6.251	5.63e-18	5.86e-16
Dihydrosphingomyelins	5	1	15.216	2.722	0.0831	1

results/006/enrichment/form-components/6-subpathway.csv



results/005_1/component-graphs/former/sub_pathway-5.pdf

Component 6



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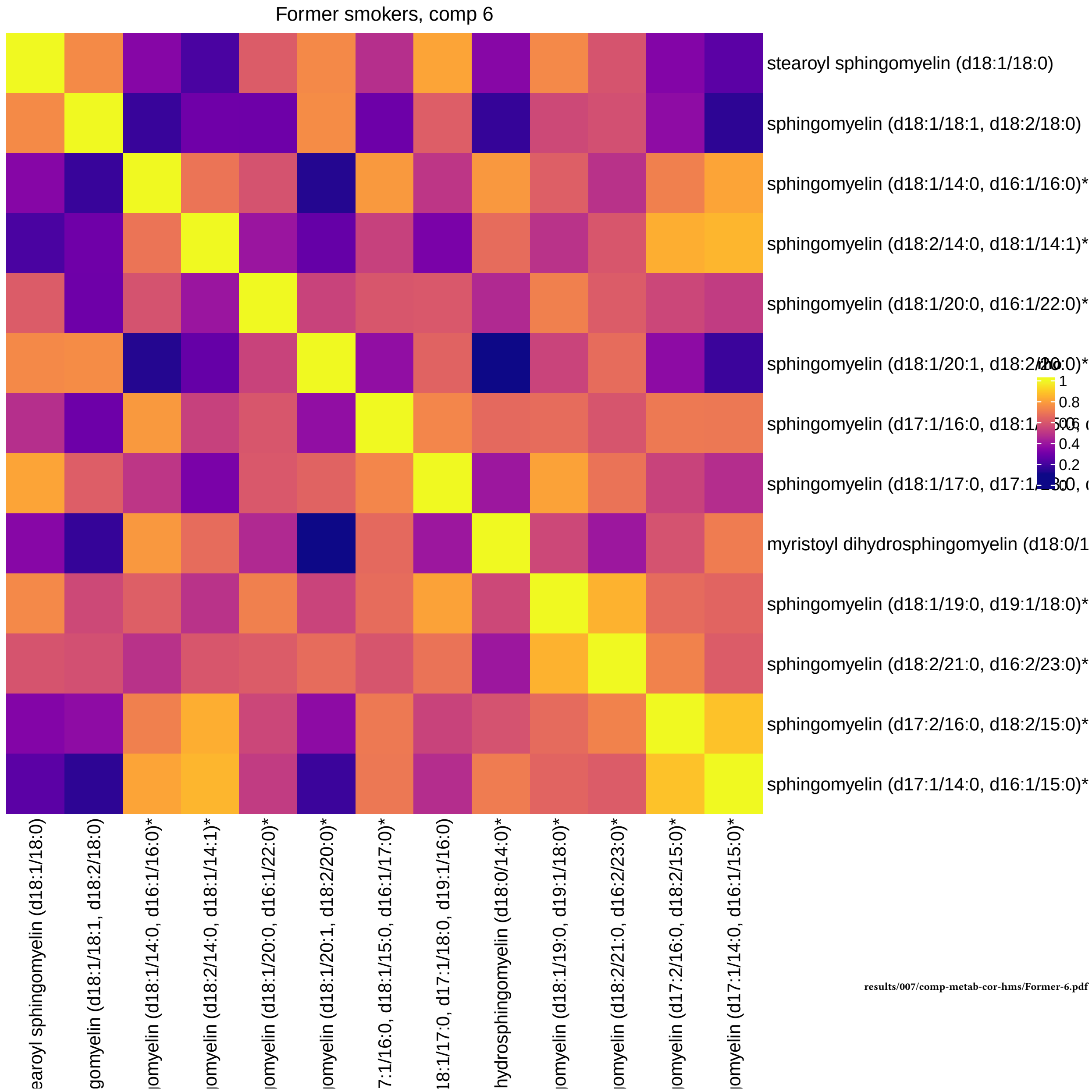


results/007/comp-metab-cor-hms/Former-6.pdf

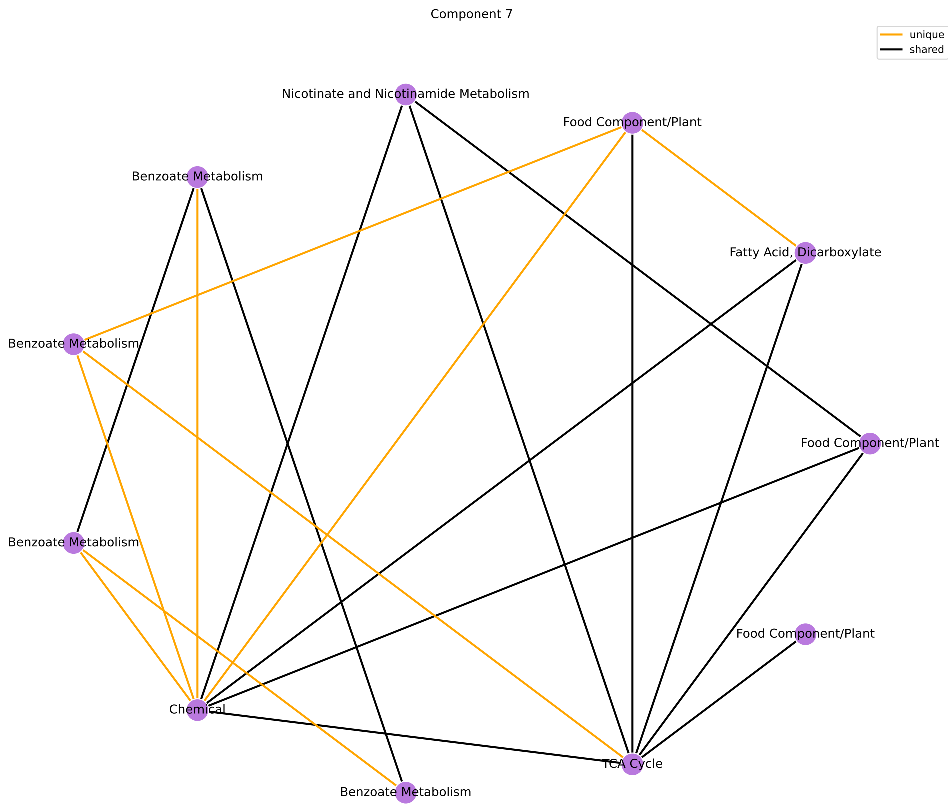
Component 7

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Xenobiotics	96	12	23.307	3.149	1.06e-08	9.54e-08
Energy	10	1	5.401	1.687	0.193	0.868
Cofactors and Vitamins	25	1	1.992	0.689	0.418	1
Amino Acid	180	1	0.21	-1.561	0.988	1

results/006/enrichment/form-components/7-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Benzoate Metabolism	21	5	20.312	3.011	3.55e-05	0.00369
Food Component/Plant	38	5	9.691	2.271	0.000708	0.0368
Chemical	18	2	6.444	1.863	0.0525	1
Nicotinate and Nicotinamide Metabolism	5	1	12.157	2.498	0.101	1
TCA Cycle	9	1	6.081	1.805	0.176	1
Tyrosine Metabolism	17	1	3.018	1.105	0.307	1
Fatty Acid, Dicarboxylate	23	1	2.179	0.779	0.392	1

results/006/enrichment/form-components/7-subpathway.csv



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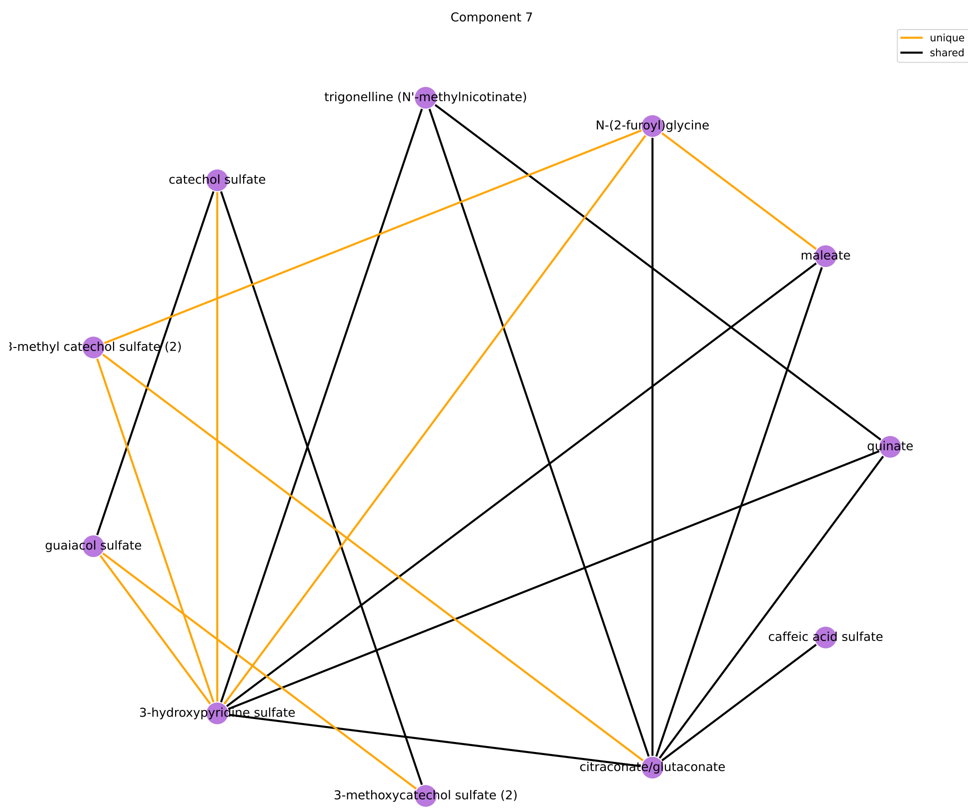

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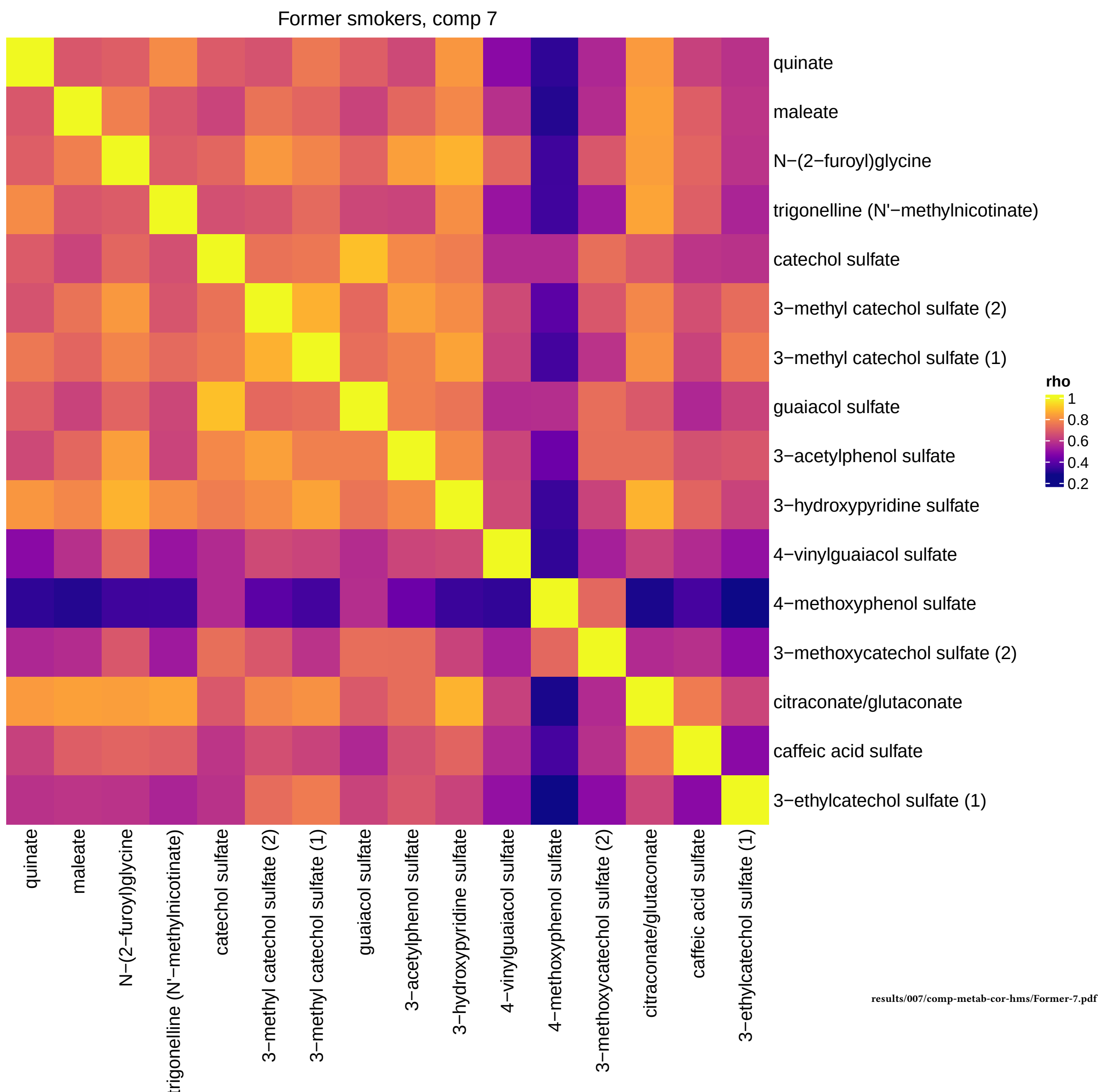

Former smokers, comp 7
quinate
maleate
N−(2−furoyl)glycine
trigonelline (N'−methylnicotinate)
catechol sulfate
3−methyl catechol sulfate (2)
3−methyl catechol sulfate (1)
guaiacol sulfate
3−acetylphenol sulfate
3−hydroxypyridine sulfate
4−vinylguaiacol sulfate
4−methoxyphenol sulfate
3−methoxycatechol sulfate (2)
citraconate/glutaconate
caffeic acid sulfate
3−ethylcatechol sulfate (1)
rho
1
0.8
0.6
0.4
0.2
results/007/comp-metab-cor-hms/Former-7.pdf

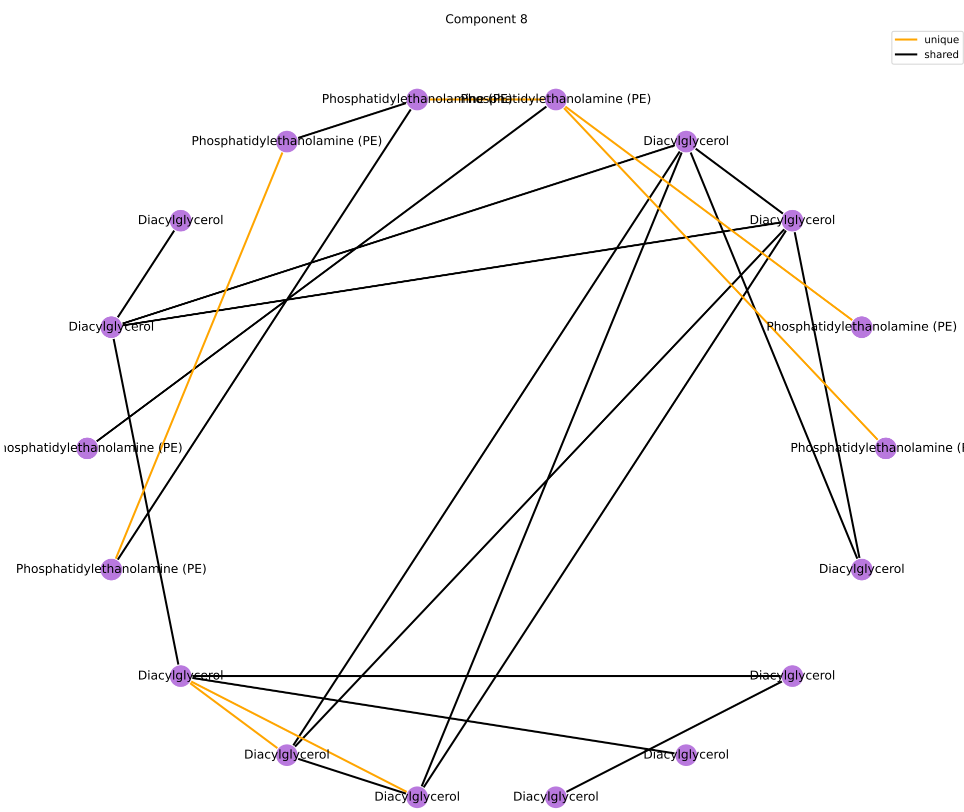
Component 8

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	19	Inf	Inf	6.53e-07	5.88e-06

results/006/enrichment/form-components/8-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Diacylglycerol	19	11	119.97	4.787	4.77e-15	4.96e-13
Phosphatidylethanolamine (PE)	11	8	168.335	5.126	4.6e-12	2.39e-10

results/006/enrichment/form-components/8-subpathway.csv



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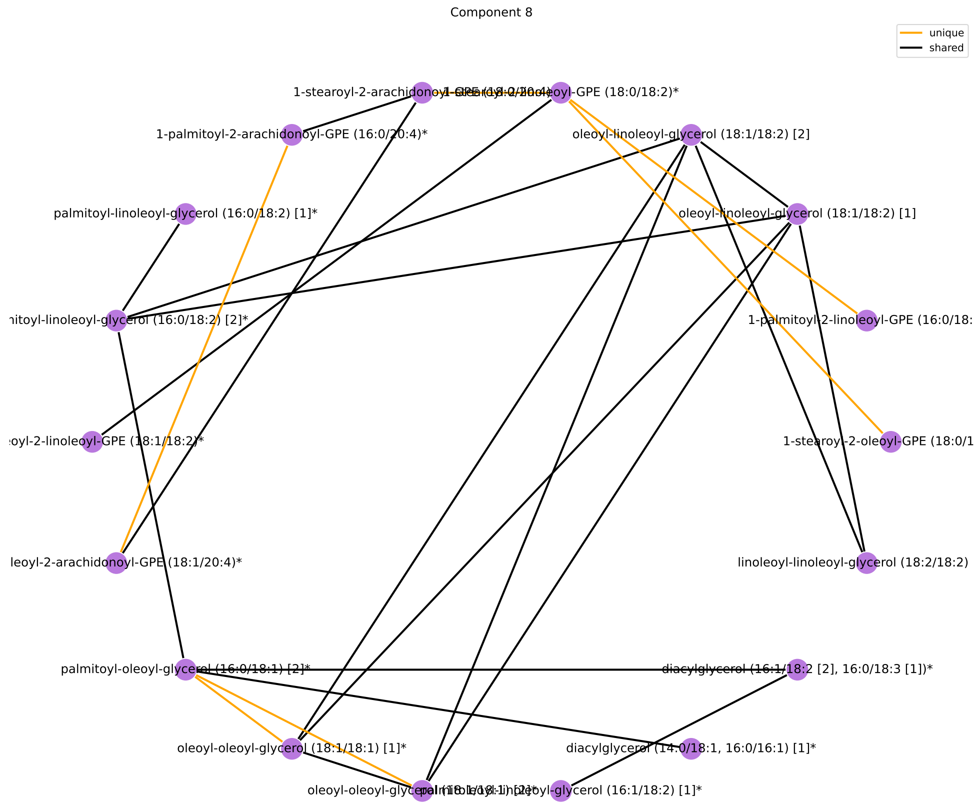

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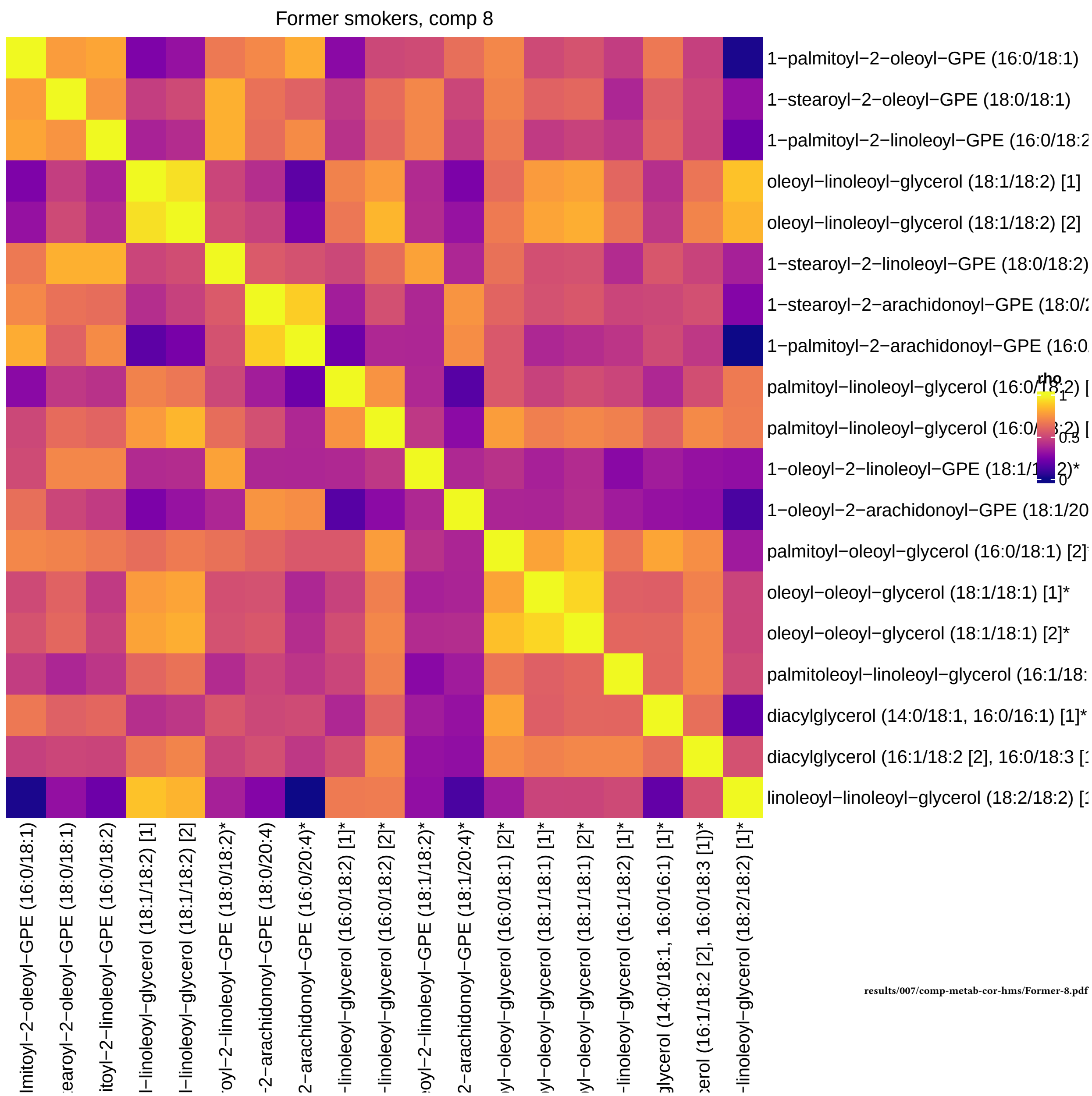


results/007/comp-metab-cor-hms/Former-8.pdf

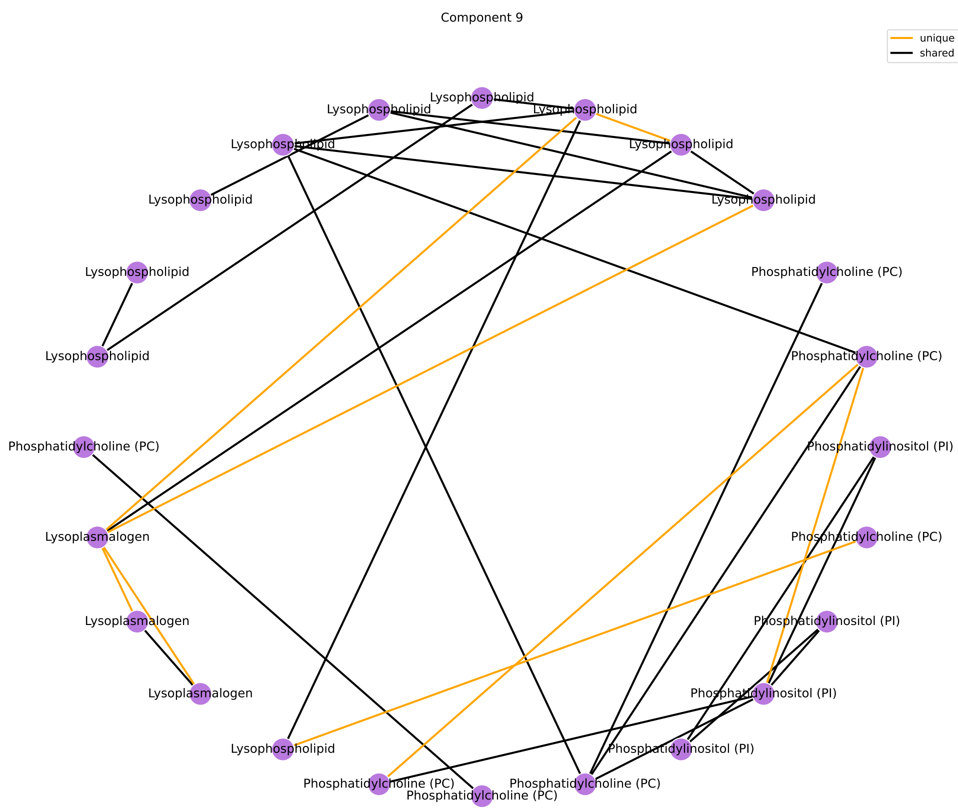
Component 9

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	30	Inf	Inf	1.33e-10	1.2e-09

results/006/enrichment/form-components/9-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lysophospholipid	24	12	38.964	3.663	2.59e-12	2.69e-10
Phosphatidylcholine (PC)	19	10	39.118	3.667	1.37e-10	7.12e-09
Phosphatidylinositol (PI)	6	5	141.28	4.951	4.04e-07	1.4e-05
Lysoplasmalogen	3	3	Inf	Inf	5.62e-05	0.00146

results/006/enrichment/form-components/9-subpathway.csv



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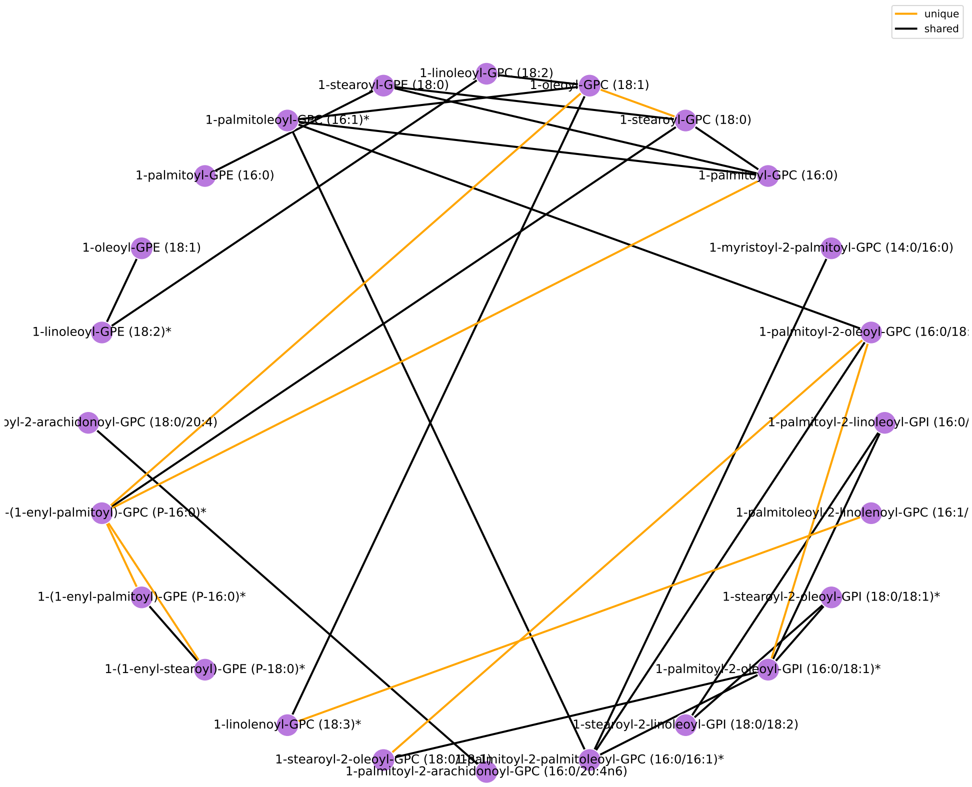
Component 9



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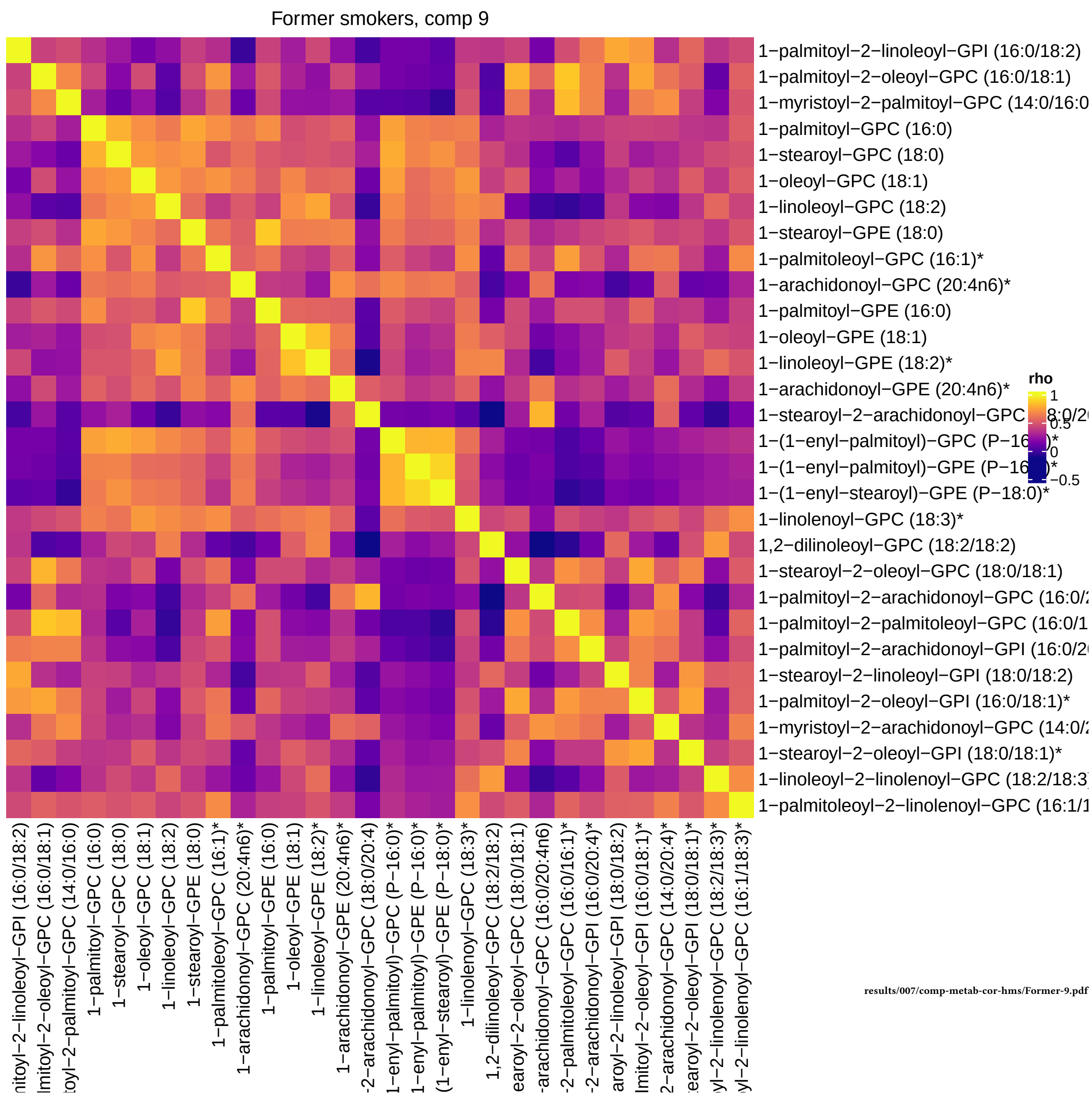


results/007/comp-metab-cor-hms/Former-9.pdf

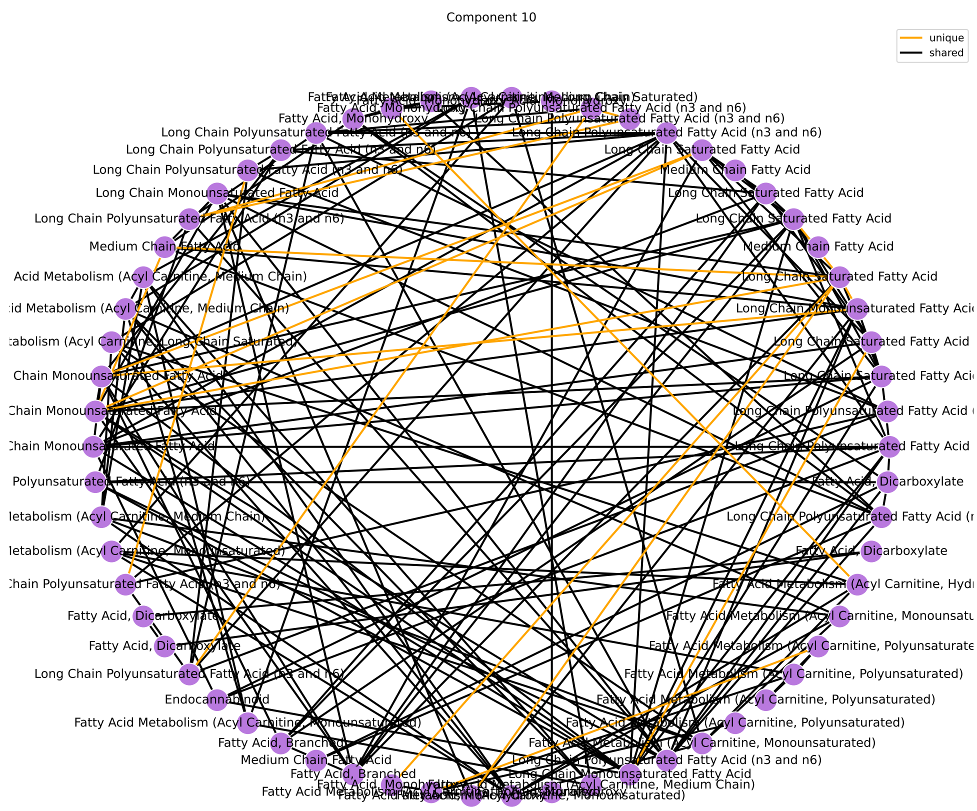
Component 10

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	69	Inf	Inf	2.09e-24	1.88e-23

results/006/enrichment/form-components/10-superpathway.csv

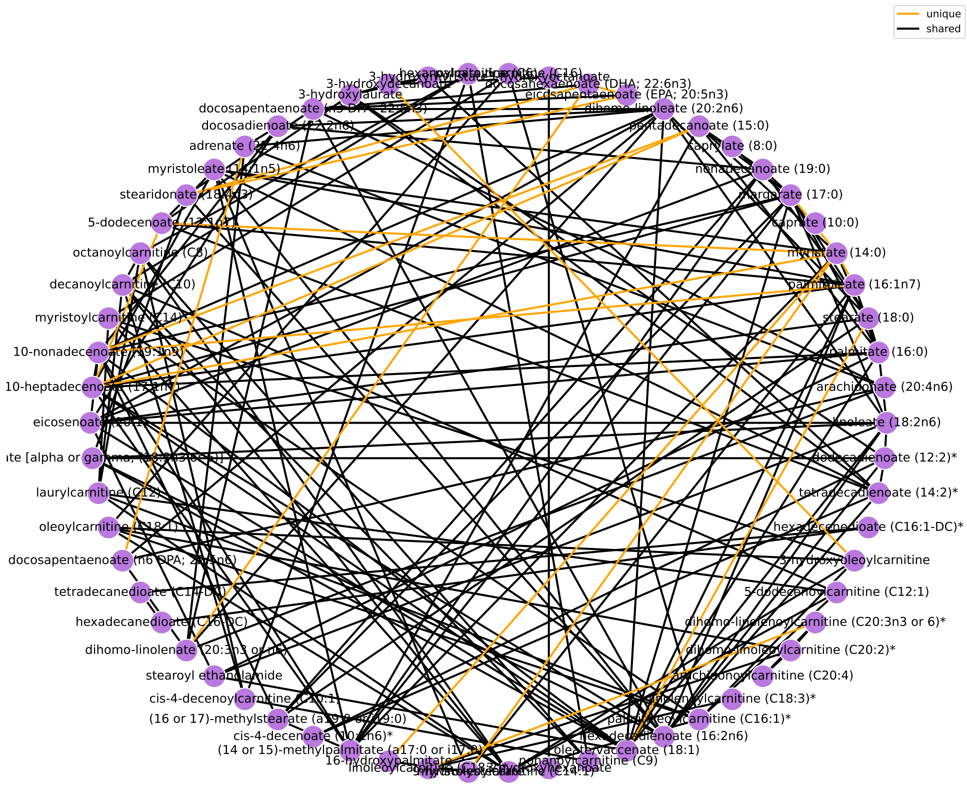
Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Long Chain Polyunsaturated Fatty Acid (n3 and n6)	14	14	Inf	Inf	7.36e-16	7.65e-14
Long Chain Monounsaturated Fatty Acid	7	6	64.656	4.169	3.02e-06	0.000105
Long Chain Saturated Fatty Acid	7	6	64.656	4.169	3.02e-06	0.000105
Fatty Acid Metabolism (Acyl Carnitine, Medium Chain)	5	5	Inf	Inf	5.46e-06	0.000114
Fatty Acid Metabolism (Acyl Carnitine, Polyunsaturated)	5	5	Inf	Inf	5.46e-06	0.000114
Fatty Acid, Monohydroxy	20	8	7.36	1.996	0.000164	0.00284
Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated)	8	5	17.702	2.874	0.000245	0.00318
Medium Chain Fatty Acid	8	5	17.702	2.874	0.000245	0.00318
Fatty Acid, Dicarboxylate	23	7	4.732	1.554	0.00288	0.0333
Fatty Acid, Branched	2	2	Inf	Inf	0.00818	0.0851
Fatty Acid Metabolism (Acyl Carnitine, Long Chain Saturated)	8	3	6.189	1.823	0.029	0.274
Glycerolipid Metabolism	2	1	10.051	2.308	0.174	1
Fatty Acid Metabolism (Acyl Carnitine, Hydroxy)	3	1	5.032	1.616	0.249	1
Endocannabinoid	5	1	2.514	0.922	0.38	1

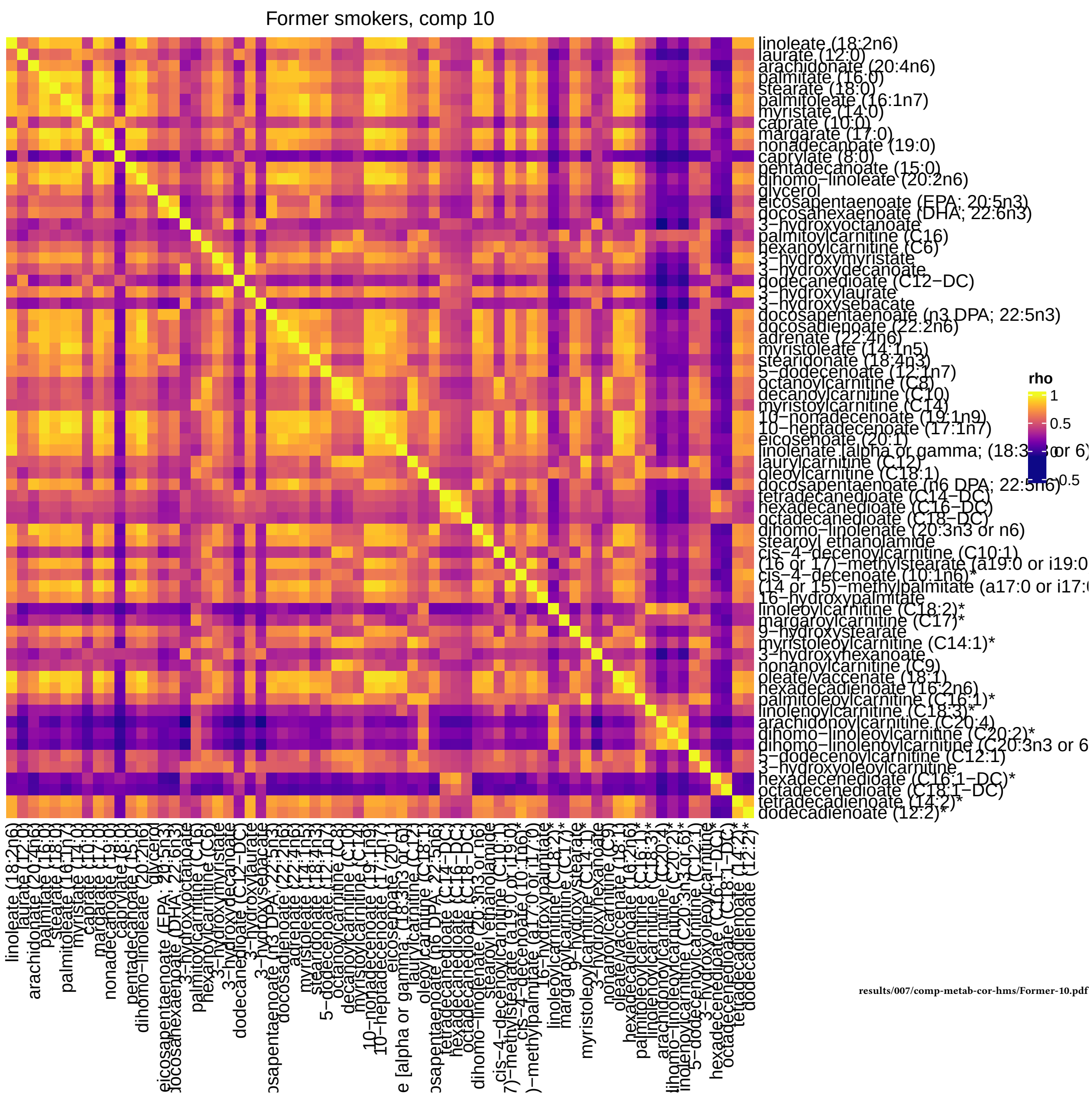
results/006/enrichment/form-components/10-subpathway.csv



results/005_1/component-graphs/former/sub_pathway-9.pdf

Component 10





results/007/comp-metab-cor-hms/Former-10.pdf

Unique To Former Smokers

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Cofactors and Vitamins	25	10	5.825	1.762	0.00015	0.00135
Partially Characterized Molecules	3	2	16.072	2.777	0.0346	0.156
Xenobiotics	96	13	1.283	0.249	0.267	0.7
Energy	10	2	2.001	0.694	0.311	0.7
Nucleotide	32	4	1.137	0.128	0.492	0.886
Lipid	363	38	0.866	-0.144	0.77	0.994
Carbohydrate	24	2	0.713	-0.338	0.773	0.994
Amino Acid	180	14	0.603	-0.506	0.969	1

results/006/enrichment/form-unique-superpathway.csv

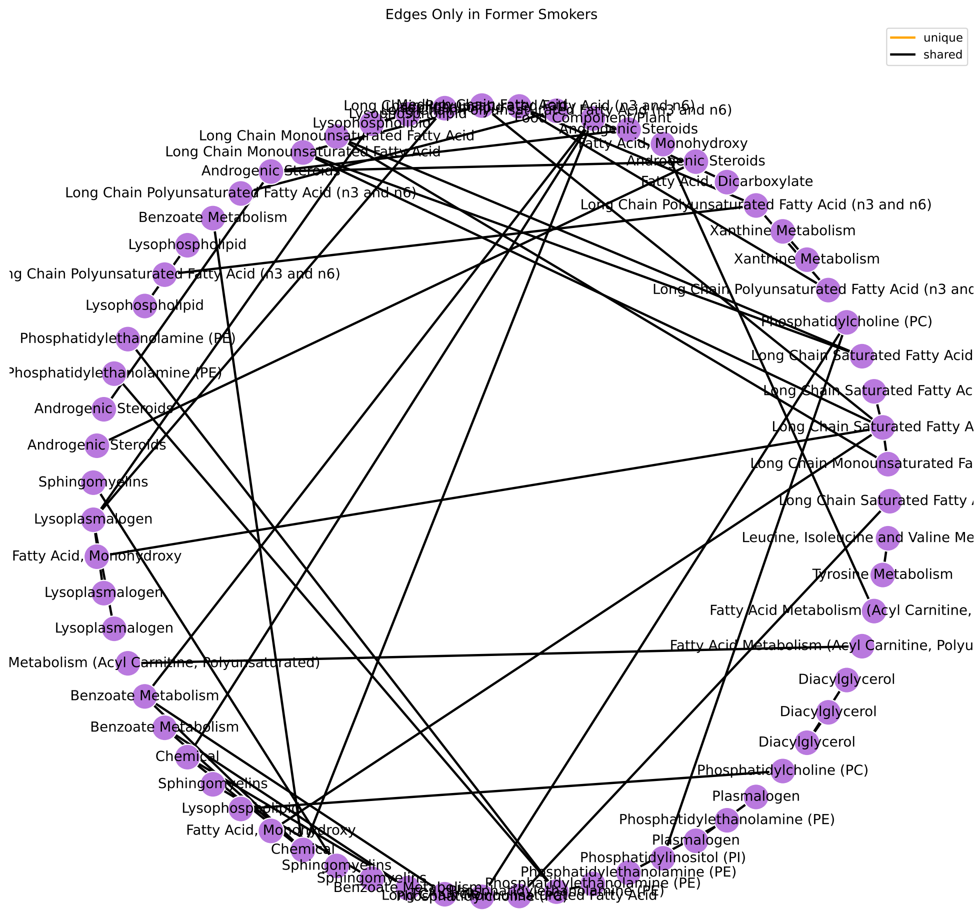
Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Dihydrosphingomyelins	5	3	12.194	2.501	0.0115	0.322
Nicotinate and Nicotinamide Metabolism	5	3	12.194	2.501	0.0115	0.322
Corticosteroids	2	2	Inf	Inf	0.0124	0.322
Vitamin B6 Metabolism	2	2	Inf	Inf	0.0124	0.322
Partially Characterized Molecules	3	2	16.072	2.777	0.0346	0.6
Pyrimidine Metabolism, Orotate containing	3	2	16.072	2.777	0.0346	0.6
Fatty Acid, Monohydroxy	20	5	2.737	1.007	0.0628	0.835
Hemoglobin and Porphyrin Metabolism	4	2	8.042	2.085	0.0642	0.835
Vitamin A Metabolism	5	2	5.361	1.679	0.0993	0.971
Fatty Acid, Dicarboxylate	23	5	2.271	0.82	0.104	0.971
Oxidative Phosphorylation	1	1	Inf	Inf	0.112	0.971
Pantothenate and CoA Metabolism	1	1	Inf	Inf	0.112	0.971
Chemical	18	4	2.321	0.842	0.133	1
Glycolysis, Gluconeogenesis, and Pyruvate Metabolism	6	2	4.019	1.391	0.138	1
Hexosylceramides (HCER)	7	2	3.212	1.167	0.18	1
Benzoate Metabolism	21	4	1.904	0.644	0.201	1
Drug - Topical Agents	2	1	7.958	2.074	0.212	1
Glycerolipid Metabolism	2	1	7.958	2.074	0.212	1
Purine Metabolism, Guanine containing	2	1	7.958	2.074	0.212	1
Alanine and Aspartate Metabolism	8	2	2.674	0.984	0.223	1
Tyrosine Metabolism	17	3	1.721	0.543	0.295	1
Glutamate Metabolism	11	2	1.776	0.574	0.355	1
Phosphatidylcholine (PC)	19	3	1.501	0.406	0.361	1
Progestin Steroids	4	1	2.654	0.976	0.379	1

Purine Metabolism, Adenine containing	4	1	2.654	0.976	0.379	1
Fatty Acid Metabolism (also BCAA Metabolism)	5	1	1.989	0.688	0.449	1
Androgenic Steroids	14	2	1.327	0.283	0.478	1
Phosphatidylinositol (PI)	6	1	1.589	0.463	0.511	1
Secondary Bile Acid Metabolism	15	2	1.223	0.201	0.516	1
Lysophospholipid	24	3	1.136	0.128	0.517	1
Glutathione Metabolism	7	1	1.323	0.28	0.567	1
Long Chain Saturated Fatty Acid	7	1	1.323	0.28	0.567	1
Methionine, Cysteine, SAM and Taurine Metabolism	17	2	1.057	0.055	0.586	1
Fatty Acid Metabolism (Acyl Carnitine, Long Chain Saturated)	8	1	1.132	0.124	0.616	1
Medium Chain Fatty Acid	8	1	1.132	0.124	0.616	1
Food Component/Plant	38	4	0.928	-0.075	0.635	1
TCA Cycle	9	1	0.99	-0.01	0.659	1
Leucine, Isoleucine and Valine Metabolism	31	3	0.843	-0.171	0.697	1
Phosphatidylethanolamine (PE)	11	1	0.79	-0.236	0.732	1
Plasmalogen	11	1	0.79	-0.236	0.732	1
Sphingomyelins	28	2	0.6	-0.511	0.843	1
Urea cycle; Arginine and Proline Metabolism	21	1	0.389	-0.944	0.921	1

results/006/enrichment/form-unique-subpathway.csv

results/005_1/sub_pathway-node-only-form-graph.pdf

results/005_1/chemical_name-node-only-form-graph.pdf



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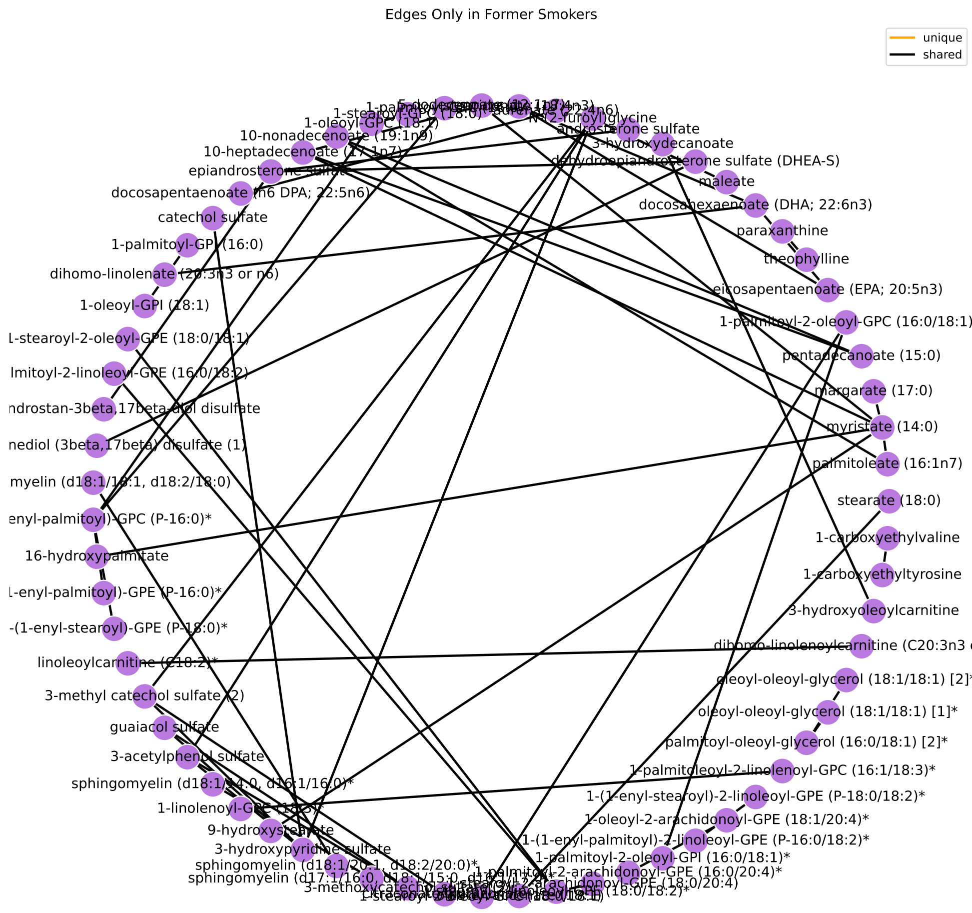


results/005_1/chemical_name-edge-only-form-graph.pdf

Intersection of Both

Super-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Lipid	363	215	4.577	1.521	2.82e-23	2.54e-22
Peptide	25	11	1.14	0.131	0.45	1
Xenobiotics	96	33	0.728	-0.317	0.934	1
Cofactors and Vitamins	25	7	0.552	-0.594	0.941	1
Energy	10	1	0.158	-1.845	0.995	1

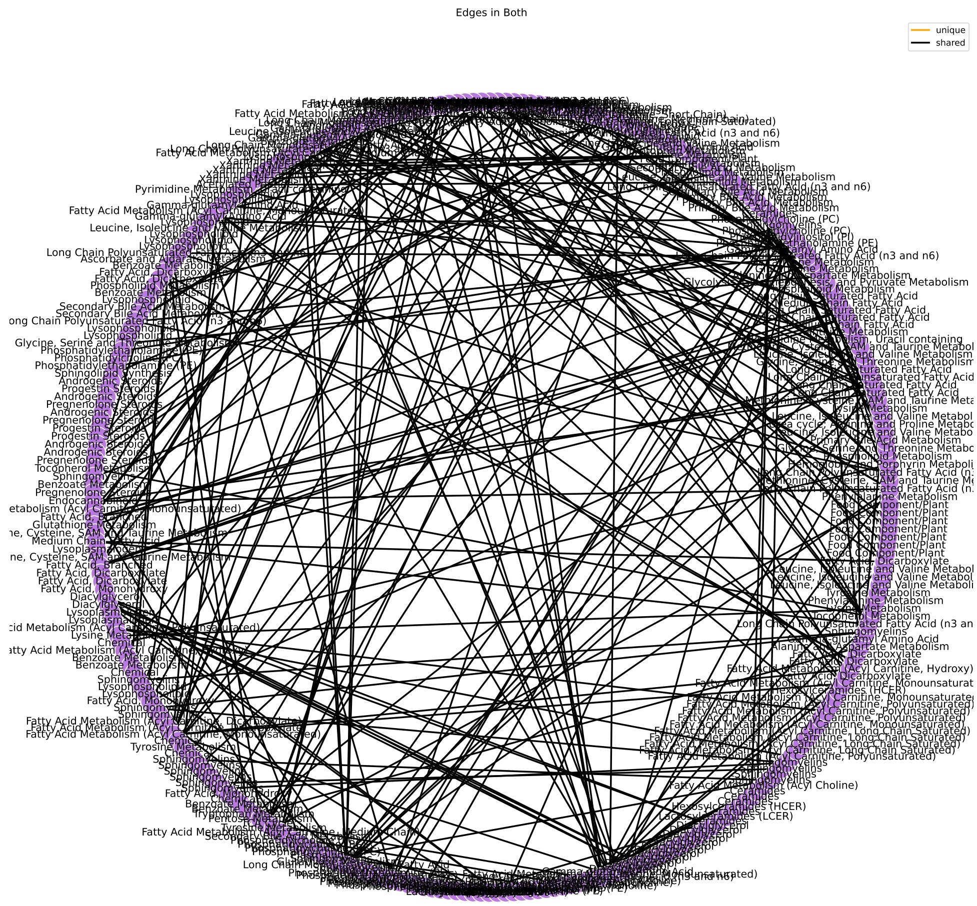
results/006/enrichment/intersection-superpathway.csv

Sub-Pathway	Expected Frequency	Observed Frequency	OR	ln(OR)	Fisher P	FDR P
Long Chain Polyunsaturated Fatty Acid (n3 and n6)	14	14	Inf	Inf	3.07e-06	0.000319
Diacylglycerol	19	17	12.903	2.557	1.29e-05	0.000671
Phosphatidylethanolamine (PE)	11	10	14.858	2.699	0.000839	0.0225
Sphingomyelins	28	20	3.786	1.331	0.000866	0.0225
Plasmalogen	11	9	6.654	1.895	0.00671	0.129
Lysophospholipid	24	16	2.989	1.095	0.00868	0.129
Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated)	8	7	10.299	2.332	0.00951	0.129
Fatty Acid Metabolism (Acyl Carnitine, Medium Chain)	5	5	Inf	Inf	0.0112	0.129
Fatty Acid Metabolism (Acyl Carnitine, Polyunsaturated)	5	5	Inf	Inf	0.0112	0.129
Long Chain Monounsaturated Fatty Acid	7	6	8.8	2.175	0.0208	0.197
Long Chain Saturated Fatty Acid	7	6	8.8	2.175	0.0208	0.197
Fatty Acid Metabolism (Acyl Choline)	4	4	Inf	Inf	0.0277	0.222
Pregnenolone Steroids	4	4	Inf	Inf	0.0277	0.222
Fatty Acid Metabolism (Acyl Carnitine, Long Chain Saturated)	8	6	4.393	1.48	0.0546	0.406
Ascorbate and Aldarate Metabolism	3	3	Inf	Inf	0.068	0.442
Lysoplasmalogen	3	3	Inf	Inf	0.068	0.442
Phosphatidylcholine (PC)	19	11	2.022	0.704	0.0995	0.595
Gamma-glutamyl Amino Acid	17	10	2.098	0.741	0.103	0.595
Androgenic Steroids	14	8	1.95	0.668	0.165	0.795
Xanthine Metabolism	14	8	1.95	0.668	0.165	0.795
Fatty Acid, Branched	2	2	Inf	Inf	0.167	0.795
Primary Bile Acid Metabolism	8	5	2.429	0.887	0.187	0.795
Progestin Steroids	4	3	4.36	1.472	0.189	0.795
Phosphatidylinositol (PI)	6	4	2.911	1.068	0.191	0.795

Phospholipid Metabolism	6	4	2.911	1.068	0.191	0.795
Ceramides	11	6	1.747	0.558	0.265	1
Glutathione Metabolism	7	4	1.937	0.661	0.307	1
Phenylalanine Metabolism	7	4	1.937	0.661	0.307	1
Fatty Acid Metabolism (Acyl Carnitine, Hydroxy)	3	2	2.899	1.064	0.365	1
Lactosylceramides (LCER)	3	2	2.899	1.064	0.365	1
Fatty Acid Metabolism (Acyl Carnitine, Short Chain)	1	1	Inf	Inf	0.409	1
Ketone Bodies	1	1	Inf	Inf	0.409	1
Medium Chain Fatty Acid	8	4	1.45	0.372	0.427	1
Fatty Acid, Dicarboxylate	23	10	1.115	0.109	0.479	1
Sphingosines	2	1	1.446	0.369	0.651	1
Fatty Acid Metabolism (Acyl Carnitine, Dicarboxylate)	5	2	0.963	-0.038	0.679	1
Tocopherol Metabolism	5	2	0.963	-0.038	0.679	1
Benzoate Metabolism	21	8	0.887	-0.12	0.684	1
Fatty Acid, Monohydroxy	20	7	0.773	-0.257	0.778	1
Leucine, Isoleucine and Valine Metabolism	31	11	0.788	-0.238	0.79	1
Acetylated Peptides	3	1	0.722	-0.326	0.794	1
Sphingolipid Synthesis	3	1	0.722	-0.326	0.794	1
Secondary Bile Acid Metabolism	15	5	0.718	-0.331	0.805	1
Glycine, Serine and Threonine Metabolism	10	3	0.616	-0.485	0.848	1
Hexosylceramides (HCER)	7	2	0.576	-0.552	0.854	1
Hemoglobin and Porphyrin Metabolism	4	1	0.48	-0.734	0.879	1
Methionine, Cysteine, SAM and Taurine Metabolism	17	5	0.596	-0.518	0.891	1
Alanine and Aspartate Metabolism	8	2	0.479	-0.736	0.904	1
Food Component/Plant	38	12	0.654	-0.425	0.916	1
Chemical	18	5	0.549	-0.6	0.921	1
Lysine Metabolism	12	3	0.477	-0.74	0.927	1
Dihydrosphingomyelins	5	1	0.36	-1.022	0.929	1
Endocannabinoid	5	1	0.36	-1.022	0.929	1
Nicotinate and Nicotinamide Metabolism	5	1	0.36	-1.022	0.929	1
Glycolysis, Gluconeogenesis, and Pyruvate Metabolism	6	1	0.287	-1.248	0.958	1
Pentose Metabolism	6	1	0.287	-1.248	0.958	1
Pyrimidine Metabolism, Uracil containing	10	2	0.358	-1.027	0.96	1

Tryptophan Metabolism	16	3	0.327	-1.118	0.985	1
Tyrosine Metabolism	17	3	0.303	-1.194	0.99	1
TCA Cycle	9	1	0.178	-1.726	0.991	1

results/006/enrichment/intersection-subpathway.csv



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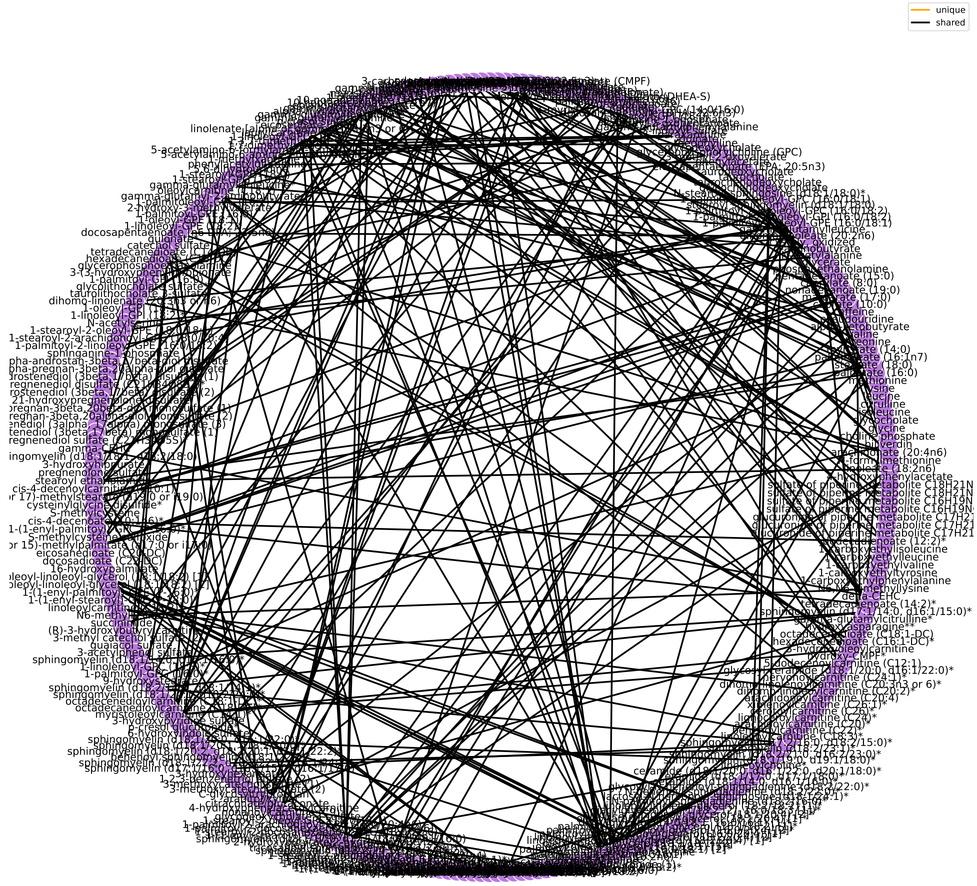
Edges in Both



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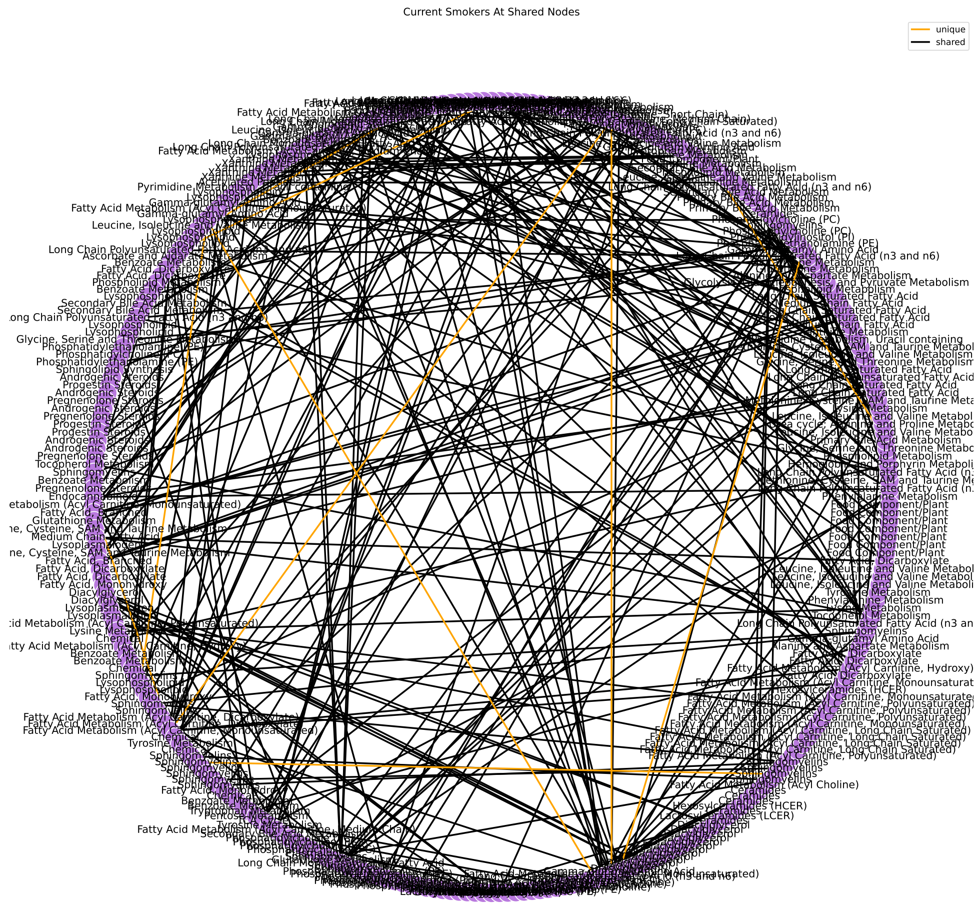

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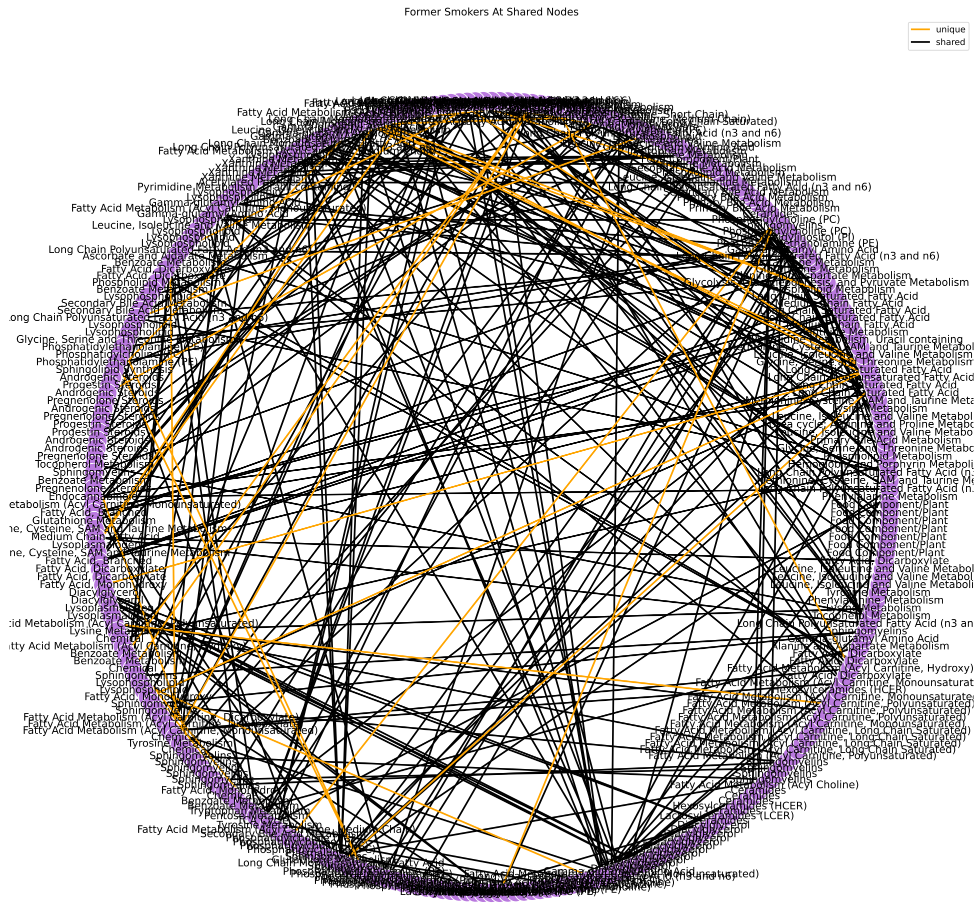

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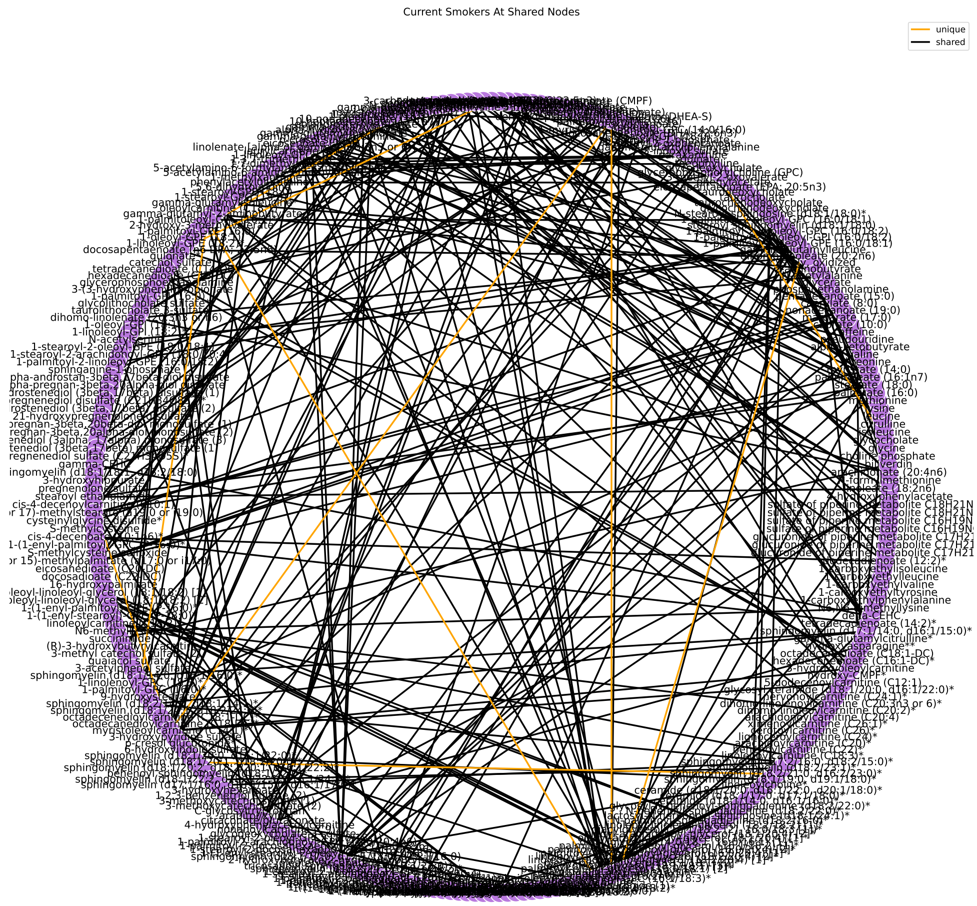

results/005_1/chemical_name-node-shared-curr-graph.pdf

Unique Edges (in orange)

Primary Bile Acid Metabolism glycocholate	Secondary Bile Acid Metabolism glycodeoxycholate
Gamma-glutamyl Amino Acid gamma-glutamylleucine	Gamma-glutamyl Amino Acid gamma-glutamyl-alpha-lysine
Primary Bile Acid Metabolism glycochenodeoxycholate	Secondary Bile Acid Metabolism glycodeoxycholate
Fatty Acid Metabolism (Acyl Carnitine, Medium Chain) hexanoylcarnitine (C6)	Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated) myristoleoylcarnitine (C14:1)*
Fatty Acid Metabolism (Acyl Carnitine, Medium Chain) hexanoylcarnitine (C6)	Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated) palmitoleoylcarnitine (C16:1)*

Long Chain Polyunsaturated Fatty Acid (n3 and n6) docosapentaenoate (n3 DPA; 22:5n3)	Long Chain Polyunsaturated Fatty Acid (n3 and n6) adrenate (22:4n6)
Long Chain Polyunsaturated Fatty Acid (n3 and n6) docosadienoate (22:2n6)	Long Chain Polyunsaturated Fatty Acid (n3 and n6) docosapentaenoate (n6 DPA; 22:5n6)
Lysophospholipid 1-palmitoyl-GPC (16:0)	Lysophospholipid 1-palmitoyl-GPE (16:0)
Xanthine Metabolism 1,7-dimethylurate	Xanthine Metabolism 5-acetylamino-6-amino-3-methyluracil
Lysophospholipid 1-palmitoleoyl-GPC (16:1)*	Lysophospholipid 1-linolenoyl-GPC (18:3)*
Lysophospholipid 1-palmitoleoyl-GPC (16:1)*	Phosphatidylcholine (PC) 1-palmitoleoyl-2-linolenoyl-GPC (16:1/18:3)*
Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated) cis-4-decenoylcarnitine (C10:1)	Fatty Acid Metabolism (Acyl Carnitine, Monounsaturated) myristoleoylcarnitine (C14:1)*
Sphingomyelins sphingomyelin (d18:1/20:1, d18:2/20:0)*	Sphingomyelins sphingomyelin (d18:2/21:0, d16:2/23:0)*
Diacylglycerol palmitoyl-oleoyl-glycerol (16:0/18:1) [2]*	Diacylglycerol palmitoyl-arachidonoyl-glycerol (16:0/20:4) [2]*
Diacylglycerol oleoyl-oleoyl-glycerol (18:1/18:1) [2]*	Diacylglycerol diacylglycerol (16:1/18:2 [2], 16:0/18:3 [1])*
Sphingomyelins sphingomyelin (d18:2/21:0, d16:2/23:0)*	Sphingomyelins sphingomyelin (d18:2/23:1)*

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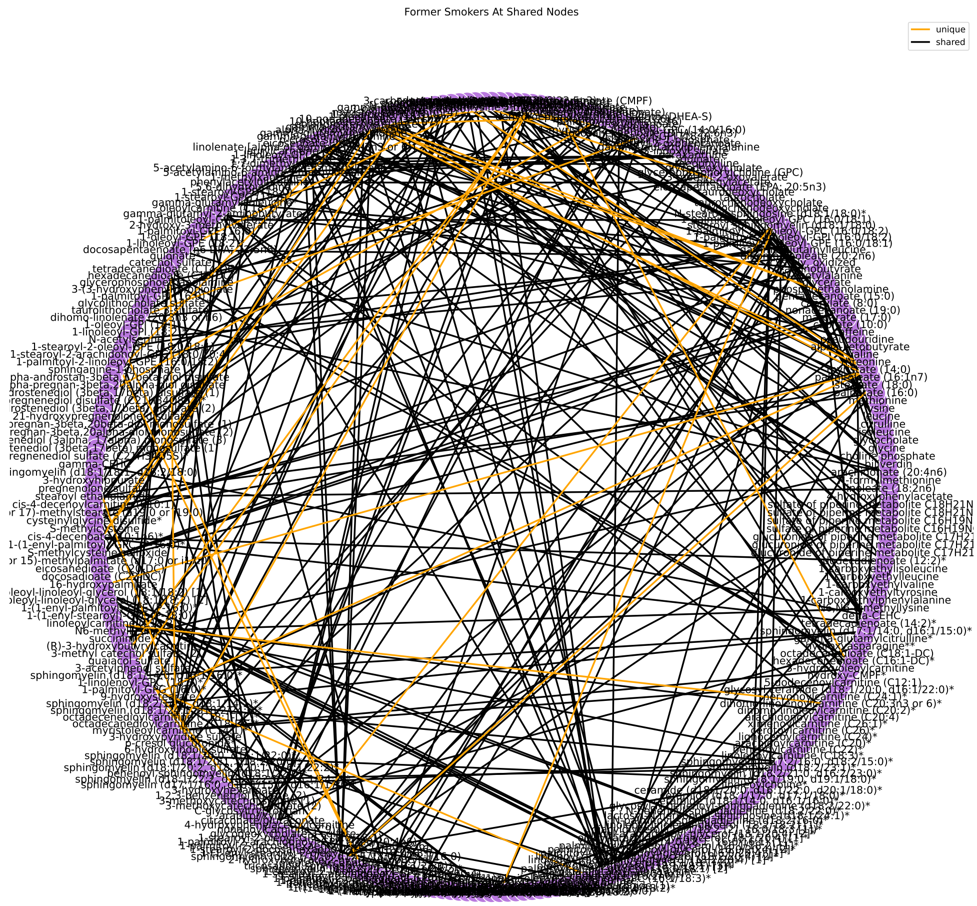


results/005_1/chemical_name-node-shared-form-graph.pdf

Unique Edges (in orange)

Long Chain Saturated Fatty Acid stearate (18:0)	Long Chain Monounsaturated Fatty Acid oleate/vaccenate (18:1)
Long Chain Monounsaturated Fatty Acid palmitoleate (16:1n7)	Long Chain Saturated Fatty Acid myristate (14:0)
Long Chain Monounsaturated Fatty Acid palmitoleate (16:1n7)	Long Chain Monounsaturated Fatty Acid 10-nonadecenoate (19:1n9)
Long Chain Saturated Fatty Acid myristate (14:0)	Long Chain Saturated Fatty Acid margarate (17:0)
Long Chain Saturated Fatty Acid myristate (14:0)	Medium Chain Fatty Acid 5-dodecenoate (12:1n7)

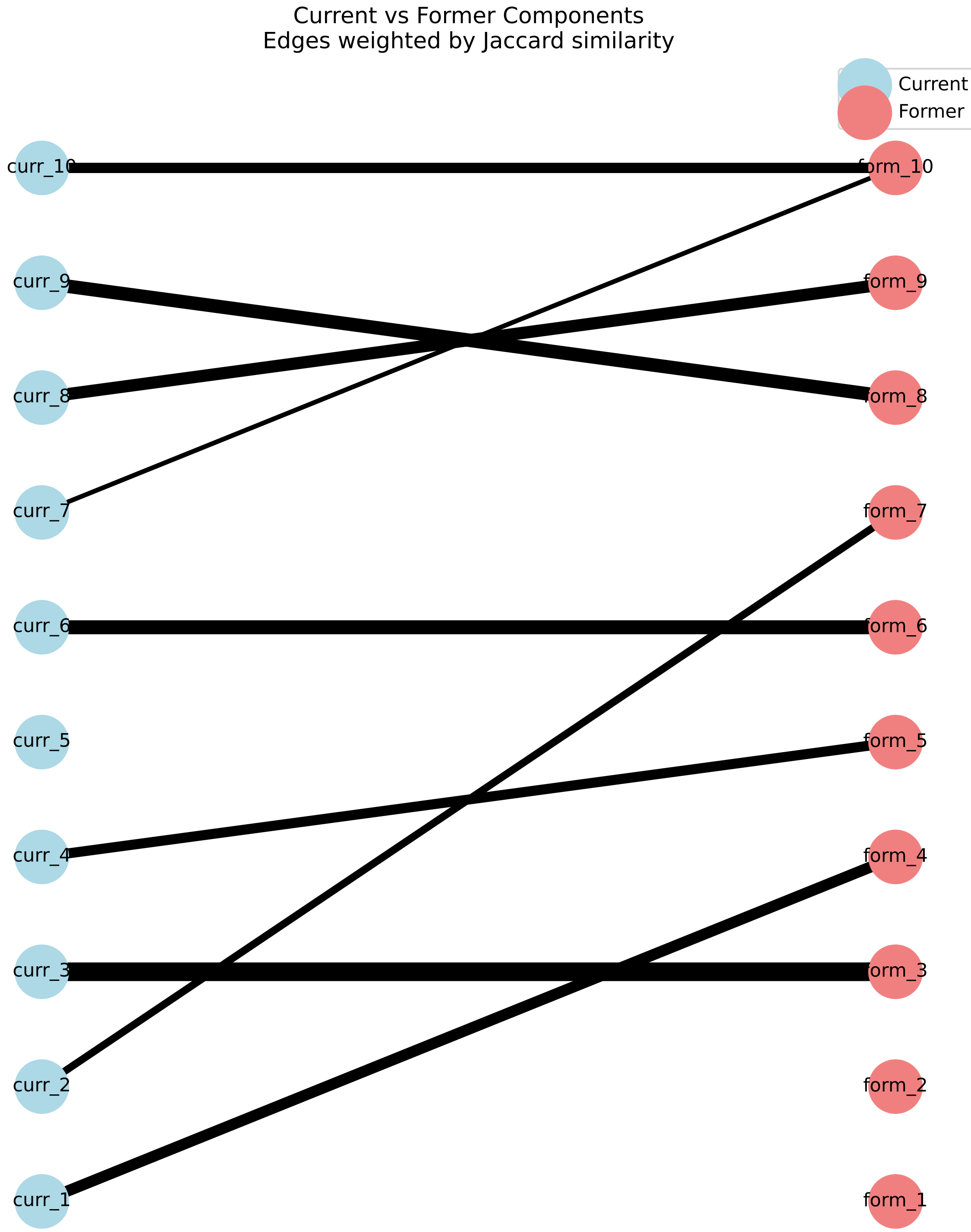
Long Chain Saturated Fatty Acid myristate (14:0)	Long Chain Monounsaturated Fatty Acid 10-heptadecenoate (17:1n7)
Long Chain Saturated Fatty Acid myristate (14:0)	Fatty Acid, Monohydroxy 16-hydroxypalmitate
Long Chain Saturated Fatty Acid myristate (14:0)	Fatty Acid, Monohydroxy 9-hydroxystearate
Long Chain Saturated Fatty Acid pentadecanoate (15:0)	Long Chain Monounsaturated Fatty Acid 10-nonadecenoate (19:1n9)
Long Chain Saturated Fatty Acid pentadecanoate (15:0)	Long Chain Monounsaturated Fatty Acid 10-heptadecenoate (17:1n7)
Phosphatidylcholine (PC) 1-palmitoyl-2-oleoyl-GPC (16:0/18:1)	Phosphatidylcholine (PC) 1-stearoyl-2-oleoyl-GPC (18:0/18:1)
Phosphatidylcholine (PC) 1-palmitoyl-2-oleoyl-GPC (16:0/18:1)	Phosphatidylinositol (PI) 1-palmitoyl-2-oleoyl-GPI (16:0/18:1)*
Long Chain Polyunsaturated Fatty Acid (n3 and n6) eicosapentaenoate (EPA; 20:5n3)	Long Chain Polyunsaturated Fatty Acid (n3 and n6) docosahexaenoate (DHA; 22:6n3)
Long Chain Polyunsaturated Fatty Acid (n3 and n6) eicosapentaenoate (EPA; 20:5n3)	Long Chain Polyunsaturated Fatty Acid (n3 and n6) stearidonate (18:4n3)
Xanthine Metabolism theophylline	Xanthine Metabolism paraxanthine
Long Chain Polyunsaturated Fatty Acid (n3 and n6) docosahexaenoate (DHA; 22:6n3)	Long Chain Polyunsaturated Fatty Acid (n3 and n6) stearidonate (18:4n3)
Long Chain Polyunsaturated Fatty Acid (n3 and n6) docosahexaenoate (DHA; 22:6n3)	Long Chain Polyunsaturated Fatty Acid (n3 and n6) dihomo-linolenate (20:3n3 or n6)
Fatty Acid, Dicarboxylate maleate	Food Component/Plant N-(2-furoyl)glycine
Androgenic Steroids dehydroepiandrosterone sulfate (DHEA-S)	Androgenic Steroids epiandrosterone sulfate
Androgenic Steroids dehydroepiandrosterone sulfate (DHEA-S)	Androgenic Steroids androstenediol (3beta,17beta) disulfate (1)
Fatty Acid, Monohydroxy 3-hydroxydecanoate	Fatty Acid Metabolism (Acyl Carnitine, Hydroxy) 3-hydroxyoleoylcarnitine
Androgenic Steroids androsterone sulfate	Androgenic Steroids epiandrosterone sulfate
Food Component/Plant N-(2-furoyl)glycine	Benzoate Metabolism 3-methyl catechol sulfate (2)
Food Component/Plant N-(2-furoyl)glycine	Chemical 3-acetylphenol sulfate
Food Component/Plant N-(2-furoyl)glycine	Chemical 3-hydroxypyridine sulfate
Long Chain Polyunsaturated Fatty Acid (n3 and n6) adrenate (22:4n6)	Long Chain Polyunsaturated Fatty Acid (n3 and n6) docosapentaenoate (n6 DPA; 22:5n6)
Medium Chain Fatty Acid 5-dodecenoate (12:1n7)	Long Chain Monounsaturated Fatty Acid 10-heptadecenoate (17:1n7)
Lysophospholipid 1-palmitoyl-GPC (16:0)	Lysoplasmalogen 1-(1-enyl-palmitoyl)-GPC (P-16:0)*

Lysophospholipid 1-stearoyl-GPC (18:0)	Lysophospholipid 1-oleoyl-GPC (18:1)
Lysophospholipid 1-oleoyl-GPC (18:1)	Lysoplasmalogen 1-(1-enyl-palmitoyl)-GPC (P-16:0)*
Androgenic Steroids epiandrosterone sulfate	Androgenic Steroids 5alpha-androstan-3beta,17beta-diol disulfate
Benzoate Metabolism catechol sulfate	Chemical 3-hydroxypyridine sulfate
Lysophospholipid 1-palmitoyl-GPI (16:0)	Lysophospholipid 1-oleoyl-GPI (18:1)
Phosphatidylethanolamine (PE) 1-stearoyl-2-oleoyl-GPE (18:0/18:1)	Phosphatidylethanolamine (PE) 1-stearoyl-2-linoleoyl-GPE (18:0/18:2)*
Phosphatidylethanolamine (PE) 1-palmitoyl-2-linoleoyl-GPE (16:0/18:2)	Phosphatidylethanolamine (PE) 1-stearoyl-2-linoleoyl-GPE (18:0/18:2)*
Sphingomyelins sphingomyelin (d18:1/18:1, d18:2/18:0)	Sphingomyelins sphingomyelin (d18:1/20:1, d18:2/20:0)*
Lysoplasmalogen 1-(1-enyl-palmitoyl)-GPC (P-16:0)*	Lysoplasmalogen 1-(1-enyl-palmitoyl)-GPE (P-16:0)*
Lysoplasmalogen 1-(1-enyl-palmitoyl)-GPC (P-16:0)*	Lysoplasmalogen 1-(1-enyl-stearoyl)-GPE (P-18:0)*
Fatty Acid Metabolism (Acyl Carnitine, Polyunsaturated) linoleoylcarnitine (C18:2)*	Fatty Acid Metabolism (Acyl Carnitine, Polyunsaturated) dihomo-linolenoylcarnitine (C20:3n3 or 6)*
Benzoate Metabolism 3-methyl catechol sulfate (2)	Chemical 3-hydroxypyridine sulfate
Benzoate Metabolism 3-methyl catechol sulfate (2)	TCA Cycle citraconate/glutaconate
Benzoate Metabolism guaiacol sulfate	Chemical 3-hydroxypyridine sulfate
Benzoate Metabolism guaiacol sulfate	Benzoate Metabolism 3-methoxycatechol sulfate (2)
Chemical 3-acetylphenol sulfate	Chemical 3-hydroxypyridine sulfate
Chemical 3-acetylphenol sulfate	Benzoate Metabolism 3-methoxycatechol sulfate (2)
Sphingomyelins sphingomyelin (d18:1/14:0, d16:1/16:0)*	Sphingomyelins sphingomyelin (d17:1/16:0, d18:1/15:0, d16:1/17:0)*
Lysophospholipid 1-linolenoyl-GPC (18:3)*	Phosphatidylcholine (PC) 1-palmitoleoyl-2-linolenoyl-GPC (16:1/18:3)*
Phosphatidylethanolamine (PE) 1-stearoyl-2-linoleoyl-GPE (18:0/18:2)*	Phosphatidylethanolamine (PE) 1-stearoyl-2-arachidonoyl-GPE (18:0/20:4)
Phosphatidylethanolamine (PE) 1-palmitoyl-2-arachidonoyl-GPE (16:0/20:4)*	Phosphatidylethanolamine (PE) 1-oleoyl-2-arachidonoyl-GPE (18:1/20:4)*
Plasmalogen 1-(1-enyl-palmitoyl)-2-linoleoyl-GPE (P-16:0/18:2)*	Plasmalogen 1-(1-enyl-stearoyl)-2-linoleoyl-GPE (P-18:0/18:2)*
Diacylglycerol palmitoyl-oleoyl-glycerol (16:0/18:1) [2]*	Diacylglycerol oleoyl-oleoyl-glycerol (18:1/18:1) [1]*

Diacylglycerol palmitoyl-oleoyl-glycerol (16:0/18:1) [2]*	Diacylglycerol oleoyl-oleoyl-glycerol (18:1/18:1) [2]*
Tyrosine Metabolism 1-carboxyethyltyrosine	Leucine, Isoleucine and Valine Metabolism 1-carboxyethylvaline

results/005/form-metabs/unique-edges-at-shared-nodes.txt

Smoker Component Pairs



Current Smoker Component Metabolites

Component 1: pseudouridine, N-acetylserine, hydroxyasparagine**, C-glycosyltryptophan, N-acetylalanine, N-formylmethionine, 5,6-dihydrouridine,

Component 2: maleate, N-(2-furoyl)glycine, citraconate/glutaconate, quinate, trigonelline (N'-methylnicotinate), caffeic acid sulfate, 3-hydroxypyridine sulfate,

Component 3: 5-acetylamino-6-formylamino-3-methyluracil, caffeine, 1,3,7-trimethylurate, paraxanthine, 1-methylxanthine, 5-acetylamino-6-amino-3-methyluracil, 1,7-dimethylurate, theophylline,

Component 4: 21-hydroxypregnenolone disulfate, androstenediol (3beta,17beta) disulfate (2), 5alpha-androstan-3beta,17beta-diol disulfate, androstenediol (3beta,17beta) monosulfate (1), androstenediol (3beta,17beta) disulfate (1), dehydroepiandrosterone sulfate (DHEA-S), pregnenediol sulfate (C21H34O5S)*, pregnenolone sulfate, pregnenediol disulfate (C21H34O8S2)*,

Component 5: gamma-glutamylmethionine, isoleucine, gamma-glutamylphenylalanine, gamma-glutamylisoleucine*, gamma-glutamyl-alpha-lysine, gamma-glutamyl-epsilon-lysine, valine, leucine, lysine, gamma-glutamylleucine, methionine,

Component 6: sphingomyelin (d18:2/14:0, d18:1/14:1)*, sphingomyelin (d18:1/18:1, d18:2/18:0), sphingomyelin (d18:1/20:1, d18:2/20:0)*, sphingomyelin (d17:1/16:0, d18:1/15:0, d16:1/17:0)*, stearoyl sphingomyelin (d18:1/18:0), sphingomyelin (d18:1/17:0, d17:1/18:0, d19:1/16:0), sphingomyelin (d18:2/18:1)*, sphingomyelin (d18:2/21:0, d16:2/23:0)*, sphingomyelin (d18:2/23:1)*, sphingomyelin (d18:1/22:2, d18:2/22:1, d16:1/24:2)*, sphingomyelin (d18:1/20:2, d18:2/20:1, d16:1/22:2)*, sphingomyelin (d18:1/14:0, d16:1/16:0)*, sphingomyelin (d17:2/16:0, d18:2/15:0)*, sphingomyelin (d18:1/19:0, d19:1/18:0)*, myristoyl dihydrosphingomyelin (d18:0/14:0)*, sphingomyelin (d17:1/14:0, d16:1/15:0)*, sphingomyelin (d18:1/20:0, d16:1/22:0)*,

Component 7: linoleoylcarnitine (C18:2)*, decanoylcarnitine (C10), arachidonoylcarnitine (C20:4), palmitoleoylcarnitine (C16:1)*, cis-4-decenoylcarnitine (C10:1), laurylcarnitine (C12), linolenoylcarnitine (C18:3)*, myristoleoylcarnitine (C14:1)*, palmitoylcarnitine (C16), 5-dodecenoylcarnitine (C12:1), 3-hydroxyoleoylcarnitine, dihomo-linoleoylcarnitine (C20:2)*, dihomo-linolenoylcarnitine (C20:3n3 or 6)*, oleoylcarnitine (C18:1), hexanoylcarnitine (C6), myristoylcarnitine (C14), nonanoylcarnitine (C9), octanoylcarnitine (C8),

Component 8: 1-palmitoleoyl-2-linolenoyl-GPC (16:1/18:3)*, 1-palmitoyl-2-oleoyl-GPC (16:0/18:1), 1-myristoyl-2-palmitoyl-GPC (14:0/16:0), 1-palmitoyl-2-oleoyl-GPI (16:0/18:1)*, 1-stearoyl-2-oleoyl-GPC (18:0/18:1), 1-(1-enyl-palmitoyl)-GPC (P-16:0)*, 1-oleoyl-GPE (18:1), 1-palmitoleoyl-GPC (16:1)*, 1-stearoyl-2-oleoyl-GPI (18:0/18:1)*, 1-oleoyl-GPC (18:1), 1-palmitoyl-2-linoleoyl-GPI (16:0/18:2), 1-linoleoyl-GPE (18:2)*, 1-linoleoyl-GPC (18:2), 1-palmitoyl-GPC (16:0), 1-linolenoyl-GPC (18:3)*, 1-stearoyl-2-linoleoyl-GPI (18:0/18:2), 1-palmitoyl-GPE (16:0), 1-palmitoyl-2-palmitoleoyl-GPC (16:0/16:1)*, 1-stearoyl-GPC (18:0), 1-stearoyl-GPE (18:0),

Component 9: oleoyl-linoleoyl-glycerol (18:1/18:2) [2], palmitoyl-linoleoyl-glycerol (16:0/18:2) [1]*, 1-stearoyl-2-arachidonoyl-GPE (18:0/20:4), oleoyl-arachidonoyl-glycerol (18:1/20:4) [2]*, 1-palmitoyl-2-arachidonoyl-GPE (16:0/20:4)*, 1-oleoyl-2-arachidonoyl-GPE (18:1/20:4)*, palmitoleoyl-linoleoyl-glycerol (16:1/18:2) [1]*, oleoyl-oleoyl-glycerol (18:1/18:1) [2]*, oleoyl-linoleoyl-glycerol (18:1/18:2) [1], palmitoyl-arachidonoyl-glycerol (16:0/20:4) [2]*, diacylglycerol (14:0/18:1, 16:0/16:1) [1]*, oleoyl-oleoyl-glycerol (18:1/18:1) [1]*, linoleoyl-linoleoyl-glycerol (18:2/18:2) [1]*, diacylglycerol (16:1/18:2 [2], 16:0/18:3 [1])*, palmitoyl-linoleoyl-glycerol (16:0/18:2) [2]*, 1-palmitoyl-2-linoleoyl-GPE (16:0/18:2), linoleoyl-arachidonoyl-glycerol (18:2/20:4) [2]*, palmitoyl-oleoyl-glycerol (16:0/18:1) [2]*, 1-stearoyl-2-oleoyl-GPE (18:0/18:1), palmitoyl-arachidonoyl-glycerol (16:0/20:4) [1]*, 1-palmitoyl-2-oleoyl-GPE (16:0/18:1),

Component 10: 16-hydroxypalmitate, 9-hydroxystearate, linolenate [alpha or gamma; (18:3n3 or 6)], docosadienoate (22:2n6), 3-hydroxyhexanoate, 3-hydroxymyristate, cis-4-decenoate (10:1n6)*, linoleate (18:2n6), myristoleate (14:1n5), margarate (17:0), 3-hydroxylaurate, 3-hydroxydecanoate, 10-nonadecenoate (19:1n9), eicosenoate (20:1), arachidonate (20:4n6), (14 or 15)-methylpalmitate (a17:0 or i17:0), myristate (14:0), palmitate (16:0), stearate (18:0), dihomo-linolenate (20:3n3 or n6), docosahexaenoate (DHA; 22:6n3), 3-hydroxyoctanoate, dodecadienoate (12:2)*, 3-hydroxybutyrate (BHBA), stearidonate (18:4n3), oleate/vaccenate (18:1), nonadecanoate (19:0), tetradecadienoate (14:2)*, pentadecanoate (15:0), (16 or 17)-methylstearate (a19:0 or i19:0), 5-dodecenoate (12:1n7), docosapentaenoate (n3 DPA; 22:5n3), palmitoleate (16:1n7), dihomo-linoleate (20:2n6), eicosapentaenoate (EPA; 20:5n3), hexadecadienoate (16:2n6), stearoyl ethanolamide, docosapentaenoate (n6 DPA; 22:5n6), 10-heptadecenoate (17:1n7), adrenate (22:4n6),

Former Smoker Component Metabolites

Component 1: glucuronide of piperine metabolite C17H21NO3 (5)*, glucuronide of piperine metabolite C17H21NO3 (3)*, piperine, sulfate of piperine metabolite C16H19NO3 (3)*, sulfate of piperine metabolite C16H19NO3 (2)*, glucuronide of piperine metabolite C17H21NO3 (4)*,

Component 2: 1-(1-enyl-stearoyl)-2-arachidonoyl-GPE (P-18:0/20:4)*, 1-(1-enyl-stearoyl)-2-oleoyl-GPE (P-18:0/18:1), 1-(1-enyl-palmitoyl)-2-arachidonoyl-GPE (P-16:0/20:4)*, 1-(1-enyl-palmitoyl)-2-oleoyl-GPE (P-16:0/18:1)*, 1-(1-enyl-palmitoyl)-2-linoleoyl-GPE (P-16:0/18:2)*, 1-(1-enyl-stearoyl)-2-linoleoyl-GPE (P-18:0/18:2)*,

Component 3: 5-acetylamino-6-formylamino-3-methyluracil, caffeine, 1,3,7-trimethylurate, paraxanthine, 1-methylxanthine, 5-acetylamino-6-amino-3-methyluracil, 1,7-dimethylurate, theophylline,

Component 4: pseudouridine, N-acetylserine, N2,N2-dimethylguanosine, hydroxyasparagine**, C-glycosyltryptophan, N-acetylalanine, N6-carbamoylthreonyladenosine, N-formylmethionine, gamma-carboxyglutamate, 2,3-dihydroxy-5-methylthio-4-pentenoate (DMTPA)*, 5,6-dihydrouridine,

Component 5: epiandrosterone sulfate, androsterone sulfate, androstenediol (3beta,17beta) disulfate (2), 21-hydroxypregnenolone disulfate, androstenediol (3alpha, 17alpha) monosulfate (3), 5alpha-androstan-3beta,17beta-diol disulfate, pregnenediol disulfate (C21H34O8S2)*, androstenediol (3beta,17beta) monosulfate (1), androstenediol (3beta,17beta) disulfate (1), androstenediol (3beta,17beta) monosulfate (2), dehydroepiandrosterone sulfate (DHEA-S), pregnenolone sulfate, pregnenediol sulfate (C21H34O5S)*,

Component 6: sphingomyelin (d18:2/14:0, d18:1/14:1)*, sphingomyelin (d18:1/18:1, d18:2/18:0), sphingomyelin (d18:1/20:1, d18:2/20:0)*, sphingomyelin (d17:1/16:0, d18:1/15:0, d16:1/17:0)*, stearoyl sphingomyelin (d18:1/18:0), sphingomyelin (d18:1/17:0, d17:1/18:0, d19:1/16:0), sphingomyelin (d18:2/21:0, d16:2/23:0)*, sphingomyelin (d17:2/16:0, d18:2/15:0)*, myristoyl dihydrosphingomyelin (d18:0/14:0)*, sphingomyelin (d18:1/14:0, d16:1/16:0)*, sphingomyelin (d18:1/19:0, d19:1/18:0)*, sphingomyelin (d17:1/14:0, d16:1/15:0)*, sphingomyelin (d18:1/20:0, d16:1/22:0)*,

Component 7: 3-methyl catechol sulfate (2), guaiacol sulfate, 4-vinylguaiacol sulfate, 3-methyl catechol sulfate (1), 3-methoxycatechol sulfate (2), maleate, N-(2-furoyl)glycine, citraconate/glutaconate, 3-ethylcatechol sulfate (1), quinate, trigonelline (N'-methylnicotinate), caffeic acid sulfate, 4-methoxyphenol sulfate, 3-hydroxypyridine sulfate, catechol sulfate, 3-acetylphenol sulfate,

Component 8: oleoyl-linoleoyl-glycerol (18:1/18:2) [2], palmitoyl-linoleoyl-glycerol (16:0/18:2) [1]*, 1-stearoyl-2-arachidonoyl-GPE (18:0/20:4), 1-palmitoyl-2-arachidonoyl-GPE (16:0/20:4)*, 1-oleoyl-2-arachidonoyl-GPE (18:1/20:4)*, palmitoleoyl-linoleoyl-glycerol (16:1/18:2) [1]*, oleoyl-oleoyl-glycerol (18:1/18:1) [2]*, oleoyl-linoleoyl-glycerol (18:1/18:2) [1], 1-stearoyl-2-linoleoyl-GPE (18:0/18:2)*, diacylglycerol (14:0/18:1, 16:0/16:1) [1]*, oleoyl-oleoyl-glycerol (18:1/18:1) [1]*, linoleoyl-linoleoyl-glycerol (18:2/18:2) [1]*, diacylglycerol (16:1/18:2 [2], 16:0/18:3 [1])*, palmitoyl-linoleoyl-glycerol (16:0/18:2) [2]*, 1-palmitoyl-2-linoleoyl-GPE (16:0/18:2), palmitoyl-oleoyl-glycerol (16:0/18:1) [2]*, 1-oleoyl-2-linoleoyl-GPE (18:1/18:2)*, 1-stearoyl-2-oleoyl-GPE (18:0/18:1), 1-palmitoyl-2-oleoyl-GPE (16:0/18:1),

Component 9: 1-palmitoleoyl-2-linolenoyl-GPC (16:1/18:3)*, 1-palmitoyl-2-oleoyl-GPC (16:0/18:1), 1-(1-enyl-stearoyl)-GPE (P-18:0)*, 1-myristoyl-2-palmitoyl-GPC (14:0/16:0), 1-(1-enyl-palmitoyl)-GPE (P-16:0)*, 1-palmitoyl-2-oleoyl-GPI (16:0/18:1)*, 1-linoleoyl-2-linolenoyl-GPC (18:2/18:3)*, 1-stearoyl-2-oleoyl-GPC (18:0/18:1), 1-(1-enyl-palmitoyl)-GPC (P-16:0)*, 1-palmitoyl-2-arachidonoyl-GPI (16:0/20:4)*, 1-oleoyl-GPE (18:1), 1-palmitoleoyl-GPC (16:1)*, 1-stearoyl-2-oleoyl-GPI (18:0/18:1)*, 1-oleoyl-GPC (18:1), 1-palmitoyl-2-linoleoyl-GPI (16:0/18:2), 1-linoleoyl-GPE (18:2)*, 1-palmitoyl-2-arachidonoyl-GPC (16:0/20:4n6), 1-stearoyl-2-arachidonoyl-GPC (18:0/20:4), 1-linoleoyl-GPC (18:2), 1-palmitoyl-GPC (16:0), 1-linolenoyl-GPC (18:3)*, 1-arachidonoyl-GPE (20:4n6)*, 1-myristoyl-2-arachidonoyl-GPC (14:0/20:4)*, 1-stearoyl-2-linoleoyl-GPI (18:0/18:2), 1-palmitoyl-GPE (16:0), 1-palmitoyl-2-palmitoleoyl-GPC (16:0/16:1)*, 1-stearoyl-GPC (18:0), 1,2-dilinoleoyl-GPC (18:2/18:2), 1-stearoyl-GPE (18:0), 1-arachidonoyl-GPC (20:4n6)*,

Component 10: 16-hydroxypalmitate, decanoylcarnitine (C10), hexadecanedioate (C16-DC), 9-hydroxystearate, caprylate (8:0), linolenate [alpha or gamma; (18:3n3 or 6)], docosadienoate (22:2n6), cis-4-decenoylcarnitine (C10:1), 3-hydroxyhexanoate, 3-hydroxymyristate, linolenoylcarnitine (C18:3)*, palmitoylcarnitine (C16), cis-4-decenoate (10:1n6)*, linoleate (18:2n6), nonanoylcarnitine (C9), hexadecenedioate (C16:1-DC)*, myristoleate (14:1n5), tetradecanedioate (C14-DC), margarate (17:0), laurylcarnitine (C12), 3-hydroxysebacate, 3-hydroxylaurate, 3-hydroxydecanoate, 5-dodecenoylcarnitine (C12:1), 10-nonadecenoate (19:1n9), dihomo-linoleoylcarnitine (C20:2)*, eicosenoate (20:1), arachidonate (20:4n6), oleoylcarnitine (C18:1), (14 or 15)-methylpalmitate (a17:0 or i17:0), myristate (14:0), myristoylcarnitine (C14), hexanoylcarnitine (C6), linoleoylcarnitine (C18:2)*, arachidonoylcarnitine (C20:4), palmitate (16:0), stearate (18:0), dihomo-linolenate (20:3n3 or n6), docosahexaenoate (DHA; 22:6n3), 3-hydroxyoctanoate, dodecadienoate (12:2)*, laurate (12:0), stearidonate (18:4n3), caprate (10:0), myristoleoylcarnitine (C14:1)*, margaroylcarnitine (C17)*, oleate/vaccenate (18:1), nonadecanoate (19:0), tetradecadienoate (14:2)*, octanoylcarnitine (C8), pentadecanoate (15:0), (16 or 17)-methylstearate (a19:0 or i19:0), dodecanedioate (C12-DC), 5-dodecenoate (12:1n7), docosapentaenoate (n3 DPA; 22:5n3), palmitoleate (16:1n7), dihomo-linoleate (20:2n6), eicosapentaenoate (EPA; 20:5n3), hexadecadienoate (16:2n6), stearoyl ethanolamide, palmitoleoylcarnitine (C16:1)*, octadecanedioate (C18-DC), docosapentaenoate (n6 DPA; 22:5n6), 10-heptadecenoate (17:1n7), 3-hydroxyoleoylcarnitine, dihomo-linolenoylcarnitine (C20:3n3 or 6)*, glycerol, adrenate (22:4n6), octadecenedioate (C18:1-DC),